PHYSICS EXERCISE BOOK SERIES

MECHANICS

of a Material Point

The Complete Exercise Book

From Absolute Zero to Mastery

A structured training course in vectors, kinematics, dynamics, work, energy, oscillations, and gravitation.

Every exercise comes with a fully worked solution.

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First edition

November 9, 2026

Preface

This book is a complete training companion for a first university course in **mechanics of a material point** (*mécanique d'un point matériel*). It assumes no prior knowledge of mechanics. It does assume that you can add, multiply, solve a linear equation, and read a graph. Everything else—vectors, derivatives, integrals, units, trigonometry—is introduced when it is needed and practised immediately in the exercises.

The organisation is deliberately simple:

- **Short theory.** Each chapter opens with a compact reminder of the definitions and results you will use. This is a toolbox, not a treatise.
- **Worked examples.** One or two examples per chapter show the standard method from beginning to end.
- **Graded exercises.** Every exercise carries a difficulty mark: ⋆ warm-up, ⋆⋆ core, ⋆⋆⋆ mastery.
- **Full solutions.** Solutions are placed at the end of each chapter, so that you can attempt the exercises first and then compare your reasoning.

How to use this book

Work actively. Reading a solution creates the illusion of understanding; producing it yourself creates understanding. A good rhythm is:

1. read the toolbox and the worked example;
2. solve the warm-up exercises with the book closed;
3. solve the core exercises, allowing yourself to look at the toolbox only;
4. attempt the mastery exercises under time pressure;
5. compare with the solutions and write down *why* any mistake happened.

A note on units and significant figures

Numerical answers are given in SI units unless stated otherwise, with three significant figures in general. In intermediate steps, keep more digits than you report. Never write a physical answer without its unit.

Acknowledgement

The structure follows the four-chapter syllabus of the course *Mécanique d'un point matériel*: vector calculus, kinematics, dynamics, and work–energy. Three mastery chapters extend the syllabus with momentum, central forces, and oscillations so that the reader can move from "passing" to "commanding" the subject.

The harder you train, the simpler the exam looks.

Notation and Conventions

Symbol	Meaning
m	mass of the material point (kg)
t	time (s)
$\vec{r} = \vec{OM}$	position vector of the point M relative to origin O
$\vec{v} = \mathrm{d}\vec{r}/\mathrm{d}t$	velocity ($\mathrm{m\,s^{-1}}$)
$\vec{a} = \mathrm{d}\vec{v}/\mathrm{d}t$	acceleration ($\mathrm{m\,s^{-2}}$)
$\vec{p} = m\vec{v}$	linear momentum ($\mathrm{kg\,m\,s^{-1}}$)
$\vec{L}_O = \vec{r} \times \vec{p}$	angular momentum about O ($\mathrm{kg\,m^2\,s^{-1}}$)
$\vec{F}$	force (N)
$\vec{g}$	gravitational acceleration, $g \simeq 9.81\ \mathrm{m\,s^{-2}}$ near Earth
W	work (J)
P	power (W)
$E_c = \frac{1}{2}mv^2$	kinetic energy (J)
E_p	potential energy (J)
$E_T = E_c + E_p$	mechanical energy (J)
ω	angular frequency ($\mathrm{rad\,s^{-1}}$)
$\vec{\Omega}$	angular velocity of a rotating frame ($\mathrm{rad\,s^{-1}}$)
G	gravitational constant, $6.674 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$

Conventions.

- Vectors are written with arrows: $\vec{v}$, $\vec{F}$, $\vec{u}_r$.
- Unit vectors: $\hat{\imath}, \hat{\jmath}, \hat{k}$ (Cartesian), $\vec{u}_r, \vec{u}_\theta$ (polar), $\vec{u}_t, \vec{u}_n$ (tangent/normal).
- A dot over a symbol means a time derivative: $\dot{x} = \mathrm{d}x/\mathrm{d}t$, $\ddot{x} = \mathrm{d}^2x/\mathrm{d}t^2$.
- The scalar product is $\vec{a} \cdot \vec{b}$; the vector product is $\vec{a} \times \vec{b}$.
- $g = 9.81\ \mathrm{m\,s^{-2}}$ is used in numerical problems unless another value is specified.

Contents

Preface i

Notation and Conventions ii

I Foundations 1

1 Starting from Zero 2
1.1 Physical quantities, units, and dimensions . . . 2
1.2 Scientific notation and significant figures . . . 3
1.3 Trigonometry you must own . . . 3
1.4 Reading graphs: slope and area . . . 3
1.5 Worked examples . . . 4
1.6 Exercises and Step-by-Step Solutions . . . 4

2 Vectors and Vector Calculus 11
2.1 Scalars and vectors . . . 11
2.2 Vector algebra . . . 11
2.3 Vector functions and their derivatives . . . 13
2.4 Differential operators . . . 13
2.5 Worked examples . . . 13
2.6 Exercises and Step-by-Step Solutions . . . 15

II Kinematics 22

3 The Language of Motion 23
3.1 Reference frame and position . . . 23
3.2 Velocity and acceleration . . . 23
3.3 Graphical interpretation . . . 23
3.4 Worked examples . . . 24
3.5 Exercises and Step-by-Step Solutions . . . 25

4 Rectilinear Motion 32
4.1 Uniform rectilinear motion . . . 32
4.2 Uniformly accelerated rectilinear motion . . . 32
4.3 Free fall . . . 32
4.4 Variable acceleration . . . 32
4.5 Worked examples . . . 34
4.6 Exercises and Step-by-Step Solutions . . . 35

5 Motion in a Plane: Projectiles and Circles 42
5.1 Projectile motion . . . 42
5.2 Circular motion . . . 43
5.3 Worked examples . . . 44
5.4 Exercises and Step-by-Step Solutions . . . 45

6 Curvilinear Coordinates 52
6.1 Polar coordinates . . . 52
6.2 Cylindrical coordinates . . . 53
6.3 Spherical coordinates . . . 53
6.4 Worked examples . . . 54
6.5 Exercises and Step-by-Step Solutions . . . 55

7 Relative and Composite Motion 63

7.1 Galilean transformation . . . 63
7.2 Composition of velocities and accelerations . . . 63
7.3 Non-inertial frames and fictitious forces . . . 65
7.4 Worked examples . . . 65
7.5 Exercises and Step-by-Step Solutions . . . 66

III Dynamics 74

8 Newton's Laws and Free-Body Diagrams 75
8.1 Newton's laws . . . 75
8.2 Standard force models . . . 75
8.3 The free-body diagram method . . . 76
8.4 Dynamics of circular motion . . . 76
8.5 Worked examples . . . 77
8.6 Exercises and Step-by-Step Solutions . . . 77

9 Forces at Work: Friction, Inclines, Pulleys, Springs 85
9.1 Solid friction . . . 85
9.2 Rough inclined plane . . . 85
9.3 Connected bodies and pulleys . . . 86
9.4 Elastic springs . . . 86
9.5 Viscous drag and terminal velocity . . . 88
9.6 Worked examples . . . 88
9.7 Exercises and Step-by-Step Solutions . . . 89

10 Dynamics in Non-Inertial Reference Frames 97
10.1 Inertial vs Non-Inertial Frames . . . 97
10.2 Composition of Accelerations and Fictitious Forces . . . 97
10.3 Worked Examples . . . 99
10.4 Exercises and Step-by-Step Solutions . . . 100

11 Angular Momentum and Central Forces 113
11.1 Angular Momentum and Torque . . . 113
11.2 Central Forces and Conservation of Angular Momentum . . . 113
11.3 The Effective Potential . . . 114
11.4 Worked Examples . . . 115
11.5 Exercises and Step-by-Step Solutions . . . 117

IV Work and Energy 129

12 Work, Power, and the Work-Energy Theorem 130
12.1 Definition of Work . . . 130
12.2 Instantaneous Power . . . 131
12.3 The Work-Energy Theorem . . . 131
12.4 Work Done by Common Forces . . . 131
12.5 Worked Examples . . . 131
12.6 Exercises and Step-by-Step Solutions . . . 133

13 Potential Energy and Conservative Forces 141
13.1 Conservative Forces and Potential Energy . . . 141
13.2 Conservation of Mechanical Energy . . . 142

13.3 Worked Examples . . . 142
13.4 Exercises and Step-by-Step Solutions . . . 143

14 1D Potential Wells, Stability, and Small Oscillations 152
14.1 Analysis of 1D Potential Energy Curves . . . 152
14.2 Permitted Regions and Turning Points . . . 153
14.3 Equilibrium and Stability Criteria . . . 153
14.4 Small Oscillations around a Stable Minimum . . . 153
14.5 Worked Examples . . . 154
14.6 Exercises and Step-by-Step Solutions . . . 155

15 Harmonic, Damped, and Driven Oscillations 162
15.1 The Simple Harmonic Oscillator . . . 162
15.2 Damped Harmonic Oscillations . . . 162
15.3 Driven (Forced) Oscillations and Resonance . . . 163
15.4 Worked Examples . . . 164
15.5 Exercises and Step-by-Step Solutions . . . 166

V Mastery 172

16 Systems of Particles, Collisions, and Synoptic Mastery 173
16.1 Center of Mass and Total Momentum . . . 173
16.2 Classification of Collisions in 1D . . . 174
16.3 Center-of-Mass Frame and Koenig's First Theorem . . . 174
16.4 Worked Examples . . . 175
16.5 Exercises and Step-by-Step Solutions . . . 176

Appendices 184

Appendices 185

A Mathematical Reference Toolbox 185
A.1 Coordinate Systems and Unit Vectors . . . 185
A.2 Vector Differential Identities . . . 186
A.3 Essential Taylor Series Approximations ($x \to 0$) . . . 186

B Physical Constants and Astronomical Data 187
B.1 Fundamental Physical Constants . . . 187
B.2 Earth, Moon, and Solar Data . . . 187

C Comprehensive Formula Sheet and Problem-Solving Protocols 189
C.1 The Five-Step Mechanics Problem-Solving Protocol . . . 189
C.2 Core Formula Quick-Reference Sheet . . . 190

PART I

Foundations

CHAPTER 1

Starting from Zero

You will learn to: use SI units correctly; convert between units; check formulas with dimensions; handle scientific notation and significant figures; use trigonometry to resolve vectors; and read the slope and area of a graph. These are the tools that every later chapter assumes.

1.1 Physical quantities, units, and dimensions

A **physical quantity** is a number together with a unit. Writing "$v = 5$" is meaningless; writing "$v = 5\ \mathrm{m\,s^{-1}}$" is a statement about the world. Mechanics uses three base dimensions:

Dimension	Symbol	SI base unit
length	L	metre (m)
mass	M	kilogram (kg)
time	T	second (s)

Derived quantities are combinations of these. For example

$$[v] = \mathrm{L\,T^{-1}}, \qquad [a] = \mathrm{L\,T^{-2}}, \qquad [F] = \mathrm{M\,L\,T^{-2}}.$$

The last one defines the newton: $1\ \mathrm{N} = 1\ \mathrm{kg\,m\,s^{-2}}$.

Dimensional homogeneity

Both sides of a correct physical equation must have the same dimensions. You may add or equate quantities only if they share dimensions. You may multiply and divide any quantities, combining their dimensions.

The most common first-year error

Adding a length to a time, or a speed to an acceleration, is always wrong. If an equation contains $x + vt$, then vt must be a length; check it before you continue.

1.1.1 Prefixes and conversions

Prefix	Symbol	Factor	Example
giga	G	10^{9}	$1\ \text{Gm} = 10^{9}\ \text{m}$
mega	M	10^{6}	$1\ \text{MPa} = 10^{6}\ \text{Pa}$
kilo	k	10^{3}	$1\ \text{km} = 10^{3}\ \text{m}$
centi	c	10^{-2}	$1\ \text{cm} = 10^{-2}\ \text{m}$
milli	m	10^{-3}	$1\ \text{mm} = 10^{-3}\ \text{m}$
micro	μ	10^{-6}	$1\ \mu\text{m} = 10^{-6}\ \text{m}$
nano	n	10^{-9}	$1\ \text{nm} = 10^{-9}\ \text{m}$

To convert units, multiply by a fraction equal to one. For example,

$$72\ \text{km}\,\text{h}^{-1} = 72\ \frac{\text{km}}{\text{h}} \times \frac{1000\ \text{m}}{1\ \text{km}} \times \frac{1\ \text{h}}{3600\ \text{s}} = 20\ \text{m}\,\text{s}^{-1}.$$

1.2 Scientific notation and significant figures

Write very large or very small numbers as $a \times 10^{n}$ with $1 \le |a| < 10$. In calculations, keep all digits in intermediate steps and round only the final answer. A reasonable rule for this book is three significant figures.

Rounding

If the first discarded digit is less than 5, round down; if it is 5 or more, round up. For example, $2.347 \to 2.35$ (three figures) and $0.004562 \to 0.00456$.

1.3 Trigonometry you must own

For a right triangle with angle θ,

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}, \qquad \cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}, \qquad \tan\theta = \frac{\text{opposite}}{\text{adjacent}}.$$

Also

$$\sin^{2}\theta + \cos^{2}\theta = 1, \qquad \tan\theta = \frac{\sin\theta}{\cos\theta}.$$

Angles are measured in radians in calculus: π rad $= 180^{\circ}$. The arc length of a circle of radius R subtending angle θ is $s = R\theta$.

θ	0	30°	45°	60°	90°
$\sin\theta$	0	$1/2$	$\sqrt{2}/2$	$\sqrt{3}/2$	1
$\cos\theta$	1	$\sqrt{3}/2$	$\sqrt{2}/2$	$1/2$	0

1.4 Reading graphs: slope and area

Two graphical operations recur constantly in mechanics.

- The **slope** of a graph of y against x is $\mathrm{d}y/\mathrm{d}x$. On a straight line $y = mx + b$, the slope is m and the intercept is b.

- The **area** under a graph of y against x, between two values of x, is $\int y\,\mathrm{d}x$. For a triangle this is $\tfrac{1}{2}\times$ base $\times$ height; for a rectangle it is base $\times$ height.

What slopes and areas mean in mechanics

On an x–t graph, the slope is velocity. On a v–t graph, the slope is acceleration and the area is displacement. On an a–t graph, the area is the change in velocity.

1.5 Worked examples

Worked example 1.1 · Dimensional check of a kinematic formula

Check the dimensions of $v^2 = v_0^2 + 2ax$ and deduce the SI unit of the product ax.

Solution 1.1. Each term must be a velocity squared. Indeed

$$[v^2] = \mathrm{L}^2\mathrm{T}^{-2}, \qquad [2ax] = [a][x] = \mathrm{L}\,\mathrm{T}^{-2}\cdot\mathrm{L} = \mathrm{L}^2\mathrm{T}^{-2}.$$

The equation is dimensionally homogeneous. The product ax therefore has the unit of velocity squared, $\mathrm{m}^2\,\mathrm{s}^{-2}$.

Worked example 1.2 · Resolving a force into components

A force of magnitude $F = 80.0$ N makes an angle of 35.0° above the positive x-axis. Find its x- and y-components.

Solution 1.2.

$$F_x = F\cos\theta = 80.0\cos 35.0^\circ = 65.5\ \mathrm{N},$$

$$F_y = F\sin\theta = 80.0\sin 35.0^\circ = 45.9\ \mathrm{N}.$$

Check: $F_x^2 + F_y^2 = (65.5)^2 + (45.9)^2 = 6400$, so $F = 80.0$ N.

1.6 Exercises and Step-by-Step Solutions

Exercise 1.1 · ⋆ Unit conversions

Convert: (a) $72\ \mathrm{km\,h^{-1}}$ to $\mathrm{m\,s^{-1}}$; (b) $1.0\ \mathrm{g\,cm^{-3}}$ to $\mathrm{kg\,m^{-3}}$; (c) 1.0 kWh to joules.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 1.1. (a) $72\ \text{km}\,\text{h}^{-1} = 72 \times 1000/3600 = 20.0\ \text{m}\,\text{s}^{-1}$.
(b) $1\ \text{g}\,\text{cm}^{-3} = 10^{-3}\ \text{kg}/(10^{-2}\ \text{m})^3 = 10^{-3}/10^{-6} = 10^{3}\ \text{kg}\,\text{m}^{-3}$.
(c) $1\ \text{kWh} = 10^{3}\ \text{W} \times 3600\ \text{s} = 3.60 \times 10^{6}\ \text{J}$.

Exercise 1.2 · ⋆ Dimensional analysis

(a) Show that $F = ma$ is dimensionally consistent. (b) The force of a spring is $F = -kx$, where x is a length. Find the dimensions and SI unit of the spring constant k. (c) Find the dimensions of the gravitational constant G in $F = GMm/r^2$.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 1.2. (a) $[ma] = \mathrm{M}\cdot\mathrm{L}\,\mathrm{T}^{-2} = \mathrm{M}\,\mathrm{L}\,\mathrm{T}^{-2} = [F]$.
(b) $k = F/x$, so $[k] = \mathrm{M}\,\mathrm{L}\,\mathrm{T}^{-2}/\mathrm{L} = \mathrm{M}\,\mathrm{T}^{-2}$, unit $\text{kg}\,\text{s}^{-2}$ (equivalently $\text{N}\,\text{m}^{-1}$).
(c) From $G = Fr^2/(Mm)$,

$$[G] = \frac{\mathrm{M}\,\mathrm{L}\,\mathrm{T}^{-2}\cdot\mathrm{L}^2}{\mathrm{M}^2} = \mathrm{L}^3\mathrm{M}^{-1}\mathrm{T}^{-2},$$

unit $\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}$.

Exercise 1.3 · ⋆ Scientific notation

Compute

$$\frac{(3.2\times10^{-3})(4.5\times10^{6})}{1.8\times10^{-2}}$$

and report the result with three significant figures.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 1.3.

$$\frac{3.2 \times 4.5}{1.8} \times 10^{-3+6+2} = 8.0 \times 10^{5}.$$

The result is 8.00×10^{5} to three significant figures.

Exercise 1.4 · Right-triangle trigonometry

In a right triangle the hypotenuse is 13.0 cm and one leg is 5.00 cm. Find the other leg, the two acute angles, and the values of $\sin\theta$, $\cos\theta$, and $\tan\theta$ for the angle opposite the 5.00 cm leg.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 1.4. The other leg is

$$b = \sqrt{13.0^2 - 5.00^2} = \sqrt{169 - 25} = \sqrt{144} = 12.0 \text{ cm}.$$

For the angle θ opposite the 5.00 cm leg,

$$\sin\theta = \frac{5.00}{13.0} = 0.385, \qquad \cos\theta = \frac{12.0}{13.0} = 0.923, \qquad \tan\theta = \frac{5.00}{12.0} = 0.417.$$

Thus $\theta = \arcsin(0.385) = 22.6^\circ$; the other acute angle is $90^\circ - 22.6^\circ = 67.4^\circ$.

Exercise 1.5 · Components and magnitude

A displacement has components $\Delta x = -3.00$ m and $\Delta y = 4.00$ m. Find its magnitude and the angle it makes with the positive x-axis. Draw the vector.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 1.5.

$$\|\Delta\vec{r}\| = \sqrt{(-3.00)^2 + (4.00)^2} = 5.00 \text{ m}.$$

The angle measured from the positive x-axis is

$$\theta = \arctan\frac{4.00}{-3.00}.$$

The point is in the second quadrant, so $\theta = 180^\circ - 53.1^\circ = 127^\circ$.

Exercise 1.6 · Slope of a graph

The position of a particle is measured as a function of time:

t (s)	0	2	4	6	8
x (m)	2.0	6.0	10.0	14.0	18.0

(a) Plot x against t. (b) Find the slope and intercept. (c) What physical quantity does the slope represent? (d) Write the equation of the line.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 1.6. (a) The points lie on a straight line.
(b) Choosing $(0, 2.0)$ and $(8.0, 18.0)$,

$$m = \frac{18.0 - 2.0}{8.0 - 0} = 2.0 \text{ m s}^{-1}, \qquad b = 2.0 \text{ m}.$$

(c) The slope is the constant velocity, $v = 2.0 \text{ m s}^{-1}$.

(d) $x(t) = 2.0t + 2.0$ with x in metres and t in seconds.

Exercise 1.7 · Area under a graph

A car starts from rest and its velocity increases linearly to $12.0\ \mathrm{m\,s^{-1}}$ in 6.00 s. It then travels at constant velocity for 4.00 s. Draw the v–t graph and find the total distance travelled.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 1.7. The v–t graph consists of a triangle (from 0 to 6 s) and a rectangle (from 6 to 10 s).

$$d = \underbrace{\tfrac{1}{2}(6.00)(12.0)}_{\text{triangle}} + \underbrace{(4.00)(12.0)}_{\text{rectangle}} = 36.0 + 48.0 = 84.0\ \text{m}.$$

Exercise 1.8 · The pendulum formula

The period of a simple pendulum is believed to be $T = 2\pi\sqrt{L/g}$, where L is a length and g an acceleration. (a) Verify the dimensions. (b) Compute T for $L = 1.00$ m and $g = 9.81\ \mathrm{m\,s^{-2}}$. (c) What happens to T if L is quadrupled?

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 1.8. (a) $[L/g] = \mathrm{L}/(\mathrm{L\,T^{-2}}) = \mathrm{T^2}$, so $\sqrt{L/g}$ has dimension T, as required.
(b)

$$T = 2\pi\sqrt{\frac{1.00}{9.81}} = 2.01\ \text{s}.$$

(c) $T \propto \sqrt{L}$. Quadrupling L multiplies T by $\sqrt{4} = 2$, so the period doubles to 4.01 s.

Exercise 1.9 · ⋆⋆⋆ **Density of a sphere**

A solid sphere has radius 2.00 cm and mass 250 g. (a) Find its density in $\mathrm{kg\,m^{-3}}$. (b) Find its weight on Earth. (c) If the same material forms a sphere of twice the radius, by what factor does the mass increase?

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 1.9. (a) The volume is

$$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(2.00\times10^{-2})^3 = 3.35\times10^{-5}\ \mathrm{m^3}.$$

The mass is 0.250 kg, hence

$$\rho = \frac{m}{V} = \frac{0.250}{3.35\times10^{-5}} = 7.46\times10^{3}\ \mathrm{kg\,m^{-3}}.$$

(b) $W = mg = 0.250(9.81) = 2.45$ N.
(c) $m = \rho V \propto r^3$. Doubling r multiplies the mass by $2^3 = 8$.

Exercise 1.10 · ⋆⋆⋆ **Linearisation of data**

A student suspects a relation $y = Ax^n$. She plots $\log y$ against $\log x$ and obtains a straight line of slope 3.0 passing through $(\log x, \log y) = (0, 0.30)$ (logarithms base 10). Find n and A.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 1.10. Taking base-10 logarithms,

$$\log y = \log A + n\log x.$$

The slope of the line is $n = 3.0$. At $\log x = 0$, $\log y = \log A = 0.30$, hence

$$A = 10^{0.30} = 2.0.$$

Therefore $y = 2.0\,x^3$.

CHAPTER 2

Vectors and Vector Calculus

You will learn to: add and subtract vectors; resolve them into components; compute scalar and vector products; differentiate vector functions; and use the gradient, divergence, and curl. These operations are the grammar of mechanics: position, velocity, force, torque, and work are all built from them.

2.1 Scalars and vectors

A **scalar** is fully described by a number and a unit: mass, time, temperature, energy. A **vector** also has a direction and a sense: position, velocity, acceleration, force, momentum.

A vector $\vec{A}$ is written in the Cartesian basis $(\hat{\imath}, \hat{\jmath}, \hat{k})$ as

$$\vec{A} = A_x\hat{\imath} + A_y\hat{\jmath} + A_z\hat{k}.$$

Its magnitude is

$$\left\|\vec{A}\right\| = \sqrt{A_x^2 + A_y^2 + A_z^2},$$

and its unit vector is $\vec{u}_A = \vec{A}/\left\|\vec{A}\right\|$.

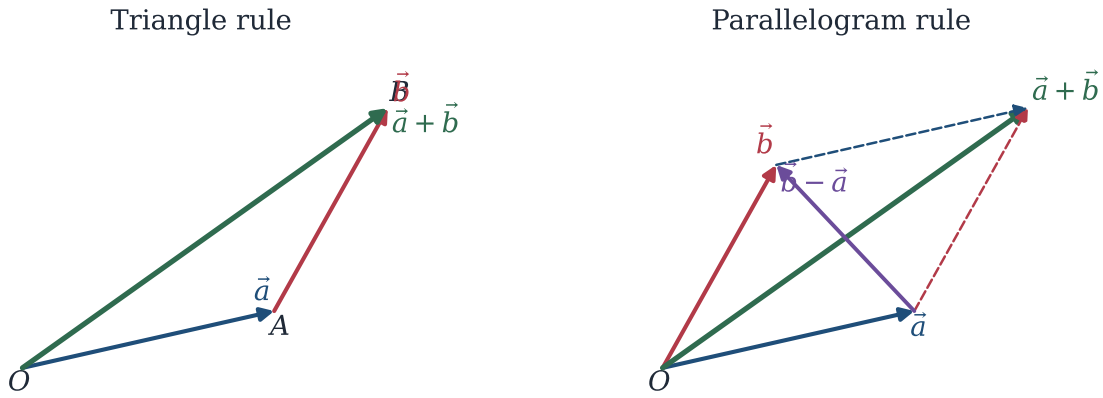


Figure 2.1: Two equivalent constructions for the sum of two vectors. The difference $\vec{b}-\vec{a}$ is the vector that must be added to $\vec{a}$ to obtain $\vec{b}$.

2.2 Vector algebra

2.2.1 Addition and subtraction

Component by component:

$$\vec{A} \pm \vec{B} = (A_x \pm B_x)\hat{\imath} + (A_y \pm B_y)\hat{\jmath} + (A_z \pm B_z)\hat{k}.$$

Graphically, use the triangle or parallelogram rule (Figure 2.1). Addition is commutative and associative.

2.2.2 Scalar (dot) product

$$\vec{A}\cdot\vec{B} = \left\|\vec{A}\right\|\left\|\vec{B}\right\|\cos\theta = A_xB_x + A_yB_y + A_zB_z.$$

The result is a scalar. In particular,

$$\vec{A}\cdot\vec{B} = 0 \iff \vec{A}\perp\vec{B}, \qquad \vec{A}\cdot\vec{A} = \left\|\vec{A}\right\|^2.$$

The projection of $\vec{B}$ onto the direction of $\vec{A}$ is $B\cos\theta = (\vec{A}\cdot\vec{B})/\left\|\vec{A}\right\|$.

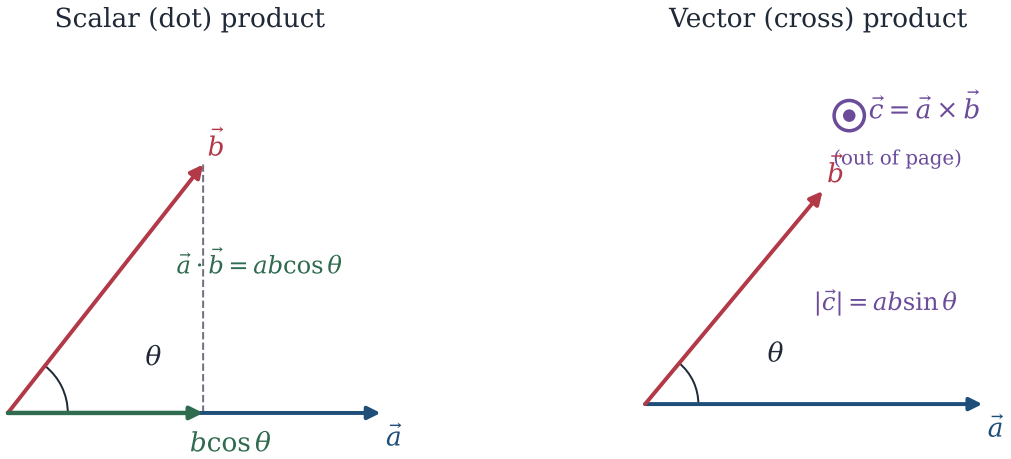


Figure 2.2: The scalar product measures projection; the vector product measures the area of the parallelogram and points along the right-hand normal.

2.2.3 Vector (cross) product

$$\vec{A}\times\vec{B} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}, \qquad \left\|\vec{A}\times\vec{B}\right\| = \left\|\vec{A}\right\|\left\|\vec{B}\right\|\sin\theta.$$

The result is a vector perpendicular to both $\vec{A}$ and $\vec{B}$, oriented by the right-hand rule (Figure 2.2). The product is anti-commutative: $\vec{A}\times\vec{B} = -(\vec{B}\times\vec{A})$. Moreover,

$$\vec{A}\times\vec{B} = \vec{0} \iff \vec{A}\parallel\vec{B}.$$

2.2.4 Triple products

The scalar triple product gives the signed volume of the parallelepiped built on the three vectors:

$$\vec{A}\cdot(\vec{B}\times\vec{C}) = \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix}.$$

The vector triple product satisfies

$$\vec{A}\times(\vec{B}\times\vec{C}) = (\vec{A}\cdot\vec{C})\vec{B} - (\vec{A}\cdot\vec{B})\vec{C}.$$

2.3 Vector functions and their derivatives

If a vector depends on time, $\vec{A}(t) = A_x(t)\hat{\imath} + A_y(t)\hat{\jmath} + A_z(t)\hat{k}$, then

$$\frac{\mathrm{d}\vec{A}}{\mathrm{d}t} = \dot{A}_x\hat{\imath} + \dot{A}_y\hat{\jmath} + \dot{A}_z\hat{k}.$$

The usual rules hold:

$$\frac{\mathrm{d}}{\mathrm{d}t}(\vec{A} + \vec{B}) = \dot{\vec{A}} + \dot{\vec{B}}, \qquad \frac{\mathrm{d}}{\mathrm{d}t}(\vec{A} \cdot \vec{B}) = \dot{\vec{A}} \cdot \vec{B} + \vec{A} \cdot \dot{\vec{B}},$$

$$\frac{\mathrm{d}}{\mathrm{d}t}(\vec{A} \times \vec{B}) = \dot{\vec{A}} \times \vec{B} + \vec{A} \times \dot{\vec{B}}.$$

Why derivatives matter

Velocity is the time derivative of the position vector and acceleration is the time derivative of velocity. Master the differentiation of vectors and kinematics becomes bookkeeping.

2.4 Differential operators

For a scalar field $f(x, y, z)$ and a vector field $\vec{V} = V_x\hat{\imath} + V_y\hat{\jmath} + V_z\hat{k}$:

$$\begin{aligned}
\operatorname{grad} f &= \frac{\partial f}{\partial x}\hat{\imath} + \frac{\partial f}{\partial y}\hat{\jmath} + \frac{\partial f}{\partial z}\hat{k}, \\
\operatorname{div} \vec{V} &= \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z}, \\
\operatorname{rot} \vec{V} &= \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ V_x & V_y & V_z \end{vmatrix}, \\
\Delta f = \operatorname{div}(\operatorname{grad} f) &= \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}.
\end{aligned}$$

Two identities are used repeatedly:

$$\operatorname{rot}(\operatorname{grad} f) = \vec{0}, \qquad \operatorname{div}(\operatorname{rot} \vec{V}) = 0.$$

The first says that every gradient field is irrotational; it is the mathematical basis of conservative forces in **??**.

2.5 Worked examples

Worked example 2.1 · Full vector algebra

Let $\vec{A} = 2\hat{\imath} - 3\hat{\jmath} + \hat{k}$ and $\vec{B} = -\hat{\imath} + 4\hat{\jmath} - 2\hat{k}$. Compute $\vec{A} + \vec{B}$, $\left\|\vec{A}\right\|$, $\vec{A} \cdot \vec{B}$, the angle between them, and $\vec{A} \times \vec{B}$.

Solution 2.1.

$$\vec{A} + \vec{B} = (2 - 1)\hat{\imath} + (-3 + 4)\hat{\jmath} + (1 - 2)\hat{k} = \hat{\imath} + \hat{\jmath} - \hat{k}.$$

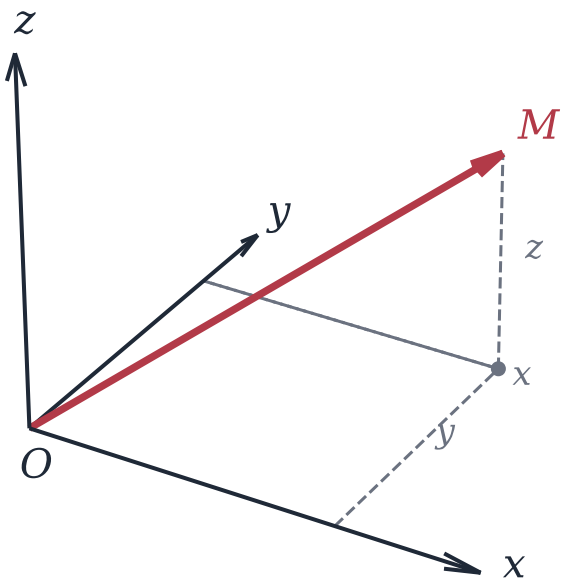


Figure 2.3: The Cartesian components of the position vector $\vec{OM}$.

$$\left\|\vec{A}\right\| = \sqrt{2^2 + (-3)^2 + 1^2} = \sqrt{14} = 3.74.$$

$$\vec{A} \cdot \vec{B} = (2)(-1) + (-3)(4) + (1)(-2) = -2 - 12 - 2 = -16.$$

Since $\vec{A} \cdot \vec{B} = \left\|\vec{A}\right\| \left\|\vec{B}\right\| \cos\theta$ and $\left\|\vec{B}\right\| = \sqrt{1 + 16 + 4} = \sqrt{21} = 4.58$,

$$\cos\theta = \frac{-16}{\sqrt{14}\sqrt{21}} = \frac{-16}{17.15} = -0.933, \qquad \theta = 159^\circ.$$

Finally,

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 2 & -3 & 1 \\ -1 & 4 & -2 \end{vmatrix} = (6-4)\hat{\imath} - (-4+1)\hat{\jmath} + (8-3)\hat{k} = 2\hat{\imath} + 3\hat{\jmath} + 5\hat{k}.$$

Check: $(2,3,5) \cdot (2,-3,1) = 4 - 9 + 5 = 0$ and $(2,3,5) \cdot (-1,4,-2) = -2 + 12 - 10 = 0$.

Worked example 2.2 · Gradient and conservative field

For $f(x,y,z) = x^2 y + y z^3$, compute $\operatorname{grad} f$ at $P(1,2,-1)$. Then find the force field $\vec{F} = -\operatorname{grad} f$ and verify that $\operatorname{rot} \vec{F} = \vec{0}$.

Solution 2.2.

$$\frac{\partial f}{\partial x} = 2xy, \qquad \frac{\partial f}{\partial y} = x^2 + z^3, \qquad \frac{\partial f}{\partial z} = 3yz^2.$$

At $(1,2,-1)$:

$$\operatorname{grad} f = (4)\hat{\imath} + (1-1)\hat{\jmath} + (3 \cdot 2 \cdot 1)\hat{k} = 4\hat{\imath} + 6\hat{k}.$$

Thus $\vec{F} = -4\hat{\imath} - 6\hat{k}$ at P. Since $\vec{F} = -\operatorname{grad} f$ is a gradient field, $\operatorname{rot} \vec{F} = -\operatorname{rot}(\operatorname{grad} f) = \vec{0}$ identically. Directly,

$$\vec{F} = (-2xy, -x^2 - z^3, -3yz^2),$$

and each component of the curl cancels.

2.6 Exercises and Step-by-Step Solutions

Exercise 2.1 · ⋆ Components and magnitude

Let $\vec{A} = 3\hat{\imath} + 4\hat{\jmath}$ and $\vec{B} = -2\hat{\imath} + 5\hat{\jmath}$. Find $\vec{A} + \vec{B}$, $\vec{A} - \vec{B}$, $\left\|\vec{A}\right\|$, $\left\|\vec{B}\right\|$, and the unit vectors $\vec{u}_A$ and $\vec{u}_B$.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 2.1.

$$\vec{A} + \vec{B} = (3-2)\hat{\imath} + (4+5)\hat{\jmath} = \hat{\imath} + 9\hat{\jmath},$$

$$\vec{A} - \vec{B} = (3+2)\hat{\imath} + (4-5)\hat{\jmath} = 5\hat{\imath} - \hat{\jmath}.$$

$$\left\|\vec{A}\right\| = \sqrt{9+16} = 5, \qquad \left\|\vec{B}\right\| = \sqrt{4+25} = \sqrt{29} = 5.39.$$

$$\vec{u}_A = \frac{3}{5}\hat{\imath} + \frac{4}{5}\hat{\jmath}, \qquad \vec{u}_B = \frac{-2}{\sqrt{29}}\hat{\imath} + \frac{5}{\sqrt{29}}\hat{\jmath}.$$

Exercise 2.2 · ⋆ **Dot product and angle**

For $\vec{A} = (1, 2, 3)$ and $\vec{B} = (4, -5, 6)$, compute $\vec{A} \cdot \vec{B}$ and the angle between the two vectors.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 2.2.

$$\vec{A} \cdot \vec{B} = 4 - 10 + 18 = 12.$$

$$\left\|\vec{A}\right\| = \sqrt{14} = 3.742, \qquad \left\|\vec{B}\right\| = \sqrt{77} = 8.775.$$

$$\cos\theta = \frac{12}{(3.742)(8.775)} = 0.3655, \qquad \theta = 68.6^\circ.$$

Exercise 2.3 · ⋆ **Cross product and area**

For $\vec{A} = (3, 1, 0)$ and $\vec{B} = (1, 2, -1)$, compute $\vec{A} \times \vec{B}$ and the area of the parallelogram spanned by the two vectors.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 2.3.

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 3 & 1 & 0 \\ 1 & 2 & -1 \end{vmatrix} = (-1 - 0)\hat{\imath} - (-3 - 0)\hat{\jmath} + (6 - 1)\hat{k} = -\hat{\imath} + 3\hat{\jmath} + 5\hat{k}.$$

The area is

$$\left\| \vec{A} \times \vec{B} \right\| = \sqrt{1 + 9 + 25} = \sqrt{35} = 5.92.$$

Exercise 2.4 · **Perpendicular and parallel conditions**

(a) Find λ such that $\vec{A} = (2, \lambda, 1)$ is perpendicular to $\vec{B} = (3, -1, 2)$. (b) Find μ such that $\vec{C} = (\mu, -1, 2)$ is parallel to $\vec{B}$.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 2.4. (a) Perpendicularity requires $\vec{A} \cdot \vec{B} = 0$:

$$(2)(3) + (\lambda)(-1) + (1)(2) = 0 \implies 8 - \lambda = 0 \implies \lambda = 8.$$

(b) Parallelism requires $\vec{C}$ to be a multiple of $\vec{B}$. The second component already matches, so the multiplier is 1; hence $\mu = 3$.

Exercise 2.5 · Scalar triple product

For $\vec{A} = (1, 0, 2)$, $\vec{B} = (2, 1, 0)$, and $\vec{C} = (0, 3, 1)$, compute $\vec{A} \cdot (\vec{B} \times \vec{C})$ and the volume of the parallelepiped they define.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 2.5.

$$\vec{B} \times \vec{C} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 2 & 1 & 0 \\ 0 & 3 & 1 \end{vmatrix} = \hat{\imath} - 2\hat{\jmath} + 6\hat{k}.$$

$$\vec{A} \cdot (\vec{B} \times \vec{C}) = (1)(1) + (0)(-2) + (2)(6) = 13.$$

The volume is $|13| = 13$ cubic units.

Exercise 2.6 · Vector triple product

Verify the identity $\vec{A} \times (\vec{B} \times \vec{C}) = (\vec{A} \cdot \vec{C})\vec{B} - (\vec{A} \cdot \vec{B})\vec{C}$ for $\vec{A} = (1, 2, 0)$, $\vec{B} = (0, 1, 3)$, and $\vec{C} = (2, 0, 1)$.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 2.6. First,

$$\vec{B}\times\vec{C}=\begin{vmatrix}\hat{\imath}&\hat{\jmath}&\hat{k}\\0&1&3\\2&0&1\end{vmatrix}=\hat{\imath}+6\hat{\jmath}-2\hat{k}.$$

Then

$$\vec{A}\times(\vec{B}\times\vec{C})=\begin{vmatrix}\hat{\imath}&\hat{\jmath}&\hat{k}\\1&2&0\\1&6&-2\end{vmatrix}=(-4)\hat{\imath}-(-2)\hat{\jmath}+(6-2)\hat{k}=-4\hat{\imath}+2\hat{\jmath}+4\hat{k}.$$

On the other hand, $\vec{A}\cdot\vec{C}=2$ and $\vec{A}\cdot\vec{B}=2$, so

$$2\vec{B}-2\vec{C}=(0,2,6)-(4,0,2)=(-4,2,4).$$

The two results agree.

Exercise 2.7 · Differentiating a vector

A particle has position $\vec{r}(t)=(3t^2,2t,t^3)$ (SI units). Find its velocity and acceleration at $t=1.00$ s, and the magnitudes of both.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 2.7.

$$\vec{v}(t)=\dot{\vec{r}}=(6t,2,3t^2),\qquad \vec{a}(t)=\ddot{\vec{r}}=(6,0,6t).$$

At $t=1$:

$$\vec{v}=(6,2,3)\ \mathrm{m\,s^{-1}},\qquad \|\vec{v}\|=\sqrt{36+4+9}=7.00\ \mathrm{m\,s^{-1}},$$
$$\vec{a}=(6,0,6)\ \mathrm{m\,s^{-2}},\qquad \|\vec{a}\|=\sqrt{36+36}=8.49\ \mathrm{m\,s^{-2}}.$$

Exercise 2.8 · Gradient

For $f(x,y,z)=x^2+y^2+z^2$, compute $\operatorname{grad} f$ at $P(1,2,2)$, its magnitude, and the unit vector in the direction of steepest increase.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 2.8.

$$\operatorname{grad} f = (2x, 2y, 2z).$$

At $P(1,2,2)$:

$$\operatorname{grad} f = (2,4,4), \qquad \|\operatorname{grad} f\| = \sqrt{4+16+16} = 6.00.$$

The direction of steepest increase is

$$\vec{u} = \frac{1}{6}(2,4,4) = \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right).$$

Exercise 2.9 · ⋆⋆⋆ **Divergence and curl**

For $\vec{V} = (x^2y,\ yz^2,\ xz)$, compute $\operatorname{div}\vec{V}$ and $\operatorname{rot}\vec{V}$, and verify explicitly that $\operatorname{div}(\operatorname{rot}\vec{V}) = 0$.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 2.9.

$$\operatorname{div}\vec{V} = \frac{\partial}{\partial x}(x^2y) + \frac{\partial}{\partial y}(yz^2) + \frac{\partial}{\partial z}(xz) = 2xy + z^2 + x.$$

For the curl,

$$\operatorname{rot}\vec{V} = \left(\frac{\partial V_z}{\partial y} - \frac{\partial V_y}{\partial z}, \frac{\partial V_x}{\partial z} - \frac{\partial V_z}{\partial x}, \frac{\partial V_y}{\partial x} - \frac{\partial V_x}{\partial y}\right) = (-2yz,\ -z,\ -x^2).$$

Then

$$\operatorname{div}(\operatorname{rot}\vec{V}) = \frac{\partial(-2yz)}{\partial x} + \frac{\partial(-z)}{\partial y} + \frac{\partial(-x^2)}{\partial z} = 0.$$

Exercise 2.10 · ⋆⋆⋆ **Recovering a potential**

Show that $\vec{F} = (y^2,\, 2xy + z,\, y)$ is conservative, then find a scalar function $\varphi(x, y, z)$ such that $\vec{F} = \operatorname{grad}\varphi$.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 2.10. Write $\vec{F} = (F_x, F_y, F_z) = (y^2,\, 2xy + z,\, y)$. Compute

$$\operatorname{rot}\vec{F} = \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z}, \frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x}, \frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}\right) = (1 - 1,\, 0 - 0,\, 2y - 2y) = \vec{0}.$$

The field is irrotational, hence conservative. To find φ:

$$\frac{\partial\varphi}{\partial x} = y^2 \implies \varphi = xy^2 + g(y, z).$$

Then

$$\frac{\partial\varphi}{\partial y} = 2xy + \frac{\partial g}{\partial y} = 2xy + z \implies \frac{\partial g}{\partial y} = z \implies g = yz + h(z).$$

Finally

$$\frac{\partial\varphi}{\partial z} = y + h'(z) = y \implies h'(z) = 0.$$

Thus

$$\varphi(x, y, z) = xy^2 + yz + C.$$

PART II

Kinematics

CHAPTER 3

The Language of Motion

You will learn to: define a reference frame; write the position vector of a particle; obtain velocity and acceleration by differentiation; eliminate time to find a trajectory; and interpret motion graphically.

3.1 Reference frame and position

Motion is always described *relative to a reference frame*. A frame consists of an origin O, a set of axes, and a clock. The position of a material point M is the vector

$$\vec{OM}(t) = \vec{r}(t) = x(t)\hat{\imath} + y(t)\hat{\jmath} + z(t)\hat{k}.$$

The **trajectory** is the set of points actually visited. It is obtained by eliminating the time t from the three parametric equations $x(t), y(t), z(t)$.

A **material point** is a body whose size is negligible compared with the distances involved. A train moving between two cities can be a point; a gymnast rotating about the bar cannot.

3.2 Velocity and acceleration

The average velocity between t and $t + \Delta t$ is

$$\vec{v}_{\text{av}} = \frac{\Delta \vec{r}}{\Delta t} = \frac{\vec{r}(t+\Delta t) - \vec{r}(t)}{\Delta t}.$$

The instantaneous velocity is the limit:

$$\vec{v}(t) = \frac{\mathrm{d}\vec{r}}{\mathrm{d}t} = \dot{x}\,\hat{\imath} + \dot{y}\,\hat{\jmath} + \dot{z}\,\hat{k}.$$

The acceleration is the derivative of velocity:

$$\vec{a}(t) = \frac{\mathrm{d}\vec{v}}{\mathrm{d}t} = \frac{\mathrm{d}^2\vec{r}}{\mathrm{d}t^2} = \ddot{x}\,\hat{\imath} + \ddot{y}\,\hat{\jmath} + \ddot{z}\,\hat{k}.$$

Speed versus velocity

Velocity is a vector. **Speed** is its magnitude, $v = \|\vec{v}\| = \sqrt{\dot{x}^2 + \dot{y}^2 + \dot{z}^2}$, and is never negative.

3.3 Graphical interpretation

For one-dimensional motion along x:

- the slope of the x–t graph is the velocity $v = \mathrm{d}x/\mathrm{d}t$;
- the slope of the v–t graph is the acceleration $a = \mathrm{d}v/\mathrm{d}t$;
- the area under the v–t graph is the displacement $\Delta x = \int v\,\mathrm{d}t$;
- the area under the a–t graph is the change in velocity $\Delta v = \int a\,\mathrm{d}t$.

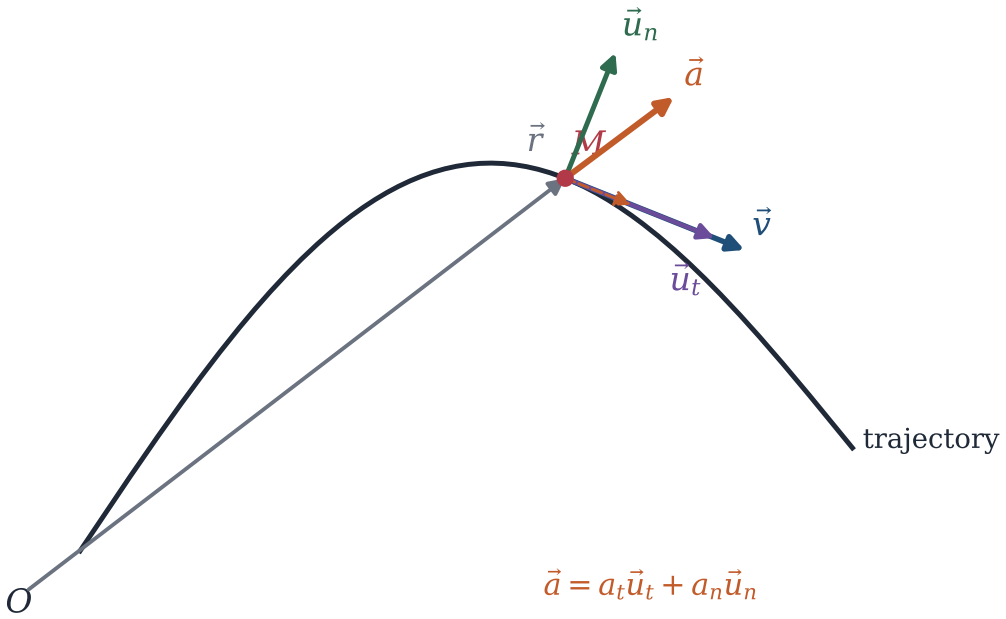


Figure 3.1: At a point of a curved trajectory, the velocity is tangent to the path. The acceleration generally has a tangential part (changing the speed) and a normal part (changing the direction).

From acceleration to position

Given $a(t)$, integrate once to get $v(t)$ and again to get $x(t)$. At each integration, use the initial conditions $v(0)$ and $x(0)$ to fix the constants.

3.4 Worked examples

Worked example 3.1 · Position, velocity, acceleration, trajectory

A particle has position $\vec{r}(t) = 2t\,\hat{\imath} + 3t^2\,\hat{\jmath} + \hat{k}$ (SI units). Find its velocity and acceleration at $t = 1.00$ s, its speed, and the equation of its trajectory.

Solution 3.1. Differentiate component by component:

$$\vec{v}(t) = \dot{\vec{r}} = 2\hat{\imath} + 6t\hat{\jmath}, \qquad \vec{a}(t) = \ddot{\vec{r}} = 6\hat{\jmath}.$$

At $t = 1$:

$$\vec{v} = 2\hat{\imath} + 6\hat{\jmath}\ \mathrm{m\,s^{-1}}, \quad v = \sqrt{4 + 36} = 6.32\ \mathrm{m\,s^{-1}},$$
$$\vec{a} = 6\hat{\jmath}\ \mathrm{m\,s^{-2}}, \quad a = 6.00\ \mathrm{m\,s^{-2}}.$$

For the trajectory, $x = 2t$ gives $t = x/2$. Then

$$y = 3t^2 = 3\left(\frac{x}{2}\right)^2 = \frac{3x^2}{4},$$

with $z = 1$. The path is a parabola in the horizontal plane $z = 1$.

Worked example 3.2 · Average versus instantaneous velocity

A particle moves along the x-axis with $x(t) = t^2 - 4t + 3$ (SI units). Find the average velocity between $t = 0$ and $t = 3.00$ s, and the instantaneous velocity at $t = 1.50$ s.

Solution 3.2.

$$x(0) = 3, \qquad x(3) = 9 - 12 + 3 = 0.$$

$$v_{\text{av}} = \frac{x(3) - x(0)}{3 - 0} = \frac{0 - 3}{3} = -1.00 \text{ m s}^{-1}.$$

The instantaneous velocity is

$$v(t) = \dot{x} = 2t - 4, \qquad v(1.50) = 2(1.50) - 4 = -1.00 \text{ m s}^{-1}.$$

The two happen to be equal here, but they are conceptually different quantities.

3.5 Exercises and Step-by-Step Solutions

Exercise 3.1 · ⋆ Straight-line kinematics

A particle has position $\vec{r}(t) = 4t\,\hat{\imath} + 3t^2\,\hat{\jmath}$ (SI units). (a) Find $\vec{v}(t)$ and $\vec{a}(t)$. (b) Find the speed at $t = 1.00$ s. (c) Find the Cartesian equation of the trajectory.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 3.1. (a)

$$\vec{v} = (4,\, 6t), \qquad \vec{a} = (0,\, 6).$$

(b) At $t = 1$: $\vec{v} = (4, 6)$, so $v = \sqrt{16 + 36} = \sqrt{52} = 7.21 \text{ m s}^{-1}$.
(c) From $x = 4t$, $t = x/4$. Then

$$y = 3\left(\frac{x}{4}\right)^2 = \frac{3x^2}{16}.$$

The trajectory is a parabola.

Exercise 3.2 · ⋆ One-dimensional motion

The position of a particle on the x-axis is $x(t) = 2t^3 - 3t^2 + 1$ (SI units). (a) Find $v(t)$ and $a(t)$. (b) Find the velocity and acceleration at $t = 2.00$ s. (c) At what times is the particle instantaneously at rest?

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 3.2. (a)

$$v(t) = 6t^2 - 6t = 6t(t-1), \qquad a(t) = 12t - 6.$$

(b) At $t = 2$:

$$v = 24 - 12 = 12.0\ \mathrm{m\,s^{-1}}, \qquad a = 24 - 6 = 18.0\ \mathrm{m\,s^{-2}}.$$

(c) $v = 0$ when $6t(t-1) = 0$, i.e. $t = 0$ and $t = 1.00$ s.

Exercise 3.3 · ⋆ **Average velocity**

A particle moves according to $x(t) = 5t^2$ (SI units). Compute the average velocity between $t = 1.00$ s and $t = 3.00$ s, and compare it with the instantaneous velocity at $t = 2.00$ s.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 3.3.

$$x(1) = 5, \qquad x(3) = 45,$$

$$v_{\mathrm{av}} = \frac{45 - 5}{3 - 1} = 20.0\ \mathrm{m\,s^{-1}}.$$

The instantaneous velocity is $v(t) = 10t$, so $v(2) = 20.0\ \mathrm{m\,s^{-1}}$. They are equal because $x \propto t^2$ is quadratic; the average over a symmetric interval equals the instantaneous value at the midpoint.

Exercise 3.4 · Uniform circular motion

A particle moves in the xy-plane with $\vec{r}(t) = R\cos(\omega t)\,\hat{\imath} + R\sin(\omega t)\,\hat{\jmath}$, where R and ω are constants. (a) Show that the trajectory is a circle. (b) Find $\vec{v}(t)$ and show that $\|\vec{v}\| = R\omega$. (c) Find $\vec{a}(t)$ and show that $\vec{a} = -\omega^2\vec{r}$.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 3.4. (a)

$$x^2 + y^2 = R^2\cos^2(\omega t) + R^2\sin^2(\omega t) = R^2,$$

a circle of radius R.

(b)

$$\vec{v} = -R\omega\sin(\omega t)\,\hat{\imath} + R\omega\cos(\omega t)\,\hat{\jmath}.$$

$$\|\vec{v}\| = R\omega\sqrt{\sin^2 + \cos^2} = R\omega.$$

(c)

$$\vec{a} = -R\omega^2\cos(\omega t)\,\hat{\imath} - R\omega^2\sin(\omega t)\,\hat{\jmath} = -\omega^2\vec{r}.$$

Thus $\|\vec{a}\| = R\omega^2$ and the acceleration points toward the centre.

Exercise 3.5 · Speed and unit tangent

A particle has position $\vec{r}(t) = 2t\,\hat{\imath} + t^2\,\hat{\jmath} + \frac{t^3}{3}\,\hat{k}$ (SI units). Find its velocity, speed, and unit tangent vector at $t = 1.00$ s.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 3.5.

$$\vec{v} = (2,\, 2t,\, t^2).$$

At $t = 1$:

$$\vec{v} = (2, 2, 1), \qquad v = \sqrt{4+4+1} = 3.00 \text{ m s}^{-1}.$$

The unit tangent vector is

$$\vec{u}_t = \frac{\vec{v}}{v} = \left(\frac{2}{3}, \frac{2}{3}, \frac{1}{3}\right).$$

Exercise 3.6 · **Eliminating time**

A particle moves with $x(t) = 2t$ and $y(t) = 4t^2 - 1$ (SI units). (a) Find the trajectory $y(x)$. (b) Find $\vec{v}$ and $\vec{a}$ at $t = 0.50$ s. (c) Is the acceleration constant?

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 3.6. (a) From $x = 2t$, $t = x/2$, hence

$$y = 4\left(\frac{x}{2}\right)^2 - 1 = x^2 - 1.$$

This is a parabola opening upward.
(b) $\vec{v} = (2, 8t)$ and $\vec{a} = (0, 8)$. At $t = 0.50$:

$$\vec{v} = (2, 4) \text{ m s}^{-1}, \qquad \|\vec{v}\| = \sqrt{20} = 4.47 \text{ m s}^{-1},$$

$$\vec{a} = (0, 8)\ \mathrm{m\,s^{-2}}.$$

(c) Yes: $\vec{a}$ is independent of time, so the acceleration is constant.

Exercise 3.7 · Integrating acceleration

A particle has acceleration $\vec{a}(t) = 6t\,\hat{\imath}$ (SI units), initial velocity $\vec{v}(0) = \hat{\imath} + 2\hat{\jmath}$, and initial position $\vec{r}(0) = \vec{0}$. Find $\vec{v}(t)$, $\vec{r}(t)$, and the position at $t = 2.00$ s.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 3.7. Integrating $\vec{a} = 6t\,\hat{\imath}$:

$$\vec{v}(t) = 3t^2\hat{\imath} + \vec{C}.$$

Using $\vec{v}(0) = \hat{\imath} + 2\hat{\jmath}$ gives $\vec{C} = \hat{\imath} + 2\hat{\jmath}$, so

$$\vec{v}(t) = (3t^2 + 1)\hat{\imath} + 2\hat{\jmath}.$$

Integrating again:

$$\vec{r}(t) = (t^3 + t)\hat{\imath} + 2t\,\hat{\jmath} + \vec{D}.$$

With $\vec{r}(0) = \vec{0}$, $\vec{D} = \vec{0}$. Thus

$$\vec{r}(t) = (t^3 + t)\hat{\imath} + 2t\,\hat{\jmath}.$$

At $t = 2$: $\vec{r} = (8 + 2)\hat{\imath} + 4\hat{\jmath} = 10\hat{\imath} + 4\hat{\jmath}$ (m).

Exercise 3.8 · From velocity to position

A particle has velocity $\vec{v}(t) = 3t^2\,\hat{\imath} + 2t\,\hat{\jmath}$ (SI units) and initial position $\vec{r}(0) = \hat{\imath}$. Find $\vec{r}(t)$ and the speed at $t = 1.00$ s.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 3.8. Integrating $\vec{v}$:

$$\vec{r}(t) = t^3\hat{\imath} + t^2\hat{\jmath} + \vec{C}.$$

With $\vec{r}(0) = \hat{\imath}$, $\vec{C} = \hat{\imath}$. Hence

$$\vec{r}(t) = (t^3 + 1)\hat{\imath} + t^2\hat{\jmath}.$$

At $t = 1$:

$$\vec{v} = (3, 2), \qquad v = \sqrt{9 + 4} = \sqrt{13} = 3.61 \text{ m s}^{-1}.$$

Exercise 3.9 · ⋆⋆⋆ **Helical motion**

A particle moves on the helix $\vec{r}(t) = a\cos(\omega t)\,\hat{\imath} + a\sin(\omega t)\,\hat{\jmath} + bt\,\hat{k}$. (a) Show that its speed is constant. (b) Find the magnitude of its acceleration. (c) Describe the direction of the acceleration.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 3.9. (a)

$$\vec{v} = -a\omega\sin(\omega t)\hat{\imath} + a\omega\cos(\omega t)\hat{\jmath} + b\hat{k}.$$

$$v^2 = a^2\omega^2\sin^2 + a^2\omega^2\cos^2 + b^2 = a^2\omega^2 + b^2 = \text{const}.$$

The speed is constant; the velocity direction changes.
(b)

$$\vec{a} = -a\omega^2\cos(\omega t)\hat{\imath} - a\omega^2\sin(\omega t)\hat{\jmath},$$

so

$$\|\vec{a}\| = a\omega^2.$$

(c) The acceleration is horizontal and points toward the z-axis. It has no vertical component: the helix is traversed at constant speed and constant vertical velocity, while the horizontal projection is uniform circular motion.

Exercise 3.10 · ⋆⋆⋆ **Acceleration on a spiral path**

A particle moves in the xy-plane with $\vec{r}(t) = R\cos(\alpha t^2)\,\hat{\imath} + R\sin(\alpha t^2)\,\hat{\jmath}$, where R and α are constants. (a) Find $\vec{v}(t)$ and $\vec{a}(t)$. (b) Identify the tangential and normal components of $\vec{a}$. (c) Verify that $a_n = v^2/R$.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 3.10. (a)

$$\vec{v} = R(2\alpha t)\big[-\sin(\alpha t^2)\hat{\imath} + \cos(\alpha t^2)\hat{\jmath}\big].$$

Differentiating again,

$$\vec{a} = R\big[-2\alpha\sin(\alpha t^2) - 4\alpha^2 t^2\cos(\alpha t^2)\big]\hat{\imath} + R\big[2\alpha\cos(\alpha t^2) - 4\alpha^2 t^2\sin(\alpha t^2)\big]\hat{\jmath}.$$

(b) The unit tangent is $\vec{u}_t = -\sin(\alpha t^2)\hat{\imath} + \cos(\alpha t^2)\hat{\jmath}$ and the unit normal is $\vec{u}_n = -\cos(\alpha t^2)\hat{\imath} - \sin(\alpha t^2)\hat{\jmath}$. Therefore

$$\vec{a} = \underbrace{2R\alpha}_{a_t}\,\vec{u}_t + \underbrace{4R\alpha^2 t^2}_{a_n}\,\vec{u}_n.$$

The tangential acceleration is constant, $a_t = 2R\alpha$; the normal acceleration $a_n = 4R\alpha^2 t^2$ grows with time and points toward the centre (along $\vec{u}_n$).
(c) The speed is $v = 2R\alpha t$, so

$$\frac{v^2}{R} = \frac{4R^2\alpha^2 t^2}{R} = 4R\alpha^2 t^2 = |a_n|\,.$$

The relation $a_n = v^2/R$ holds, as it must for circular motion of radius R.

CHAPTER 4

Rectilinear Motion

You will learn to: solve uniform and uniformly accelerated rectilinear motion; handle free fall; read motion graphs; and integrate a variable acceleration.

4.1 Uniform rectilinear motion

When the acceleration is zero, the velocity is constant:

$$a = 0, \qquad v = v_0, \qquad x(t) = x_0 + v_0 t.$$

The x–t graph is a straight line of slope v_0.

4.2 Uniformly accelerated rectilinear motion

When a is constant,

$$v(t) = v_0 + at,$$
$$x(t) = x_0 + v_0 t + \frac{1}{2} a t^2,$$

and, eliminating t,

$$v^2 - v_0^2 = 2a(x - x_0).$$

The last relation is useful when time is neither given nor asked for.

4.3 Free fall

Near the Earth's surface, neglecting air resistance, a body falls with constant acceleration $\vec{g}$ directed downward. Taking the y-axis upward,

$$a_y = -g, \qquad v_y = v_{0y} - gt, \qquad y = y_0 + v_{0y} t - \frac{1}{2} g t^2.$$

With $g = 9.81\ \mathrm{m\,s^{-2}}$.

Signs

Choose a positive direction once and keep it for the whole problem. If upward is positive, then g enters as $-g$, a downward velocity is negative, and a displacement below the starting point is negative.

4.4 Variable acceleration

If a is not constant, integrate:

$$a(t) = \frac{\mathrm{d}v}{\mathrm{d}t} \implies v(t) = v(0) + \int_0^t a(t')\,\mathrm{d}t',$$

$$v(t) = \frac{\mathrm{d}x}{\mathrm{d}t} \implies x(t) = x(0) + \int_0^t v(t')\,\mathrm{d}t'.$$

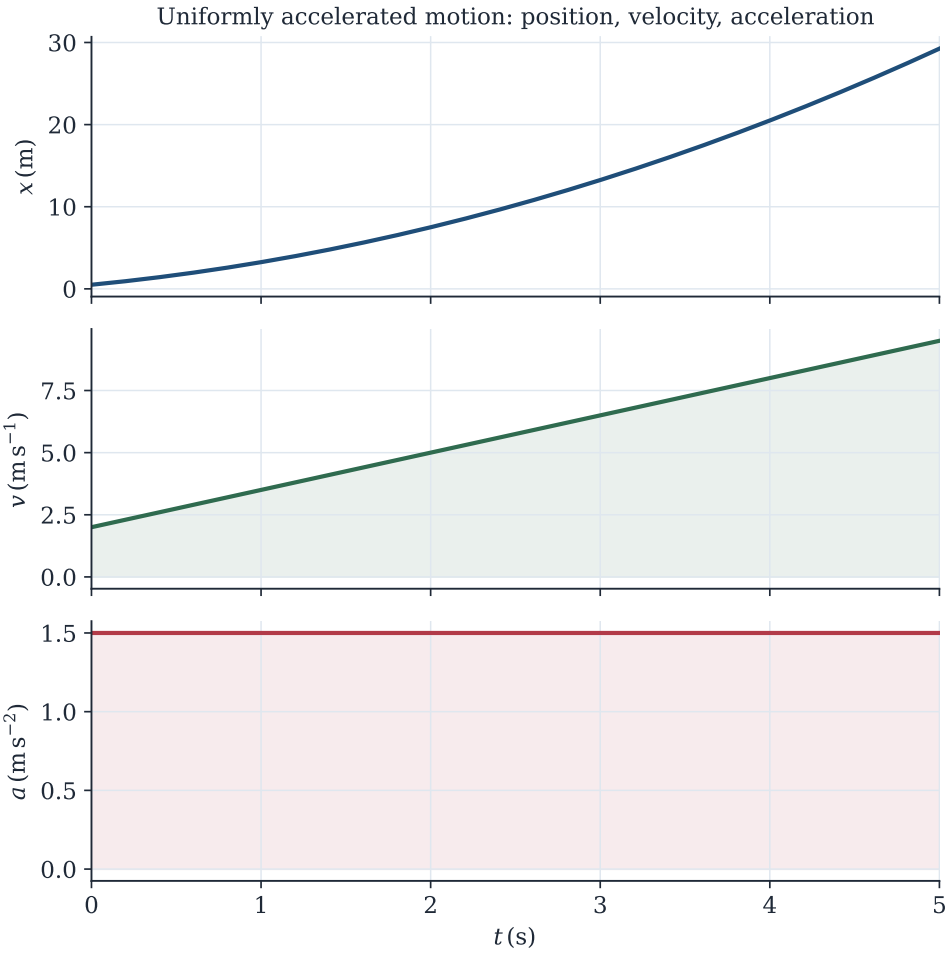


Figure 4.1: Uniformly accelerated motion with $x_0 = 0.5$ m, $v_0 = 2.0\ \mathrm{m\,s^{-1}}$, and $a = 1.5\ \mathrm{m\,s^{-2}}$. The slope of $x(t)$ is $v(t)$; the slope of $v(t)$ is a; the area under $v(t)$ is the displacement.

Two alternative forms are often useful:

$$a = \frac{\mathrm{d}v}{\mathrm{d}t} = v\frac{\mathrm{d}v}{\mathrm{d}x}.$$

The second form is the key to problems where the acceleration is given as a function of position or speed.

Choosing the right equation

- Constant acceleration and time known: use $v = v_0 + at$.
- Constant acceleration, time unknown: use $v^2 = v_0^2 + 2a\Delta x$.
- Acceleration depends on v: separate variables in $\mathrm{d}v/a(v) = \mathrm{d}t$.
- Acceleration depends on x: use $v\,\mathrm{d}v = a(x)\,\mathrm{d}x$.

4.5 Worked examples

Worked example 4.1 · Uniform acceleration

A car accelerates uniformly from $10.0\ \mathrm{m\,s^{-1}}$ to $30.0\ \mathrm{m\,s^{-1}}$ over a distance of 200 m. Find its acceleration and the time taken.

Solution 4.1. Use $v^2 - v_0^2 = 2a\Delta x$:

$$a = \frac{v^2 - v_0^2}{2\Delta x} = \frac{30.0^2 - 10.0^2}{2(200)} = \frac{800}{400} = 2.00\ \mathrm{m\,s^{-2}}.$$

Then

$$t = \frac{v - v_0}{a} = \frac{30.0 - 10.0}{2.00} = 10.0\ \mathrm{s}.$$

Worked example 4.2 · Vertical throw

A ball is thrown vertically upward from the ground with $v_0 = 20.0\ \mathrm{m\,s^{-1}}$. Neglecting air resistance, find its maximum height, the total time before it returns to the ground, and its velocity at $t = 3.00$ s.

Solution 4.2. Take upward as positive, so $a = -g = -9.81\ \mathrm{m\,s^{-2}}$.
At the highest point $v = 0$:

$$0 = v_0^2 - 2gh \;\Longrightarrow\; h = \frac{v_0^2}{2g} = \frac{400}{19.62} = 20.4\ \mathrm{m}.$$

The time to reach the top is $t_{\mathrm{up}} = v_0/g = 2.04$ s; the total time is twice this, 4.08 s.
At $t = 3.00$ s:

$$v = 20.0 - 9.81(3.00) = -9.43\ \mathrm{m\,s^{-1}}.$$

The negative sign means the ball is moving downward.

4.6 Exercises and Step-by-Step Solutions

Exercise 4.1 · ⋆ **From zero to one hundred**

A car accelerates uniformly from rest to 100 $\mathrm{km\,h^{-1}}$ in 8.00 s. Find its acceleration and the distance covered.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 4.1. First convert:

$$100\ \mathrm{km\,h^{-1}} = \frac{100}{3.6} = 27.8\ \mathrm{m\,s^{-1}}.$$

$$a = \frac{v}{t} = \frac{27.8}{8.00} = 3.47\ \mathrm{m\,s^{-2}}.$$

$$d = \frac{1}{2}at^2 = \frac{1}{2}(3.47)(8.00)^2 = 111\ \mathrm{m}.$$

Equivalently $d = v^2/(2a) = 111$ m.

Exercise 4.2 · ⋆ **Free fall**

A stone is dropped from rest from a height of 45.0 m. Neglecting air resistance, find the time to reach the ground and the impact speed.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 4.2. Taking downward as positive,

$$h = \frac{1}{2}gt^2 \implies t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{90.0}{9.81}} = 3.03 \text{ s}.$$

$$v = gt = 9.81(3.03) = 29.7 \text{ m s}^{-1}.$$

Exercise 4.3 · ⋆ **Reading a position function**

A particle moves on the x-axis with $x(t) = 3t^2 - 12t + 7$ (SI units). (a) Find $v(t)$ and $a(t)$. (b) Find the time and value of the minimum position. (c) Sketch $x(t)$ and describe the motion.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 4.3. (a)

$$v(t) = 6t - 12, \qquad a(t) = 6.$$

(b) $v = 0$ at $t = 2.00$ s. Since $a > 0$, this is a minimum:

$$x(2) = 3(4) - 12(2) + 7 = -5.00 \text{ m}.$$

(c) For $t < 2$ the particle moves in the negative x direction; at $t = 2$ it stops instantaneously; for $t > 2$ it moves in the positive direction with constant acceleration 6 m s^{-2}. The graph is an upward parabola with vertex at $(2, -5)$.

Exercise 4.4 · **Catching up**

Car A travels at a constant 20.0 m s^{-1}. Car B starts from rest from the same point 2.00 s later and accelerates uniformly at 4.00 m s^{-2}. When and where does B catch A? How fast is B moving at that moment?

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 4.4. Let t be measured from the moment car A passes the reference point. Then

$$x_A = 20.0t,$$

while car B starts at $t = 2.00$ s:

$$x_B = \frac{1}{2}(4.00)(t - 2.00)^2 = 2.00(t - 2.00)^2.$$

Set $x_A = x_B$:

$$20.0t = 2.00(t^2 - 4.00t + 4.00) \implies 2t^2 - 28t + 8 = 0 \implies t^2 - 14t + 4 = 0.$$

$$t = \frac{14 \pm \sqrt{196 - 16}}{2} = \frac{14 \pm 13.42}{2}.$$

The physical solution is $t = 13.7$ s (the other root is before B starts). Then

$$x = 20.0(13.7) = 274 \text{ m}.$$

B's speed is

$$v_B = 4.00(13.7 - 2.00) = 46.8 \text{ m s}^{-1}.$$

Exercise 4.5 · Vertical throw

A ball is thrown vertically upward from the ground with $v_0 = 15.0$ m s^{-1}. Find its maximum height, the time it returns to the ground, and its velocity at $t = 1.00$ s and $t = 2.50$ s.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 4.5. Upward positive, $a = -9.81\ \mathrm{m\,s^{-2}}$.

$$h = \frac{v_0^2}{2g} = \frac{225}{19.62} = 11.5\ \mathrm{m}.$$

$$t_{\text{return}} = \frac{2v_0}{g} = \frac{30.0}{9.81} = 3.06\ \mathrm{s}.$$

At $t = 1.00$:

$$v = 15.0 - 9.81 = 5.19\ \mathrm{m\,s^{-1}}.$$

At $t = 2.50$:

$$v = 15.0 - 9.81(2.50) = 15.0 - 24.5 = -9.53\ \mathrm{m\,s^{-1}}.$$

Exercise 4.6 · Braking distance

A train travelling at $30.0\ \mathrm{m\,s^{-1}}$ brakes uniformly and stops in 500 m. Find the deceleration and the stopping time.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 4.6. With $v = 0$ and $v_0 = 30.0\ \mathrm{m\,s^{-1}}$,

$$a = \frac{0 - v_0^2}{2d} = -\frac{900}{1000} = -0.900\ \mathrm{m\,s^{-2}}.$$

The deceleration magnitude is $0.900\ \text{m}\,\text{s}^{-2}$.

$$t = \frac{v - v_0}{a} = \frac{-30.0}{-0.900} = 33.3\ \text{s}.$$

Exercise 4.7 · Velocity graph

A particle moves along a straight line. Its velocity is $4.00\ \text{m}\,\text{s}^{-1}$ from $t = 0$ to $t = 3.00$ s, then decreases linearly to zero at $t = 7.00$ s. (a) Draw the v–t graph. (b) Find the total displacement. (c) Find the acceleration during the second phase.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 4.7. (a) The graph is horizontal at $4.00\ \text{m}\,\text{s}^{-1}$ from 0 to 3 s, then a straight line down to zero at 7 s.
(b) The area is a rectangle plus a triangle:

$$\Delta x = (3.00)(4.00) + \frac{1}{2}(4.00)(4.00) = 12.0 + 8.00 = 20.0\ \text{m}.$$

(c) During $3 < t < 7$,

$$a = \frac{0 - 4.00}{7.00 - 3.00} = -1.00\ \text{m}\,\text{s}^{-2}.$$

Exercise 4.8 · Time-dependent acceleration

A particle starts with $v(0) = 1.00\ \text{m}\,\text{s}^{-1}$ and $x(0) = 0$, and has acceleration $a(t) = 2t\ \text{m}\,\text{s}^{-2}$. Find $v(t)$, $x(t)$, and the position at $t = 2.00$ s.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 4.8. Integrating $a = 2t$:

$$v(t) = t^2 + C_1.$$

With $v(0) = 1.00$, $C_1 = 1.00$, so $v(t) = t^2 + 1.00$.
Integrating again:

$$x(t) = \frac{t^3}{3} + t + C_2.$$

With $x(0) = 0$, $C_2 = 0$. Thus

$$x(t) = \frac{t^3}{3} + t.$$

At $t = 2.00$:

$$x = \frac{8.00}{3} + 2.00 = 4.67 \text{ m}.$$

Exercise 4.9 · ⋆⋆⋆ **Linear drag**

A particle moves on a straight line with acceleration $a = -kv$, where $k > 0$ is a constant. At $t = 0$, $v = v_0$ and $x = 0$. (a) Find $v(t)$. (b) Find $x(t)$. (c) Show that the particle travels a finite maximum distance even though it never stops in finite time.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 4.9. (a) Separate variables:

$$\frac{\mathrm{d}v}{\mathrm{d}t} = -kv \implies \frac{\mathrm{d}v}{v} = -k\,\mathrm{d}t.$$

Integrating from 0 to t:

$$\ln\frac{v}{v_0} = -kt \implies v(t) = v_0 e^{-kt}.$$

(b)

$$x(t) = \int_0^t v_0 e^{-kt'}\,\mathrm{d}t' = \frac{v_0}{k}\left(1 - e^{-kt}\right).$$

(c) As $t \to \infty$, $x \to v_0/k$, a finite limit. But $v = v_0 e^{-kt}$ is never exactly zero for any finite t. The particle asymptotically approaches the position $x = v_0/k$ without ever reaching it in finite time.

Exercise 4.10 · ⋆⋆⋆ **Dropped and thrown**

A ball is dropped from rest from height h at $t = 0$. At the same instant, a second ball is thrown vertically upward from the ground with speed v_0. (a) Find the time at which the two balls meet. (b) Find the height at which they meet. (c) For $h = 15.0$ m and $v_0 = 10.0\ \mathrm{m\,s^{-1}}$, compute the meeting time and height.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 4.10. Take upward as positive, with the ground at $y = 0$. Then

$$y_A = h - \frac{1}{2}gt^2, \qquad y_B = v_0 t - \frac{1}{2}gt^2.$$

(a) Setting $y_A = y_B$:

$$h - \frac{1}{2}gt^2 = v_0 t - \frac{1}{2}gt^2 \implies t = \frac{h}{v_0}.$$

(b) Substituting back,

$$y = h - \frac{1}{2}g\left(\frac{h}{v_0}\right)^2 = h - \frac{gh^2}{2v_0^2}.$$

(c) For $h = 15.0$ m and $v_0 = 10.0\ \mathrm{m\,s^{-1}}$:

$$t = \frac{15.0}{10.0} = 1.50\ \mathrm{s},$$

$$y = 15.0 - \frac{9.81(15.0)^2}{2(10.0)^2} = 15.0 - \frac{2207}{200} = 15.0 - 11.0 = 3.96\ \mathrm{m}.$$

CHAPTER 5

Motion in a Plane: Projectiles and Circles

You will learn to: analyse projectile motion with constant acceleration; compute range, maximum height, and time of flight; describe circular motion with angular velocity and centripetal acceleration; and handle non-uniform circular motion.

5.1 Projectile motion

Take the y-axis vertical and neglect air resistance. The only acceleration is

$$a_x = 0, \qquad a_y = -g.$$

With initial speed v_0 at angle α above the horizontal:

$$x(t) = (v_0 \cos\alpha)t, \qquad y(t) = (v_0 \sin\alpha)t - \frac{1}{2}gt^2.$$

Eliminating t,

$$y(x) = x\tan\alpha - \frac{gx^2}{2v_0^2\cos^2\alpha}.$$

The trajectory is a parabola.

For a projectile launched and landing at the same height:

$$T = \frac{2v_0\sin\alpha}{g}, \qquad R = \frac{v_0^2\sin 2\alpha}{g}, \qquad H = \frac{v_0^2\sin^2\alpha}{2g}.$$

The range is maximum at $\alpha = 45^\circ$, where $R_{\max} = v_0^2/g$.

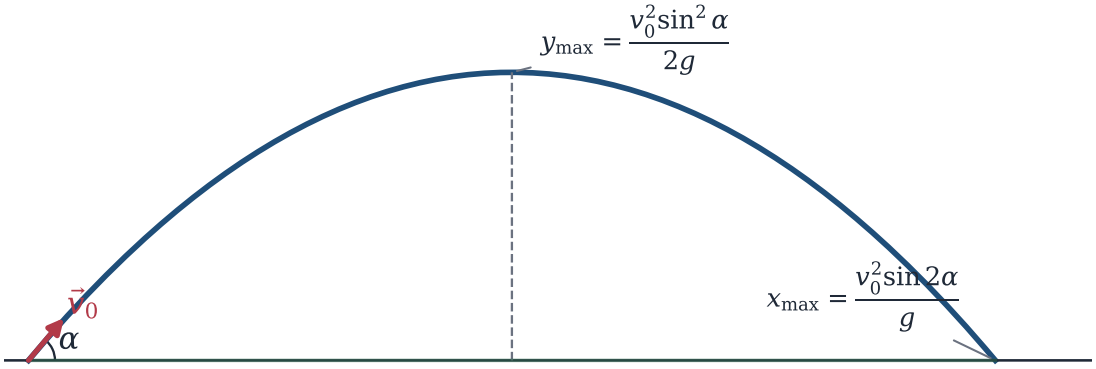


Figure 5.1: Projectile motion: the velocity is tangent to the parabola, while the acceleration is always vertically downward.

Two frequent mistakes

The horizontal velocity never changes (no air resistance), and the vertical acceleration is g even at the highest point. "$a = 0$ at the top" is false: only $v_y = 0$ there.

5.2 Circular motion

For a particle moving on a circle of radius R, let $\theta(t)$ be its angular position. Define

$$\omega = \dot{\theta} \quad (\mathrm{rad\,s^{-1}}), \qquad \alpha = \dot{\omega} = \ddot{\theta} \quad (\mathrm{rad\,s^{-2}}).$$

Then

$$v = R\omega, \qquad a_t = R\alpha, \qquad a_n = \frac{v^2}{R} = R\omega^2.$$

The tangential acceleration a_t changes the speed; the normal (centripetal) acceleration a_n changes the direction and always points toward the centre. The total acceleration has magnitude

$$a = \sqrt{a_t^2 + a_n^2}.$$

The period of uniform circular motion is $T = 2\pi/\omega$.

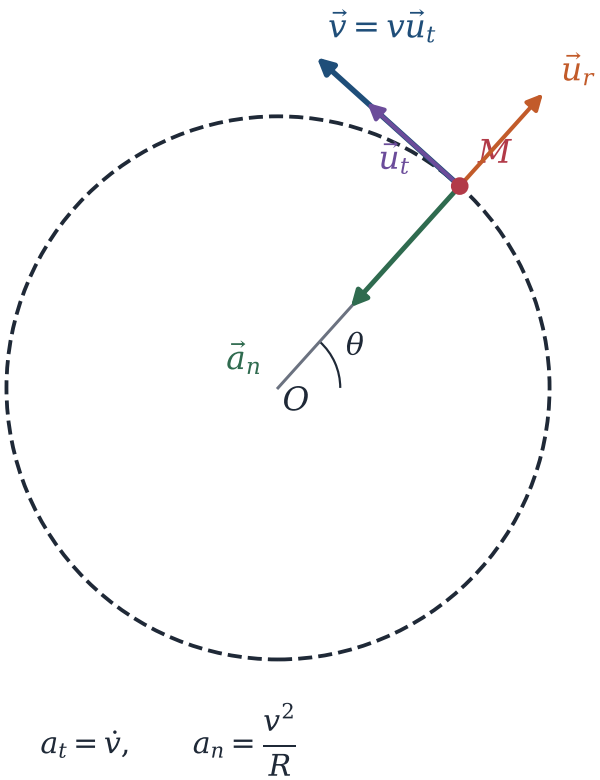


Figure 5.2: Circular motion. The velocity is tangential; the centripetal acceleration points toward the centre.

Intrinsic decomposition

For any curved path, with $\vec{u}_t$ the unit tangent and $\vec{u}_n$ the unit normal pointing toward the centre of curvature,

$$\vec{a} = \frac{\mathrm{d}v}{\mathrm{d}t}\vec{u}_t + \frac{v^2}{R}\vec{u}_n.$$

For a circle, R is constant; for a general curve, R is the local radius of curvature.

5.3 Worked examples

Worked example 5.1 · Full projectile analysis

A projectile is launched from the ground with $v_0 = 25.0\ \mathrm{m\,s^{-1}}$ at $\alpha = 40.0^\circ$. Find the time of flight, the range, the maximum height, and the velocity at $t = 1.50$ s.

Solution 5.1.

$$v_{0x} = 25.0\cos 40.0^\circ = 19.2\ \mathrm{m\,s^{-1}}, \qquad v_{0y} = 25.0\sin 40.0^\circ = 16.1\ \mathrm{m\,s^{-1}}.$$

Time of flight:

$$T = \frac{2v_{0y}}{g} = \frac{32.1}{9.81} = 3.28\ \mathrm{s}.$$

Range:

$$R = v_{0x}T = (19.2)(3.28) = 62.9\ \mathrm{m},$$

or directly $R = v_0^2 \sin 80^\circ / g = 62.7$ m.
Maximum height:

$$H = \frac{v_{0y}^2}{2g} = \frac{258}{19.62} = 13.2\ \mathrm{m}.$$

At $t = 1.50$ s:

$$v_x = 19.2\ \mathrm{m\,s^{-1}}, \qquad v_y = 16.1 - 9.81(1.50) = 1.36\ \mathrm{m\,s^{-1}},$$

$$v = \sqrt{19.2^2 + 1.36^2} = 19.2\ \mathrm{m\,s^{-1}}.$$

The projectile is still rising slightly.

Worked example 5.2 · Non-uniform circular motion

A particle moves on a circle of radius $R = 0.500$ m with angular position $\theta(t) = 2t^2$ (rad). At $t = 1.00$ s, find the angular velocity, angular acceleration, speed, tangential and normal accelerations, and the total acceleration.

Solution 5.2.

$$\omega = \dot{\theta} = 4t, \qquad \alpha = \dot{\omega} = 4.$$

At $t = 1$: $\omega = 4.00\ \mathrm{rad\,s^{-1}}$, $\alpha = 4.00\ \mathrm{rad\,s^{-2}}$.

$$v = R\omega = 0.500(4.00) = 2.00\ \mathrm{m\,s^{-1}},$$
$$a_t = R\alpha = 0.500(4.00) = 2.00\ \mathrm{m\,s^{-2}},$$
$$a_n = R\omega^2 = 0.500(16.0) = 8.00\ \mathrm{m\,s^{-2}}.$$

$$a = \sqrt{2.00^2 + 8.00^2} = \sqrt{68.0} = 8.25\ \text{m s}^{-2}.$$

5.4 Exercises and Step-by-Step Solutions

Exercise 5.1 · ⋆ **Basic projectile**

A ball is launched from the ground with $v_0 = 20.0\ \text{m s}^{-1}$ at 30.0° above the horizontal. Find the time of flight, the range, and the maximum height.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 5.1.

$$v_{0x} = 20.0\cos 30.0^\circ = 17.3\ \text{m s}^{-1}, \qquad v_{0y} = 20.0\sin 30.0^\circ = 10.0\ \text{m s}^{-1}.$$

$$T = \frac{2v_{0y}}{g} = \frac{20.0}{9.81} = 2.04\ \text{s}.$$

$$R = v_{0x}T = (17.3)(2.04) = 35.3\ \text{m}.$$

$$H = \frac{v_{0y}^2}{2g} = \frac{100}{19.62} = 5.10\ \text{m}.$$

Exercise 5.2 · ⋆ **Horizontal launch**

A ball is thrown horizontally from a cliff of height 20.0 m with speed $15.0\ \text{m s}^{-1}$. Find the time to reach the ground, the horizontal distance travelled, and the impact speed.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 5.2. Vertical motion (downward positive):

$$h = \frac{1}{2}gt^2 \implies t = \sqrt{\frac{2(20.0)}{9.81}} = 2.02 \text{ s}.$$

Horizontal distance:

$$x = v_0 t = 15.0(2.02) = 30.3 \text{ m}.$$

Impact velocity:

$$v_x = 15.0, \qquad v_y = gt = 19.8 \text{ m s}^{-1},$$
$$v = \sqrt{15.0^2 + 19.8^2} = 24.9 \text{ m s}^{-1}.$$

Exercise 5.3 · ⋆ **Uniform circular motion**

A point moves on a circle of radius 2.00 m with period 0.500 s. Find its angular velocity, speed, and centripetal acceleration.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 5.3.

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{0.500} = 12.6 \text{ rad s}^{-1}.$$
$$v = R\omega = 2.00(12.6) = 25.1 \text{ m s}^{-1}.$$

$$a_n = \frac{v^2}{R} = \frac{631}{2.00} = 316\ \text{m s}^{-2}.$$

Exercise 5.4 · Range and maximum height

A projectile launched at 45.0° has a range of 100 m. Find its initial speed, maximum height, and time of flight.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 5.4. For $\alpha = 45°$, $\sin 2\alpha = 1$, so

$$R = \frac{v_0^2}{g} \implies v_0 = \sqrt{gR} = \sqrt{9.81(100)} = 31.3\ \text{m s}^{-1}.$$

$$H = \frac{v_0^2 \sin^2 45°}{2g} = \frac{gR(1/2)}{2g} = \frac{R}{4} = 25.0\ \text{m}.$$

$$T = \frac{2v_0 \sin 45°}{g} = \frac{2(31.3)(0.707)}{9.81} = 4.51\ \text{s}.$$

Exercise 5.5 · Clearing a wall

A projectile is launched with $v_0 = 30.0\ \text{m s}^{-1}$ at 50.0°. A wall stands 60.0 m away. Does the projectile clear a wall of height 10.0 m? Find its height and velocity when it reaches the wall.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 5.5.

$$v_x = 30.0\cos 50.0^\circ = 19.3\ \text{m s}^{-1}, \qquad v_{0y} = 30.0\sin 50.0^\circ = 23.0\ \text{m s}^{-1}.$$

Time to reach the wall:

$$t = \frac{x}{v_x} = \frac{60.0}{19.3} = 3.11\ \text{s}.$$

Height at that time:

$$y = 23.0(3.11) - \frac{1}{2}(9.81)(3.11)^2 = 71.5 - 47.4 = 24.1\ \text{m}.$$

This is well above 10.0 m, so the projectile clears the wall.
Velocity at the wall:

$$v_x = 19.3, \qquad v_y = 23.0 - 9.81(3.11) = -7.51\ \text{m s}^{-1},$$

$$v = \sqrt{19.3^2 + 7.51^2} = 20.7\ \text{m s}^{-1},$$

directed 21.3° below the horizontal (the projectile is descending).

Exercise 5.6 · Speed at the top

A projectile is launched at angle θ above the horizontal. At the highest point its speed is exactly half its initial speed. Find θ.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 5.6. At the highest point $v_y = 0$, so the speed is $v_x = v_0\cos\theta$. The condition $v = v_0/2$ gives

$$v_0\cos\theta = \frac{v_0}{2} \implies \cos\theta = \frac{1}{2} \implies \theta = 60.0^\circ.$$

Exercise 5.7 · Two angles, one range

A projectile has initial speed 30.0 m s^{-1} and must reach a range of 80.0 m. Find the two possible launch angles and the corresponding times of flight.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 5.7.

$$R = \frac{v_0^2 \sin 2\theta}{g} \implies \sin 2\theta = \frac{gR}{v_0^2} = \frac{9.81(80.0)}{(30.0)^2} = 0.872.$$

Thus

$$2\theta = 60.7^\circ \text{ or } 119.3^\circ,$$

so

$$\theta_1 = 30.4^\circ, \qquad \theta_2 = 59.6^\circ.$$

The times of flight are

$$T_1 = \frac{2(30.0)\sin 30.4^\circ}{9.81} = 3.10 \text{ s},$$

$$T_2 = \frac{2(30.0)\sin 59.6^\circ}{9.81} = 5.28 \text{ s}.$$

The lower angle gives the shorter flight and the smaller maximum height.

Exercise 5.8 · Angular kinematics

A particle moves on a circle of radius $R = 0.500$ m with angular position $\theta(t) = 2t^2$ rad. At $t = 1.00$ s, find ω, α, v, a_t, a_n, and the total acceleration.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 5.8.

$$\omega = \dot{\theta} = 4t, \qquad \alpha = \dot{\omega} = 4.$$

At $t = 1.00$ s:

$$\omega = 4.00 \text{ rad}\,\text{s}^{-1}, \qquad \alpha = 4.00 \text{ rad}\,\text{s}^{-2}.$$

$$v = R\omega = 0.500(4.00) = 2.00 \text{ m}\,\text{s}^{-1},$$

$$a_t = R\alpha = 0.500(4.00) = 2.00 \text{ m}\,\text{s}^{-2},$$

$$a_n = R\omega^2 = 0.500(16.0) = 8.00 \text{ m}\,\text{s}^{-2}.$$

$$a = \sqrt{a_t^2 + a_n^2} = \sqrt{4.00 + 64.0} = 8.25 \text{ m}\,\text{s}^{-2}.$$

Exercise 5.9 · ★★★ **Deriving the range formula**

Starting from $x = v_0 \cos\alpha\, t$ and $y = v_0 \sin\alpha\, t - \tfrac{1}{2}gt^2$, derive

$$y = x\tan\alpha - \frac{gx^2}{2v_0^2\cos^2\alpha},$$

then show that for equal launch and landing heights the range is $R = v_0^2 \sin 2\alpha / g$. Deduce that the maximum range occurs at 45° and that complementary angles give the same range.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 5.9. From $x = v_0\cos\alpha\, t$,

$$t = \frac{x}{v_0\cos\alpha}.$$

Substituting into y:

$$y = v_0\sin\alpha\frac{x}{v_0\cos\alpha} - \frac{1}{2}g\frac{x^2}{v_0^2\cos^2\alpha} = x\tan\alpha - \frac{gx^2}{2v_0^2\cos^2\alpha}.$$

At landing, $y = 0$ and $x = R$:

$$R\tan\alpha = \frac{gR^2}{2v_0^2\cos^2\alpha}.$$

For $R \neq 0$,

$$R = \frac{2v_0^2\sin\alpha\cos\alpha}{g} = \frac{v_0^2\sin 2\alpha}{g}.$$

This is maximum when $\sin 2\alpha = 1$, i.e. $2\alpha = 90^\circ$ and $\alpha = 45^\circ$, giving $R_{\max} = v_0^2/g$. Since $\sin(180^\circ - 2\alpha) = \sin 2\alpha$, the angles α and $90^\circ - \alpha$ give the same range.

Exercise 5.10 · ⋆⋆⋆ Hitting a target

A projectile is launched from the origin with speed $v_0 = 25.0\ \mathrm{m\,s^{-1}}$ and must pass through the point $(x, y) = (30.0, 10.0)$ m. Using $u = \tan\alpha$, show that

$$\frac{gx^2}{2v_0^2}u^2 - xu + \left(y + \frac{gx^2}{2v_0^2}\right) = 0,$$

and find the two possible launch angles.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 5.10. With $u = \tan\alpha$ and $\cos^2\alpha = 1/(1+u^2)$,

$$y = xu - \frac{gx^2}{2v_0^2}(1+u^2).$$

Rearranging,

$$\frac{gx^2}{2v_0^2}u^2 - xu + \left(y + \frac{gx^2}{2v_0^2}\right) = 0.$$

For $x = 30.0$ m, $y = 10.0$ m, $v_0 = 25.0\ \mathrm{m\,s^{-1}}$:

$$A = \frac{9.81(30.0)^2}{2(25.0)^2} = 7.063,$$

so

$$7.063u^2 - 30.0u + 17.063 = 0.$$

$$u = \frac{30.0 \pm \sqrt{900 - 4(7.063)(17.063)}}{2(7.063)} = \frac{30.0 \pm 20.44}{14.126}.$$

Thus

$$u_1 = 3.57 \implies \alpha_1 = 74.4^\circ, \qquad u_2 = 0.677 \implies \alpha_2 = 34.1^\circ.$$

Both trajectories pass through the target; the low-angle shot arrives sooner.

CHAPTER 6

Curvilinear Coordinates

You will learn to: express position, velocity, and acceleration in polar, cylindrical, and spherical coordinates; differentiate the moving unit vectors; and recognise the centripetal and Coriolis-like terms that appear in curvilinear coordinates.

6.1 Polar coordinates

In the plane, a point M is located by its distance r from the origin and the angle θ measured from the x-axis. The associated unit vectors are $\vec{u}_r$ (radial, pointing away from O) and $\vec{u}_\theta$ (perpendicular to $\vec{u}_r$, in the direction of increasing θ).

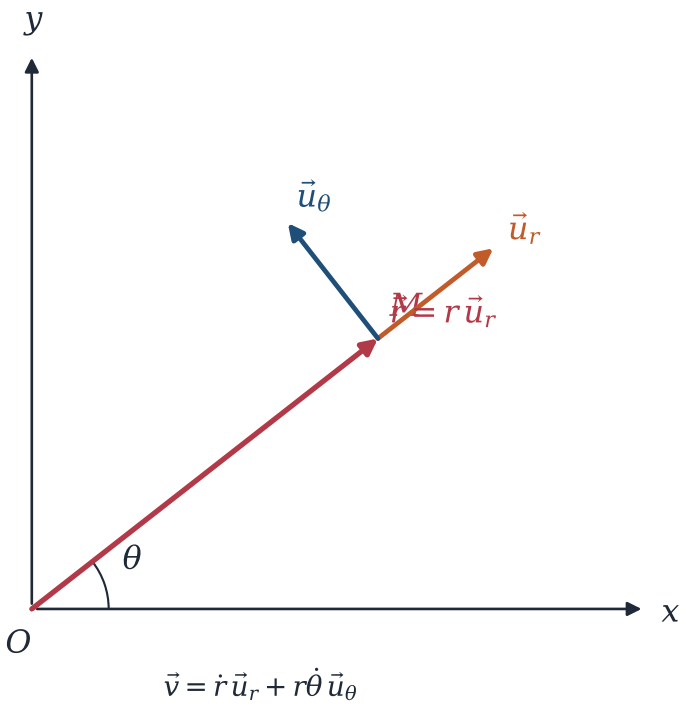


Figure 6.1: Polar coordinates and the local basis $(\vec{u}_r, \vec{u}_\theta)$. The basis rotates with the point.

The position vector is simply

$$\vec{r} = r\,\vec{u}_r.$$

Unlike the Cartesian basis, the polar basis moves:

$$\frac{\mathrm{d}\vec{u}_r}{\mathrm{d}t} = \dot{\theta}\,\vec{u}_\theta, \qquad \frac{\mathrm{d}\vec{u}_\theta}{\mathrm{d}t} = -\dot{\theta}\,\vec{u}_r.$$

Differentiating $\vec{r} = r\vec{u}_r$ gives

$$\vec{v} = \dot{r}\,\vec{u}_r + r\dot{\theta}\,\vec{u}_\theta,$$

and differentiating again gives

$$\boxed{\vec{a} = \left(\ddot{r} - r\dot{\theta}^2\right)\vec{u}_r + \left(r\ddot{\theta} + 2\dot{r}\dot{\theta}\right)\vec{u}_\theta.}$$

The term $-r\dot{\theta}^2\vec{u}_r$ is the centripetal acceleration; the term $2\dot{r}\dot{\theta}\,\vec{u}_\theta$ is sometimes called the Coriolis term in polar coordinates.

Circular motion as a special case

For a circle, $r = R$ is constant, so $\dot{r} = \ddot{r} = 0$ and

$$\vec{a} = -R\dot{\theta}^2\,\vec{u}_r + R\ddot{\theta}\,\vec{u}_\theta = -\frac{v^2}{R}\vec{u}_r + a_t\,\vec{u}_\theta.$$

This recovers the centripetal and tangential accelerations of Chapter 5.

6.2 Cylindrical coordinates

In three dimensions, cylindrical coordinates (ρ, θ, z) use the planar polar coordinates in the xy-plane and the Cartesian z:

$$\vec{r} = \rho\,\vec{u}_\rho + z\hat{k}.$$

The velocity and acceleration are

$$\vec{v} = \dot{\rho}\,\vec{u}_\rho + \rho\dot{\theta}\,\vec{u}_\theta + \dot{z}\,\hat{k},$$

$$\vec{a} = \left(\ddot{\rho} - \rho\dot{\theta}^2\right)\vec{u}_\rho + \left(\rho\ddot{\theta} + 2\dot{\rho}\dot{\theta}\right)\vec{u}_\theta + \ddot{z}\,\hat{k}.$$

6.3 Spherical coordinates

Let r be the distance from the origin, θ the polar angle measured from the positive z-axis, and ϕ the azimuthal angle in the xy-plane. Then

$$x = r\sin\theta\cos\phi, \quad y = r\sin\theta\sin\phi, \quad z = r\cos\theta.$$

The position vector is $\vec{r} = r\,\vec{u}_r$, and

$$\vec{v} = \dot{r}\,\vec{u}_r + r\dot{\theta}\,\vec{u}_\theta + r\sin\theta\,\dot{\phi}\,\vec{u}_\phi.$$

The acceleration is

$$\begin{aligned}\vec{a} = {} & \left(\ddot{r} - r\dot{\theta}^2 - r\sin^2\theta\,\dot{\phi}^2\right)\vec{u}_r \\ & + \left(r\ddot{\theta} + 2\dot{r}\dot{\theta} - r\sin\theta\cos\theta\,\dot{\phi}^2\right)\vec{u}_\theta \\ & + \left(r\sin\theta\,\ddot{\phi} + 2\dot{r}\sin\theta\,\dot{\phi} + 2r\cos\theta\,\dot{\theta}\dot{\phi}\right)\vec{u}_\phi.\end{aligned}$$

Notation varies

Many books call the azimuthal angle θ and the polar angle ϕ. Always check the definition at the start of a problem. In this book, θ is the polar angle from z and ϕ is the azimuth in the xy-plane.

Cylindrical (ρ, θ, z) Spherical (r, θ, ϕ)

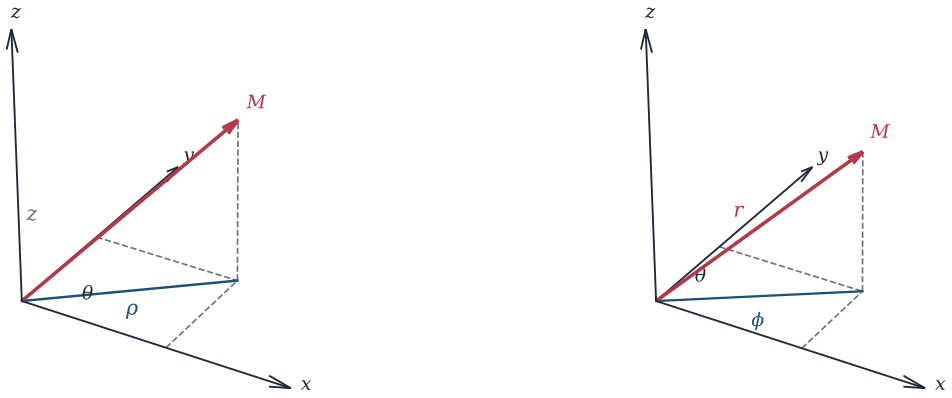


Figure 6.2: Cylindrical coordinates (ρ, θ, z) and spherical coordinates (r, θ, ϕ). In the spherical convention used here, θ is measured from the z-axis.

How to use curvilinear coordinates

1. Write down $r(t)$, $\theta(t)$, and (in 3D) $\phi(t)$ or $z(t)$.
2. Differentiate once to get the general velocity formula.
3. Differentiate again for acceleration, or use the boxed formulas.
4. Substitute the time of interest and compute magnitudes with $\|\vec{v}\|^2 = v_r^2 + v_\theta^2 + v_\phi^2$.

6.4 Worked examples

Worked example 6.1 · Polar kinematics

A particle moves in the plane with $r(t) = 2t$ (m) and $\theta(t) = t^2$ (rad), where t is in seconds. Find its velocity and acceleration at $t = 1.00$ s.

Solution 6.1. We have

$$r = 2t, \quad \dot{r} = 2, \quad \ddot{r} = 0; \qquad \theta = t^2, \quad \dot{\theta} = 2t, \quad \ddot{\theta} = 2.$$

At $t = 1$: $r = 2$, $\dot{r} = 2$, $\dot{\theta} = 2$, $\ddot{\theta} = 2$, $\ddot{r} = 0$.
Velocity:

$$\vec{v} = \dot{r}\,\vec{u}_r + r\dot{\theta}\,\vec{u}_\theta = 2\,\vec{u}_r + 2(2)\,\vec{u}_\theta = 2\,\vec{u}_r + 4\,\vec{u}_\theta \ \mathrm{m\,s^{-1}},$$

$$v = \sqrt{2^2 + 4^2} = \sqrt{20} = 4.47 \ \mathrm{m\,s^{-1}}.$$

Acceleration:

$$a_r = \ddot{r} - r\dot{\theta}^2 = 0 - 2(4) = -8.00 \ \mathrm{m\,s^{-2}},$$

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = 2(2) + 2(2)(2) = 4 + 8 = 12.0 \ \mathrm{m\,s^{-2}},$$

$$\vec{a} = -8.00\,\vec{u}_r + 12.0\,\vec{u}_\theta \ \mathrm{m\,s^{-2}}, \qquad a = \sqrt{64 + 144} = 14.4 \ \mathrm{m\,s^{-2}}.$$

Worked example 6.2 · Motion on a latitude circle

A particle moves on the sphere of radius R at constant polar angle $\theta = \pi/3$, with azimuth $\phi = \omega t$. Find its speed and acceleration.

Solution 6.2. Here $r = R$ is constant, $\dot r = \ddot r = 0$; θ is constant, so $\dot\theta = \ddot\theta = 0$; and $\dot\phi = \omega$, $\ddot\phi = 0$.
Velocity:

$$\vec v = r\sin\theta\,\dot\phi\,\vec u_\phi = R\frac{\sqrt3}{2}\omega\,\vec u_\phi, \qquad v = \frac{\sqrt3}{2}R\omega.$$

Acceleration:

$$a_r = -r\sin^2\theta\,\dot\phi^2 = -R\frac{3}{4}\omega^2,$$

$$a_\theta = -r\sin\theta\cos\theta\,\dot\phi^2 = -R\frac{\sqrt3}{2}\frac12\omega^2 = -\frac{\sqrt3}{4}R\omega^2,$$

$$a_\phi = 0.$$

Thus

$$\vec a = -\frac34 R\omega^2\,\vec u_r - \frac{\sqrt3}{4}R\omega^2\,\vec u_\theta, \qquad a = \sqrt{\frac{9}{16}+\frac{3}{16}}\,R\omega^2 = \frac{\sqrt3}{2}R\omega^2.$$

Note that $a = v^2/\rho$, where $\rho = R\sin\theta$ is the radius of the latitude circle.

6.5 Exercises and Step-by-Step Solutions

Exercise 6.1 · ⋆ Coordinate conversion

Convert the Cartesian point $(2, -2, 3)$ to cylindrical and spherical coordinates.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 6.1. Cylindrical:

$$\rho = \sqrt{x^2+y^2} = \sqrt{4+4} = \sqrt8 = 2.83,$$

$$\theta = \arctan\frac{y}{x} = \arctan(-1) = -45.0^\circ$$

(the point is in the fourth quadrant), and $z = 3.00$.
Spherical:

$$r = \sqrt{x^2+y^2+z^2} = \sqrt{4+4+9} = \sqrt{17} = 4.12,$$

$$\theta = \arccos\frac{z}{r} = \arccos\frac{3}{4.12} = 43.3^\circ,$$

and $\phi = -45.0^\circ$.

Exercise 6.2 · ⋆ **Uniform circular motion in polar form**

A particle moves on a circle of radius 3.00 m with $\theta(t) = 2.00t$ rad. Find its velocity and acceleration at $t = 1.00$ s using polar coordinates.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 6.2. With $r = R = 3.00$ constant, $\dot r = \ddot r = 0$, and $\dot\theta = 2.00$, $\ddot\theta = 0$:

$$\vec v = R\dot\theta\,\vec u_\theta = 3.00(2.00)\,\vec u_\theta = 6.00\,\vec u_\theta \ \mathrm{m\,s^{-1}},$$

$$v = 6.00 \ \mathrm{m\,s^{-1}}.$$

$$a_r = -R\dot\theta^2 = -3.00(4.00) = -12.0 \ \mathrm{m\,s^{-2}},$$

$$a_\theta = R\ddot\theta = 0.$$

Thus

$$\vec a = -12.0\,\vec u_r \ \mathrm{m\,s^{-2}}, \qquad a = 12.0 \ \mathrm{m\,s^{-2}}$$

toward the centre.

Exercise 6.3 · ⋆ **Polar kinematics**

A particle has $r(t) = t^2$ (m) and $\theta(t) = t$ (rad). Find $\vec v$ and $\vec a$ at $t = 2.00$ s, and their magnitudes.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 6.3.

$$r = t^2 \implies \dot{r} = 2t,\ \ddot{r} = 2; \qquad \theta = t \implies \dot{\theta} = 1,\ \ddot{\theta} = 0.$$

At $t = 2$: $r = 4$, $\dot{r} = 4$, $\dot{\theta} = 1$.

$$\vec{v} = 4\,\vec{u}_r + 4(1)\,\vec{u}_\theta = 4\,\vec{u}_r + 4\,\vec{u}_\theta \ \mathrm{m\,s^{-1}}, \qquad v = 4\sqrt{2} = 5.66 \ \mathrm{m\,s^{-1}}.$$

$$a_r = \ddot{r} - r\dot{\theta}^2 = 2 - 4(1) = -2.00 \ \mathrm{m\,s^{-2}},$$

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = 0 + 2(4)(1) = 8.00 \ \mathrm{m\,s^{-2}}.$$

$$\vec{a} = -2.00\,\vec{u}_r + 8.00\,\vec{u}_\theta \ \mathrm{m\,s^{-2}}, \qquad a = \sqrt{4 + 64} = 8.25 \ \mathrm{m\,s^{-2}}.$$

Exercise 6.4 · **Radial oscillation**

A particle moves with $r(t) = 2 + \cos t$ (m) and $\theta(t) = t$ (rad). Find $\vec{v}$ and $\vec{a}$ at $t = \pi/2$.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 6.4.

$$r = 2 + \cos t \implies \dot{r} = -\sin t,\ \ddot{r} = -\cos t.$$

At $t = \pi/2$: $r = 2$, $\dot{r} = -1$, $\ddot{r} = 0$; and $\theta = t$ gives $\dot{\theta} = 1$, $\ddot{\theta} = 0$.

$$\vec{v} = \dot{r}\,\vec{u}_r + r\dot{\theta}\,\vec{u}_\theta = -\vec{u}_r + 2\vec{u}_\theta \ \mathrm{m\,s^{-1}}, \qquad v = \sqrt{1 + 4} = \sqrt{5} = 2.24 \ \mathrm{m\,s^{-1}}.$$

$$a_r = \ddot{r} - r\dot{\theta}^2 = 0 - 2(1) = -2.00\ \mathrm{m\,s^{-2}},$$
$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = 0 + 2(-1)(1) = -2.00\ \mathrm{m\,s^{-2}}.$$
$$\vec{a} = -2.00\,\vec{u}_r - 2.00\,\vec{u}_\theta\ \mathrm{m\,s^{-2}}, \qquad a = \sqrt{8} = 2.83\ \mathrm{m\,s^{-2}}.$$

Exercise 6.5 · Cylindrical motion

A particle has $\rho = 2.00$ m (constant), $\theta = 3.00t$ rad, and $z = 4.00t^2$ m. Find its velocity and acceleration at $t = 1.00$ s.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 6.5.

$$\rho = 2,\ \dot{\rho} = 0,\ \ddot{\rho} = 0; \qquad \theta = 3t,\ \dot{\theta} = 3,\ \ddot{\theta} = 0; \qquad z = 4t^2,\ \dot{z} = 8t,\ \ddot{z} = 8.$$

At $t = 1$: $\dot{z} = 8.00$, $\ddot{z} = 8.00$.

$$\vec{v} = \rho\dot{\theta}\,\vec{u}_\theta + \dot{z}\,\hat{k} = 2(3)\vec{u}_\theta + 8\hat{k} = 6.00\,\vec{u}_\theta + 8.00\,\hat{k}\ \mathrm{m\,s^{-1}},$$
$$v = \sqrt{36 + 64} = 10.0\ \mathrm{m\,s^{-1}}.$$
$$a_\rho = -\rho\dot{\theta}^2 = -2(9) = -18.0\ \mathrm{m\,s^{-2}},$$
$$a_\theta = \rho\ddot{\theta} + 2\dot{\rho}\dot{\theta} = 0,$$
$$a_z = \ddot{z} = 8.00\ \mathrm{m\,s^{-2}}.$$
$$\vec{a} = -18.0\,\vec{u}_\rho + 8.00\,\hat{k}\ \mathrm{m\,s^{-2}}, \qquad a = \sqrt{324 + 64} = 19.7\ \mathrm{m\,s^{-2}}.$$

Exercise 6.6 · Motion on the equator

A particle moves on the equator of a sphere of radius R with azimuth $\phi = \omega t$. Use spherical coordinates to find its speed and acceleration.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 6.6. On the equator, $\theta = \pi/2$ is constant, so $\dot\theta = \ddot\theta = 0$, and $r = R$ is constant. With $\dot\phi = \omega$,

$$\vec v = R\sin\frac{\pi}{2}\,\omega\,\vec u_\phi = R\omega\,\vec u_\phi, \qquad v = R\omega.$$

The acceleration components are

$$a_r = -R\sin^2\frac{\pi}{2}\,\omega^2 = -R\omega^2,$$

$$a_\theta = -R\sin\frac{\pi}{2}\cos\frac{\pi}{2}\,\omega^2 = 0, \qquad a_\phi = 0.$$

Thus

$$\vec a = -R\omega^2\,\vec u_r,$$

which is the familiar centripetal acceleration toward the centre of the sphere.

Exercise 6.7 · Derivatives of the polar basis

Starting from $\vec u_r = \cos\theta\,\hat\imath + \sin\theta\,\hat\jmath$ and $\vec u_\theta = -\sin\theta\,\hat\imath + \cos\theta\,\hat\jmath$, prove that

$$\frac{\mathrm d\vec u_r}{\mathrm dt} = \dot\theta\,\vec u_\theta, \qquad \frac{\mathrm d\vec u_\theta}{\mathrm dt} = -\dot\theta\,\vec u_r.$$

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 6.7. Differentiate $\vec{u}_r$:

$$\frac{\mathrm{d}\vec{u}_r}{\mathrm{d}t} = -\sin\theta\,\dot{\theta}\,\hat{\imath} + \cos\theta\,\dot{\theta}\,\hat{\jmath} = \dot{\theta}(-\sin\theta\,\hat{\imath} + \cos\theta\,\hat{\jmath}) = \dot{\theta}\,\vec{u}_\theta.$$

Similarly,

$$\frac{\mathrm{d}\vec{u}_\theta}{\mathrm{d}t} = -\cos\theta\,\dot{\theta}\,\hat{\imath} - \sin\theta\,\dot{\theta}\,\hat{\jmath} = -\dot{\theta}(\cos\theta\,\hat{\imath} + \sin\theta\,\hat{\jmath}) = -\dot{\theta}\,\vec{u}_r.$$

Exercise 6.8 · **Circular motion from the general formula**

Starting from the general polar acceleration, set $r = R =$ constant and recover

$$\vec{a} = -\frac{v^2}{R}\vec{u}_r + a_t\,\vec{u}_\theta.$$

Identify v and a_t in terms of θ.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 6.8. For $r = R$, $\dot{r} = \ddot{r} = 0$. The general formulas give

$$\vec{a} = -R\dot{\theta}^2\,\vec{u}_r + R\ddot{\theta}\,\vec{u}_\theta.$$

The speed is $v = R\dot{\theta}$, so

$$R\dot{\theta}^2 = \frac{(R\dot{\theta})^2}{R} = \frac{v^2}{R}.$$

The tangential acceleration is

$$a_t = \frac{\mathrm{d}v}{\mathrm{d}t} = R\ddot{\theta}.$$

Hence

$$\vec{a} = -\frac{v^2}{R}\vec{u}_r + R\ddot{\theta}\,\vec{u}_\theta = -\frac{v^2}{R}\vec{u}_r + a_t\,\vec{u}_\theta.$$

Exercise 6.9 · ★★★ **Conservation of areal velocity**

For a central force the transverse acceleration vanishes: $a_\theta = 0$. Show that this implies

$$r^2\dot{\theta} = \text{constant}.$$

Interpret this constant geometrically as twice the areal velocity.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 6.9. The transverse acceleration is

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = \frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}t}\left(r^2\dot{\theta}\right).$$

If $a_\theta = 0$, then

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(r^2\dot{\theta}\right) = 0 \implies r^2\dot{\theta} = h = \text{constant}.$$

In polar coordinates the area swept by the radius vector in time $\mathrm{d}t$ is

$$\mathrm{d}A = \frac{1}{2}r^2\,\mathrm{d}\theta,$$

so the areal velocity is

$$\frac{\mathrm{d}A}{\mathrm{d}t} = \frac{1}{2}r^2\dot{\theta} = \frac{h}{2},$$

which is constant. This is Kepler's second law.

Exercise 6.10 · ⋆⋆⋆ **Logarithmic spiral**

A particle moves with $r(t) = ae^{bt}$ and $\theta(t) = ct$, where a, b, c are constants. Find the radial and transverse components of velocity and acceleration. Show that the acceleration makes a constant angle with the radial direction.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 6.10. With $r = ae^{bt}$,

$$\dot{r} = ab\,e^{bt}, \qquad \ddot{r} = ab^2 e^{bt},$$

and with $\theta = ct$, $\dot{\theta} = c$, $\ddot{\theta} = 0$. Therefore

$$v_r = \dot{r} = ab\,e^{bt}, \qquad v_\theta = r\dot{\theta} = ac\,e^{bt},$$

so

$$\vec{v} = ae^{bt}\left(b\,\vec{u}_r + c\,\vec{u}_\theta\right).$$

For the acceleration,

$$a_r = \ddot{r} - r\dot{\theta}^2 = ae^{bt}\left(b^2 - c^2\right),$$

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = 2abc\,e^{bt}.$$

Thus

$$\vec{a} = ae^{bt}\left[\left(b^2 - c^2\right)\vec{u}_r + 2bc\,\vec{u}_\theta\right].$$

The ratio $a_\theta/a_r = 2bc/(b^2 - c^2)$ is independent of time, so the acceleration makes a constant angle with the radial direction. The curve is a logarithmic spiral.

CHAPTER 7

Relative and Composite Motion

You will learn to: transform velocities and accelerations between reference frames; use the composition laws; analyse motion in a rotating frame; and identify the Coriolis and centrifugal terms.

7.1 Galilean transformation

Let $R(O, x, y, z)$ be a fixed frame and $R'(O', x', y', z')$ a frame moving with constant velocity $\vec{V}$ relative to R, without rotation. Then

$$\vec{r} = \vec{r}' + \vec{V}t, \qquad \vec{v} = \vec{v}' + \vec{V}, \qquad \vec{a} = \vec{a}'.$$

The acceleration is the same in both frames. Such frames are **inertial**: Newton's laws have the same form in all of them. This is Galilean relativity.

7.2 Composition of velocities and accelerations

For a general moving frame (translating and rotating), let

- $\vec{v}_a$ be the absolute velocity (relative to R);
- $\vec{v}_r$ the relative velocity (relative to R');
- $\vec{v}_e$ the entrainment (drag) velocity, i.e. the velocity in R of the point of R' coinciding with M.

Then

$$\boxed{\vec{v}_a = \vec{v}_r + \vec{v}_e.}$$

For a frame rotating with angular velocity $\vec{\Omega}$,

$$\vec{v}_e = \vec{v}(O'/R) + \vec{\Omega} \times \vec{O'M}.$$

The acceleration composition law is

$$\boxed{\vec{a}_a = \vec{a}_r + \vec{a}_e + \vec{a}_c,}$$

where

$$\vec{a}_e = \vec{a}(O'/R) + \frac{\mathrm{d}\vec{\Omega}}{\mathrm{d}t} \times \vec{O'M} + \vec{\Omega} \times \left(\vec{\Omega} \times \vec{O'M}\right),$$

and the **Coriolis acceleration** is

$$\vec{a}_c = 2\vec{\Omega} \times \vec{v}_r.$$

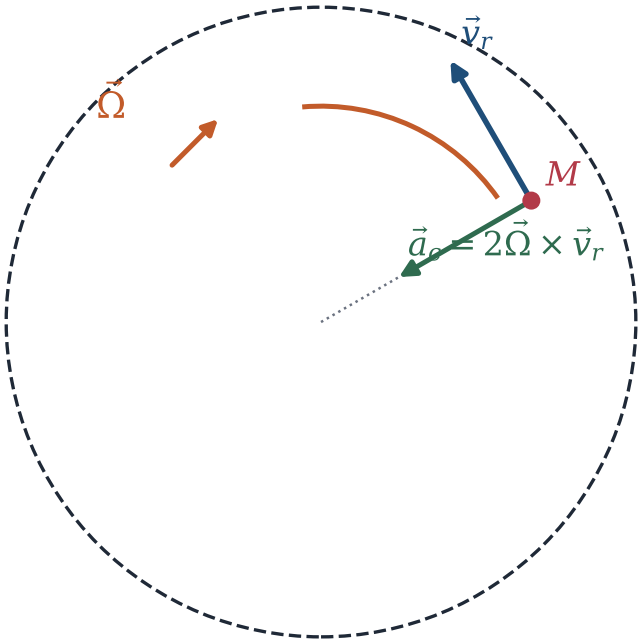


Figure 7.1: A point M moving in a rotating frame. The Coriolis acceleration is perpendicular to both the angular velocity and the relative velocity.

7.3 Non-inertial frames and fictitious forces

Newton's second law in an inertial frame is $m\vec{a}_a = \vec{F}$. Using the composition law,

$$m\vec{a}_r = \vec{F} - m\vec{a}_e - m\vec{a}_c.$$

The last two terms behave like additional forces in the rotating frame:

$$\vec{F}_{\text{Coriolis}} = -2m\vec{\Omega} \times \vec{v}_r, \qquad \vec{F}_{\text{centrifugal}} = -m\vec{\Omega} \times (\vec{\Omega} \times \vec{r}\,').$$

They are called **fictitious** or **inertial** forces because they disappear in an inertial frame.

Boat problems

In a river, the entrainment velocity is the current. The boat's velocity relative to the water is $\vec{v}_r$; the velocity relative to the bank is $\vec{v}_a = \vec{v}_r + \vec{v}_e$. Draw the vector triangle before computing.

7.4 Worked examples

Worked example 7.1 · Crossing a river

A boat has speed $5.00\ \mathrm{m\,s^{-1}}$ relative to the water. A river of width 100 m flows at $3.00\ \mathrm{m\,s^{-1}}$. (a) If the boat heads straight across, find the crossing time and the downstream drift. (b) At what angle upstream must it head to land directly opposite? Find the crossing time.

Solution 7.1. (a) Heading straight across, the cross-stream component is $v_{r,\perp} = 5.00\ \mathrm{m\,s^{-1}}$, so

$$t = \frac{d}{v_{r,\perp}} = \frac{100}{5.00} = 20.0\ \mathrm{s}.$$

The downstream drift is

$$x = v_e t = 3.00(20.0) = 60.0\ \mathrm{m}.$$

The speed relative to the bank is $\sqrt{5.00^2 + 3.00^2} = 5.83\ \mathrm{m\,s^{-1}}$.
(b) To cancel the current, the upstream component must satisfy

$$v_r \sin\theta = v_e \implies 5.00 \sin\theta = 3.00 \implies \theta = 36.9^\circ$$

upstream from the perpendicular direction. The cross-stream component is

$$v_r \cos\theta = 5.00 \cos 36.9^\circ = 4.00\ \mathrm{m\,s^{-1}},$$

so

$$t = \frac{100}{4.00} = 25.0\ \mathrm{s}.$$

Worked example 7.2 · Motion on a rotating disk

A disk rotates with constant angular velocity $\Omega = 2.00\ \mathrm{rad\,s^{-1}}$. A bead moves radially outward along a groove at constant relative speed $v_r = 0.500\ \mathrm{m\,s^{-1}}$. At the instant when $r = 1.00$ m, find the absolute velocity and acceleration of the bead.

Solution 7.2. Take $\vec{\Omega} = \Omega\hat{k}$ and $\vec{v}_r = v_r\vec{u}_r$. The entrainment velocity is

$$v_e = \Omega r = 2.00(1.00) = 2.00\ \mathrm{m\,s^{-1}}$$

in the tangential direction. Thus

$$\vec{v}_a = 0.500\,\vec{u}_r + 2.00\,\vec{u}_\theta, \qquad v_a = \sqrt{0.25 + 4.00} = 2.06\ \mathrm{m\,s^{-1}}.$$

The relative acceleration is zero (constant radial speed). The entrainment acceleration is

$$\vec{a}_e = \vec{\Omega} \times (\vec{\Omega} \times \vec{r}) = -\Omega^2 r\,\vec{u}_r = -4.00\,\vec{u}_r\ \mathrm{m\,s^{-2}}.$$

The Coriolis acceleration is

$$\vec{a}_c = 2\vec{\Omega} \times \vec{v}_r = 2(2.00)(0.500)\,\vec{u}_\theta = 2.00\,\vec{u}_\theta\ \mathrm{m\,s^{-2}}.$$

Therefore

$$\vec{a}_a = -4.00\,\vec{u}_r + 2.00\,\vec{u}_\theta\ \mathrm{m\,s^{-2}}, \qquad a_a = \sqrt{16 + 4} = 4.47\ \mathrm{m\,s^{-2}}.$$

7.5 Exercises and Step-by-Step Solutions

Exercise 7.1 · ⋆ **Walking on a train**

A train moves at 20.0 $\mathrm{m\,s^{-1}}$ relative to the ground. A passenger walks at 2.00 $\mathrm{m\,s^{-1}}$ relative to the train. Find the passenger's velocity relative to the ground when walking (a) forward and (b) backward.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 7.1. (a) Forward:

$$v = 20.0 + 2.00 = 22.0\ \mathrm{m\,s^{-1}}.$$

(b) Backward:

$$v = 20.0 - 2.00 = 18.0\ \mathrm{m\,s^{-1}}.$$

Exercise 7.2 · ⋆ **Rain and walking**

Rain falls vertically at 8.00 $\mathrm{m\,s^{-1}}$. A person walks horizontally at 6.00 $\mathrm{m\,s^{-1}}$. Find the magnitude and direction of the rain's velocity relative to the person.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 7.2. The velocity of the rain relative to the person is

$$\vec{v}_{rp} = \vec{v}_{\text{rain}} - \vec{v}_{\text{person}}.$$

With the rain vertical and the person horizontal,

$$v_{rp} = \sqrt{8.00^2 + 6.00^2} = 10.0 \text{ m s}^{-1}.$$

The angle from the vertical satisfies

$$\tan\theta = \frac{6.00}{8.00} = 0.750 \implies \theta = 36.9^\circ.$$

The rain appears to come from ahead, at 36.9° from the vertical.

Exercise 7.3 · ⋆ **Straight across a river**

A boat has speed 4.00 m s^{-1} relative to the water. The river is 200 m wide and flows at 2.00 m s^{-1}. If the boat heads straight across, find the crossing time, the downstream drift, and the speed relative to the bank.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 7.3. Heading straight across, the cross-stream speed is $4.00\ \mathrm{m\,s^{-1}}$:

$$t = \frac{200}{4.00} = 50.0\ \mathrm{s}.$$

The drift is

$$x = v_e t = 2.00(50.0) = 100\ \mathrm{m}.$$

The speed relative to the bank is

$$v = \sqrt{4.00^2 + 2.00^2} = \sqrt{20.0} = 4.47\ \mathrm{m\,s^{-1}}.$$

Exercise 7.4 · Landing directly opposite

Repeat the river of the previous exercise, but now find the heading angle required to land directly opposite the starting point, and the corresponding crossing time.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 7.4. To land directly opposite, the upstream component of the boat's velocity must cancel the current:

$$4.00\sin\theta = 2.00 \implies \sin\theta = 0.500 \implies \theta = 30.0^\circ.$$

The cross-stream component is

$$4.00\cos 30.0^\circ = 3.46\ \mathrm{m\,s^{-1}},$$

so

$$t = \frac{200}{3.46} = 57.7\ \mathrm{s}.$$

Exercise 7.5 · Relative velocity of two cars

Car A travels east at $25.0\ \mathrm{m\,s^{-1}}$; car B travels north at $15.0\ \mathrm{m\,s^{-1}}$. Find the velocity of B relative to A: magnitude and direction.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 7.5. Take east as $+x$ and north as $+y$:

$$\vec{v}_A = (25.0, 0), \qquad \vec{v}_B = (0, 15.0).$$

$$\vec{v}_{B/A} = \vec{v}_B - \vec{v}_A = (-25.0,\ 15.0)\ \mathrm{m\,s^{-1}}.$$

$$\left\|\vec{v}_{B/A}\right\| = \sqrt{625 + 225} = \sqrt{850} = 29.2\ \mathrm{m\,s^{-1}}.$$

The direction is

$$\theta = \arctan\frac{15.0}{-25.0}.$$

The vector points northwest; measured from east, $\theta = 180^\circ - 31.0^\circ = 149^\circ$.

Exercise 7.6 · Bead on a rotating disk

A disk rotates at $\Omega = 3.00\ \mathrm{rad\,s^{-1}}$. A bead moves radially outward at constant relative speed $v_r = 2.00\ \mathrm{m\,s^{-1}}$. At $r = 0.500$ m, find the absolute velocity and the Coriolis and centrifugal accelerations.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 7.6. The entrainment velocity is

$$v_e = \Omega r = 3.00(0.500) = 1.50\ \mathrm{m\,s^{-1}}$$

tangential. Thus

$$\vec{v}_a = 2.00\,\vec{u}_r + 1.50\,\vec{u}_\theta, \qquad v_a = \sqrt{4.00 + 2.25} = 2.50\ \mathrm{m\,s^{-1}}.$$

The Coriolis acceleration has magnitude

$$a_c = 2\Omega v_r = 2(3.00)(2.00) = 12.0\ \mathrm{m\,s^{-2}}$$

in the tangential direction. The centrifugal acceleration has magnitude

$$\Omega^2 r = (3.00)^2(0.500) = 4.50\ \mathrm{m\,s^{-2}}$$

directed outward. (The entrainment acceleration $\vec{\Omega} \times (\vec{\Omega} \times \vec{r})$ points inward with this magnitude.)

Exercise 7.7 · Coriolis on Earth

At latitude $45.0°$, an object moves horizontally eastward at $100\ \mathrm{m\,s^{-1}}$. Taking the Earth's angular velocity as $\Omega = 7.29 \times 10^{-5}\ \mathrm{rad\,s^{-1}}$, estimate the magnitude of the horizontal Coriolis acceleration.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 7.7. The horizontal Coriolis acceleration at latitude λ for a horizontal velocity has magnitude

$$a_C = 2\Omega v \sin\lambda.$$

Thus

$$a_C = 2(7.29 \times 10^{-5})(100)\sin 45.0° = 1.03 \times 10^{-2}\ \mathrm{m\,s^{-2}}.$$

This is small but not negligible over long times; it is what causes the deflection of long-range projectiles and cyclones.

Exercise 7.8 · Galilean invariance

Show that Newton's second law $m\vec{a} = \vec{F}$ is unchanged under the Galilean transformation $\vec{r} = \vec{r}\,' + \vec{V}t$ with constant $\vec{V}$.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 7.8. Under $\vec{r} = \vec{r}' + \vec{V}t$ with $\vec{V}$ constant,

$$\vec{v} = \vec{v}' + \vec{V}, \qquad \vec{a} = \frac{\mathrm{d}\vec{v}'}{\mathrm{d}t} + \frac{\mathrm{d}\vec{V}}{\mathrm{d}t} = \vec{a}'.$$

Therefore

$$m\vec{a}' = m\vec{a} = \vec{F}.$$

The equation of motion has the same form in both frames, so the frames are inertial.

Exercise 7.9 · ⋆⋆⋆ **Deriving the composition law**

A frame R' rotates with angular velocity $\vec{\Omega}(t)$ relative to R. Starting from $\vec{r} = \vec{r}'$ (common origin), derive

$$\vec{v}_a = \vec{v}_r + \vec{\Omega} \times \vec{r}',$$

and then

$$\vec{a}_a = \vec{a}_r + 2\vec{\Omega} \times \vec{v}_r + \vec{\Omega} \times (\vec{\Omega} \times \vec{r}') + \frac{\mathrm{d}\vec{\Omega}}{\mathrm{d}t} \times \vec{r}'.$$

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 7.9. Let $\vec{r}'$ have constant components in R'. The time derivative in R is

$$\left(\frac{\mathrm{d}\vec{r}'}{\mathrm{d}t}\right)_R = \left(\frac{\mathrm{d}\vec{r}'}{\mathrm{d}t}\right)_{R'} + \vec{\Omega} \times \vec{r}' = \vec{v}_r + \vec{\Omega} \times \vec{r}',$$

which gives the velocity law. Applying the same operator to $\vec{v}_a = \vec{v}_r + \vec{\Omega}\times\vec{r}'$:

$$\vec{a}_a = \left(\frac{\mathrm{d}\vec{v}_r}{\mathrm{d}t}\right)_R + \frac{\mathrm{d}\vec{\Omega}}{\mathrm{d}t}\times\vec{r}' + \vec{\Omega}\times\left(\frac{\mathrm{d}\vec{r}'}{\mathrm{d}t}\right)_R.$$

Now

$$\left(\frac{\mathrm{d}\vec{v}_r}{\mathrm{d}t}\right)_R = \vec{a}_r + \vec{\Omega}\times\vec{v}_r,$$

and

$$\left(\frac{\mathrm{d}\vec{r}'}{\mathrm{d}t}\right)_R = \vec{v}_r + \vec{\Omega}\times\vec{r}'.$$

Substituting,

$$\vec{a}_a = \vec{a}_r + \vec{\Omega}\times\vec{v}_r + \frac{\mathrm{d}\vec{\Omega}}{\mathrm{d}t}\times\vec{r}' + \vec{\Omega}\times\vec{v}_r + \vec{\Omega}\times(\vec{\Omega}\times\vec{r}').$$

The two $\vec{\Omega}\times\vec{v}_r$ terms combine to $2\vec{\Omega}\times\vec{v}_r$, giving the stated result.

Exercise 7.10 · ⋆⋆⋆ **Ball thrown in an accelerating train**

A train accelerates at $2.00\ \mathrm{m\,s^{-2}}$. A passenger throws a ball vertically upward with speed $5.00\ \mathrm{m\,s^{-1}}$ relative to the train. (a) Find the time of flight. (b) Find where the ball lands relative to the thrower in the train. (c) Explain the result using the composition law.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 7.10. (a) In the train frame, the ball moves vertically with acceleration $-g$. Its time of flight is

$$t = \frac{2v_0}{g} = \frac{2(5.00)}{9.81} = 1.02\ \mathrm{s}.$$

(b) During this time the train moves forward a distance

$$s = \frac{1}{2}at^2 = \frac{1}{2}(2.00)(1.02)^2 = 1.04\ \mathrm{m}.$$

In the train frame the ball lands 1.04 m behind the thrower. (c) In the ground frame the ball keeps the horizontal velocity it had at launch (the train's velocity at that instant) because no horizontal force acts on it. The train, however, continues to accelerate. The train therefore moves ahead of the ball, so in the train frame the ball appears to be deflected backward. This is the effect of the fictitious force $-m\vec{a}_{\text{train}}$ in the non-inertial frame.

PART III

Dynamics

CHAPTER 8

Newton's Laws and Free-Body Diagrams

You will learn to: state Newton's three laws; draw a correct free-body diagram; write the equations of motion along convenient axes; and solve equilibrium and circular-motion problems.

8.1 Newton's laws

1. **Inertia.** In an inertial frame, a body remains at rest or in uniform rectilinear motion unless acted on by a net external force.
2. **Fundamental principle of dynamics.**

$$\sum \vec{F}_{\text{ext}} = \frac{\mathrm{d}\vec{p}}{\mathrm{d}t} = \frac{\mathrm{d}(m\vec{v})}{\mathrm{d}t}.$$

   For constant mass,

$$\boxed{\sum \vec{F}_{\text{ext}} = m\vec{a}.}$$

3. **Action–reaction.** If A exerts a force on B, then B exerts on A a force equal in magnitude and opposite in direction:

$$\vec{F}_{B\to A} = -\vec{F}_{A\to B}.$$

The three laws are valid in an **inertial frame**. A frame accelerating or rotating relative to an inertial frame requires the fictitious forces of Chapter 7.

8.2 Standard force models

Force	Expression	Direction
Weight	$\vec{W} = m\vec{g}$	vertical, downward
Normal reaction	$\vec{N}$	perpendicular to the surface, away from it
Tension	T	along the string, pulling the body
Spring (Hooke)	$\vec{F} = -k(l - l_0)\vec{u}$	toward the equilibrium position
Solid friction	$f \le \mu_s N$, $f_k = \mu_k N$	opposite to the tendency of sliding
Viscous drag	$\vec{f} = -k\vec{v}$	opposite to the velocity
Universal gravity	$\vec{F} = -GMm\,\vec{u}_r/r^2$	attractive, toward the other mass

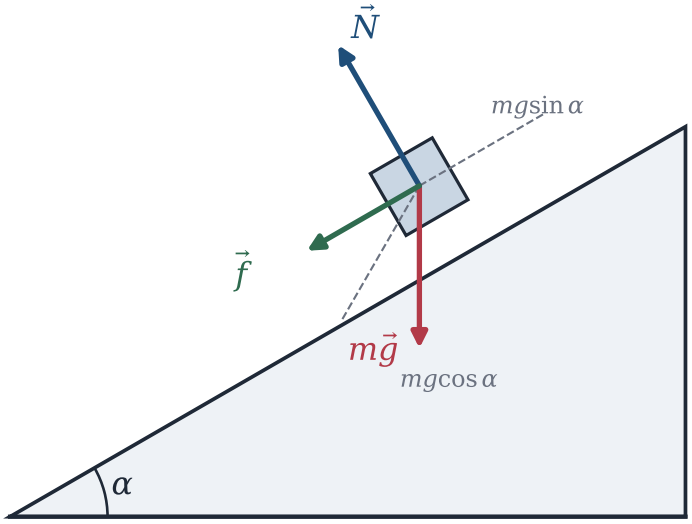


Figure 8.1: Free-body diagram for a block on an inclined plane. The weight is decomposed into components parallel and perpendicular to the plane.

8.3 The free-body diagram method

Five steps that solve most problems

1. Isolate the body of interest.
2. Draw every external force acting *on* it (not forces it exerts on other bodies).
3. Choose axes: along the motion and perpendicular to it.
4. Decompose the forces along the axes.
5. Apply $\sum F_x = ma_x$, $\sum F_y = ma_y$. If the body does not accelerate in a direction, the sum of forces in that direction is zero.

Two classic errors

Do not put both the weight $m\vec{g}$ and its components $mg\sin\alpha$, $mg\cos\alpha$ on the same diagram: choose one representation. Do not include the force that the body exerts on something else; the diagram is only about forces *on* the body.

8.4 Dynamics of circular motion

For a particle moving on a circle of radius R, project Newton's second law on the inward radial direction and on the tangent:

$$\sum F_{\text{radial}} = m\frac{v^2}{R}, \qquad \sum F_{\text{tangential}} = m\frac{\mathrm{d}v}{\mathrm{d}t}.$$

The radial equation is not a new law; it is $m\vec{a}$ written in the normal and tangential directions.

8.5 Worked examples

Worked example 8.1 · Block on a frictionless incline

A block of mass $m = 5.00$ kg rests on a frictionless plane inclined at $\alpha = 30.0^\circ$. Find its acceleration and the normal reaction.

Solution 8.1. Take axes parallel and perpendicular to the plane. The only forces are the weight $m\vec{g}$ and the normal $\vec{N}$. The weight components are $mg\sin\alpha$ down the slope and $mg\cos\alpha$ into the plane.
Perpendicular to the plane there is no acceleration:

$$N - mg\cos\alpha = 0 \implies N = mg\cos\alpha.$$

Parallel to the plane:

$$mg\sin\alpha = ma \implies a = g\sin\alpha.$$

Numerically,

$$a = 9.81\sin 30.0^\circ = 4.91\ \text{m}\,\text{s}^{-2},$$

$$N = 5.00(9.81)\cos 30.0^\circ = 42.5\ \text{N}.$$

Worked example 8.2 · Apparent weight in an elevator

A person of mass $m = 70.0$ kg stands on a scale in an elevator. Find the scale reading when the elevator (a) is at rest, (b) accelerates upward at $2.00\ \text{m}\,\text{s}^{-2}$, (c) accelerates downward at $2.00\ \text{m}\,\text{s}^{-2}$, and (d) is in free fall.

Solution 8.2. The scale reads the normal force N. Take upward as positive. Newton's second law:

$$N - mg = ma \implies N = m(g + a).$$

(a) $a = 0$:

$$N = 70.0(9.81) = 687\ \text{N}.$$

(b) $a = +2.00\ \text{m}\,\text{s}^{-2}$:

$$N = 70.0(9.81 + 2.00) = 827\ \text{N}.$$

(c) $a = -2.00\ \text{m}\,\text{s}^{-2}$:

$$N = 70.0(9.81 - 2.00) = 547\ \text{N}.$$

(d) $a = -g$:

$$N = 70.0(9.81 - 9.81) = 0.$$

The person feels weightless.

8.6 Exercises and Step-by-Step Solutions

Exercise 8.1 · ⋆ Newton's second law

A net force of 30.0 N acts on a 10.0 kg body initially at rest. Find its acceleration, its speed after 4.00 s, and the distance travelled.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 8.1.

$$a = \frac{F}{m} = \frac{30.0}{10.0} = 3.00\ \text{m s}^{-2}.$$

$$v = at = 3.00(4.00) = 12.0\ \text{m s}^{-1}.$$

$$d = \frac{1}{2}at^2 = \frac{1}{2}(3.00)(4.00)^2 = 24.0\ \text{m}.$$

Exercise 8.2 · ⋆ **Mass and weight**

A 60.0 kg astronaut travels to the Moon, where $g_{\text{Moon}} = 1.62\ \text{m s}^{-2}$. Find her mass and weight on Earth and on the Moon.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 8.2. Mass is an intrinsic property, so $m = 60.0$ kg everywhere.
On Earth:

$$W_E = mg_E = 60.0(9.81) = 589\ \text{N}.$$

On the Moon:

$$W_M = mg_M = 60.0(1.62) = 97.2\ \text{N}.$$

Exercise 8.3 · ⋆ **Elevator scale**

A 75.0 kg person stands on a scale in an elevator. Find the reading when the elevator (a) is at rest, (b) accelerates upward at 2.00 $\mathrm{m\,s^{-2}}$, (c) accelerates downward at 2.00 $\mathrm{m\,s^{-2}}$, and (d) is in free fall.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 8.3. With upward positive, $N - mg = ma$, so $N = m(g + a)$.
(a) $N = 75.0(9.81) = 736$ N.
(b) $N = 75.0(9.81 + 2.00) = 886$ N.
(c) $N = 75.0(9.81 - 2.00) = 586$ N.
(d) $N = 0$.

Exercise 8.4 · **Sliding down an incline**

A block of mass 8.00 kg slides from rest down a frictionless incline of angle 25.0°. Find its acceleration, the normal reaction, and the time to slide 10.0 m.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 8.4.

$$a = g\sin 25.0^\circ = 9.81(0.4226) = 4.15\ \mathrm{m\,s^{-2}}.$$

$$N = mg\cos 25.0^\circ = 8.00(9.81)(0.9063) = 71.1\ \mathrm{N}.$$

From $s = \frac{1}{2}at^2$,

$$t = \sqrt{\frac{2s}{a}} = \sqrt{\frac{20.0}{4.15}} = 2.20 \text{ s}.$$

Exercise 8.5 · Atwood machine

Two masses $m_1 = 3.00$ kg and $m_2 = 5.00$ kg hang from a massless string over a frictionless pulley. Find the acceleration of the masses and the tension in the string.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 8.5. Let a be the acceleration of m_2 downward (and of m_1 upward). For m_2:

$$m_2 g - T = m_2 a.$$

For m_1:

$$T - m_1 g = m_1 a.$$

Adding,

$$(m_2 - m_1)g = (m_1 + m_2)a,$$

$$a = \frac{(5.00 - 3.00)(9.81)}{8.00} = 2.45 \text{ m s}^{-2}.$$

Then

$$T = m_1(g + a) = 3.00(9.81 + 2.45) = 36.8 \text{ N}.$$

Exercise 8.6 · Two blocks in contact

A horizontal force $F = 20.0$ N pushes a block $m_1 = 2.00$ kg which is in contact with a block $m_2 = 3.00$ kg on a frictionless horizontal surface. Find the acceleration and the contact force between the blocks. What is the contact force if the force is applied to m_2 instead?

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 8.6. The two blocks move together:

$$a = \frac{F}{m_1 + m_2} = \frac{20.0}{5.00} = 4.00\ \mathrm{m\,s^{-2}}.$$

For m_2, the only horizontal force is the contact force N_c:

$$N_c = m_2 a = 3.00(4.00) = 12.0\ \mathrm{N}.$$

If the force is applied to m_2, the contact force acts on m_1:

$$N_c' = m_1 a = 2.00(4.00) = 8.00\ \mathrm{N}.$$

Exercise 8.7 · Car on a flat curve

A car rounds a flat curve of radius 50.0 m. The coefficient of static friction between the tyres and the road is $\mu_s = 0.600$. Find the maximum speed before skidding.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 8.7. The static friction provides the centripetal force:

$$f_s = m\frac{v^2}{R}.$$

At the limit $f_s = \mu_s N = \mu_s mg$, so

$$\mu_s mg = m\frac{v_{\max}^2}{R} \implies v_{\max} = \sqrt{\mu_s g R}.$$

$$v_{\max} = \sqrt{(0.600)(9.81)(50.0)} = \sqrt{294} = 17.2 \text{ m s}^{-1}.$$

Exercise 8.8 · Conical pendulum

A ball of mass 0.500 kg is attached to a string of length 1.00 m and moves in a horizontal circle, the string making 30.0° with the vertical. Find the tension and the speed of the ball.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 8.8. Let $\theta = 30.0°$ and let $r = L\sin\theta$ be the radius of the circle. Vertical equilibrium:

$$T\cos\theta = mg \implies T = \frac{mg}{\cos\theta}.$$

Horizontal (radial) equation:

$$T\sin\theta = m\frac{v^2}{r}.$$

Substituting T and r:

$$mg\tan\theta = m\frac{v^2}{L\sin\theta} \implies v^2 = gL\sin\theta\tan\theta.$$

Numerically,

$$T = \frac{0.500(9.81)}{\cos 30.0°} = 5.66 \text{ N},$$

$$v = \sqrt{9.81(1.00)\sin 30.0°\tan 30.0°} = \sqrt{2.83} = 1.68 \text{ m s}^{-1}.$$

Exercise 8.9 · Table and hanging mass

A block $m_1 = 4.00$ kg on a frictionless horizontal table is connected by a string over a pulley to a hanging mass $m_2 = 2.00$ kg. Find the acceleration and the tension.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 8.9. For the hanging mass:

$$m_2 g - T = m_2 a.$$

For the block on the table:

$$T = m_1 a.$$

Adding,

$$m_2 g = (m_1 + m_2)a \implies a = \frac{2.00(9.81)}{6.00} = 3.27\ \mathrm{m\,s^{-2}}.$$

$$T = m_1 a = 4.00(3.27) = 13.1\ \mathrm{N}.$$

Exercise 8.10 · ★★★ **Time-dependent force**

A body of mass m starts from rest under a force $F(t) = F_0(1 - t/T)$ along a straight line, where F_0 and T are positive constants. Find $v(t)$ and $x(t)$. Find the speed and position at $t = T$, and describe the motion for $t > T$.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 8.10.

$$a(t) = \frac{F(t)}{m} = \frac{F_0}{m}\left(1 - \frac{t}{T}\right).$$

Integrating with $v(0) = 0$:

$$v(t) = \frac{F_0}{m}\left(t - \frac{t^2}{2T}\right).$$

Integrating again with $x(0) = 0$:

$$x(t) = \frac{F_0}{m}\left(\frac{t^2}{2} - \frac{t^3}{6T}\right).$$

At $t = T$:

$$v(T) = \frac{F_0}{m}\left(T - \frac{T}{2}\right) = \frac{F_0 T}{2m},$$

$$x(T) = \frac{F_0}{m}\left(\frac{T^2}{2} - \frac{T^2}{6}\right) = \frac{F_0 T^2}{3m}.$$

For $t > T$ the force becomes negative: the body decelerates, reverses direction after stopping, and then accelerates in the opposite direction.

Exercise 8.11 · ⋆⋆⋆ **Vertical loop**

A small block of mass m slides inside a vertical circular loop of radius R. At the top of the loop, find the minimum speed for which it remains in contact with the track. If its speed at the top is $\sqrt{2gR}$, find the normal force there.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 8.11. At the top of the loop, both the weight and the normal force point downward (toward the centre):

$$N + mg = m\frac{v^2}{R}.$$

The block remains in contact as long as $N \geq 0$, so

$$m\frac{v^2}{R} \geq mg \implies v \geq \sqrt{gR}.$$

Thus $v_{\min} = \sqrt{gR}$.
If $v = \sqrt{2gR}$,

$$N = m\frac{2gR}{R} - mg = 2mg - mg = mg.$$

CHAPTER 9

Forces at Work: Friction, Inclines, Pulleys, Springs

You will learn to: apply the laws of solid friction; solve rough-incline problems; handle connected bodies and pulleys; use Hooke's law; and analyse motion with viscous drag.

9.1 Solid friction

When a body is pressed against a surface, the contact force has a normal component N and a tangential component f called friction.

- **Static friction** prevents sliding. It adjusts itself up to a maximum:

$$f_s \le \mu_s N.$$

  Sliding begins when the applied tangential force exceeds $\mu_s N$.

- **Kinetic friction** acts while the body slides, opposite to the velocity:

$$f_k = \mu_k N.$$

  Usually $\mu_k < \mu_s$.

The coefficients μ_s and μ_k are dimensionless and depend on the pair of surfaces.

Angle of friction and angle of repose

The maximum angle of an incline on which a body can rest without sliding is the *angle of repose*:

$$\tan \alpha_c = \mu_s.$$

For $\alpha < \alpha_c$, static friction can hold the body; for $\alpha > \alpha_c$, it slides.

9.2 Rough inclined plane

For a body sliding down a plane of angle α with kinetic friction,

$$N = mg\cos\alpha, \qquad f_k = \mu_k N = \mu_k mg\cos\alpha,$$

and along the plane

$$ma = mg\sin\alpha - \mu_k mg\cos\alpha,$$

so

$$\boxed{a = g\left(\sin\alpha - \mu_k \cos\alpha\right).}$$

If the body is pushed up the plane, friction reverses and $a = -g(\sin\alpha + \mu_k\cos\alpha)$.

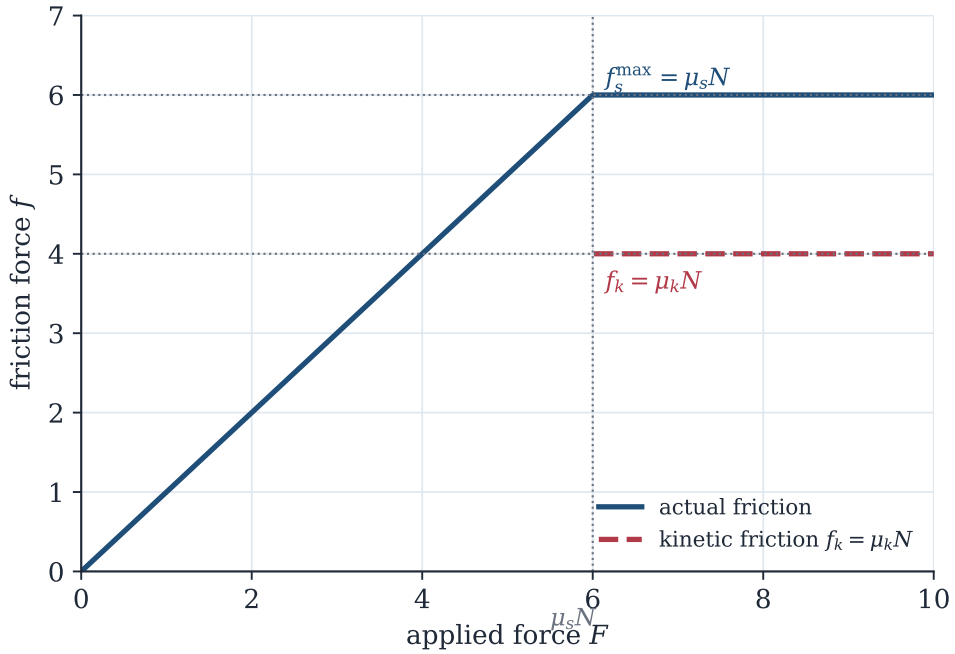


Figure 9.1: The friction force as a function of the applied force. The static friction rises to $\mu_s N$; once sliding starts, the kinetic friction drops to $\mu_k N$.

9.3 Connected bodies and pulleys

For an ideal (massless, frictionless) pulley, the tension is the same on both sides of the string. The masses have the same magnitude of acceleration, possibly in opposite directions. The method is always:

1. choose a positive direction of motion;
2. write Newton's second law for each body separately;
3. use the constraint (same string length) to relate accelerations;
4. solve the resulting linear system.

9.4 Elastic springs

For an ideal spring of stiffness k and natural length l_0,

$$\vec{F} = -k(l - l_0)\,\vec{u},$$

where $\vec{u}$ points from the support to the free end. If $x = l - l_0$ is the extension,

$$F_x = -kx.$$

At static equilibrium, the spring force balances the other forces.

For springs in **series**, the same force acts on both and

$$\frac{1}{k_{\text{eq}}} = \frac{1}{k_1} + \frac{1}{k_2}.$$

For springs in **parallel**, the extensions are equal and

$$k_{\text{eq}} = k_1 + k_2.$$

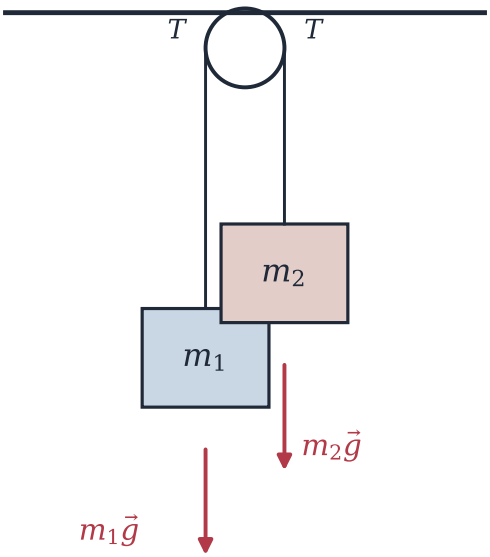


Figure 9.2: The Atwood machine. The heavier mass descends; both masses have the same magnitude of acceleration.

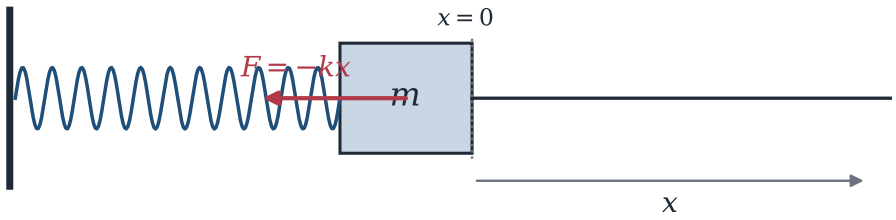


Figure 9.3: A mass attached to a horizontal spring. The restoring force is proportional to the displacement from equilibrium and points opposite to it.

9.5 Viscous drag and terminal velocity

For slow motion in a fluid, a common model is

$$\vec{f} = -k\vec{v}.$$

A body falling from rest under gravity then obeys

$$m\frac{\mathrm{d}v}{\mathrm{d}t} = mg - kv.$$

The solution is

$$v(t) = v_L\left(1 - e^{-t/\tau}\right), \qquad v_L = \frac{mg}{k}, \qquad \tau = \frac{m}{k}.$$

The velocity approaches the terminal velocity v_L exponentially.

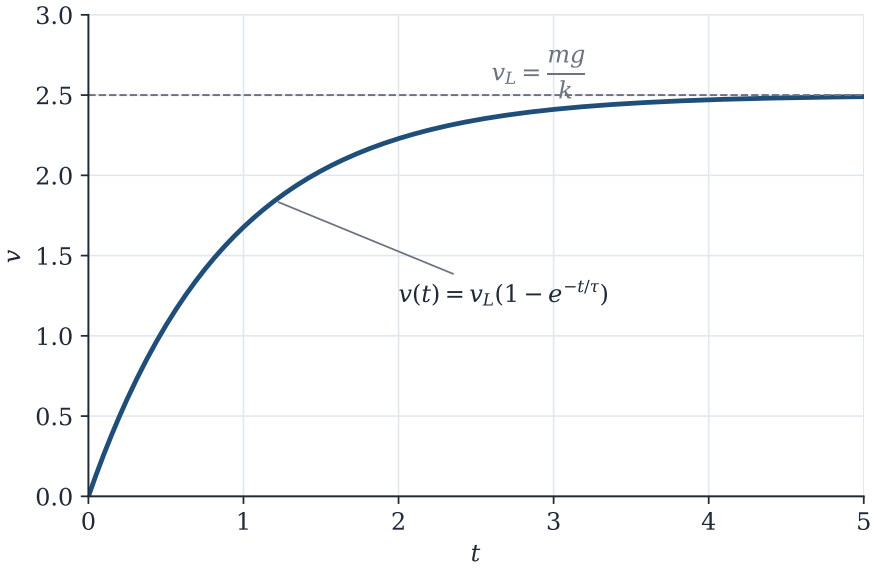


Figure 9.4: Approach to terminal velocity for linear viscous drag.

9.6 Worked examples

Worked example 9.1 · Rough incline

A block of mass $m = 10.0$ kg is placed on a plane inclined at $\alpha = 30.0°$. The coefficients are $\mu_s = 0.600$ and $\mu_k = 0.250$. (a) Does the block slide? (b) If it slides, find its acceleration.

Solution 9.1. The component of weight down the plane is

$$mg\sin\alpha = 10.0(9.81)\sin 30.0° = 49.1 \text{ N}.$$

The normal reaction is

$$N = mg\cos\alpha = 10.0(9.81)\cos 30.0° = 85.0 \text{ N}.$$

The maximum static friction is

$$f_s^{\max} = \mu_s N = 0.600(85.0) = 51.0 \text{ N}.$$

Since $49.1 < 51.0$, the block does not slide.

If the block were already sliding, the kinetic friction would be

$$f_k = \mu_k N = 0.250(85.0) = 21.3 \text{ N},$$

and the acceleration down the plane would be

$$a = g(\sin\alpha - \mu_k \cos\alpha) = 9.81(0.500 - 0.250(0.866)) = 2.78 \text{ m s}^{-2}.$$

Worked example 9.2 · Spring and mass

A mass $m = 0.500$ kg is attached to a horizontal spring of stiffness $k = 200$ N m^{-1} on a frictionless surface. The mass is pulled 0.100 m from equilibrium and released. (a) Find the spring force and acceleration at the instant of release. (b) Find the acceleration when the mass passes through equilibrium.

Solution 9.2. (a) At release, $x = 0.100$ m:

$$F_x = -kx = -200(0.100) = -20.0 \text{ N}.$$

$$a_x = \frac{F_x}{m} = \frac{-20.0}{0.500} = -40.0 \text{ m s}^{-2}.$$

The minus sign means the acceleration points back toward equilibrium.
(b) At equilibrium $x = 0$, so $F_x = 0$ and $a_x = 0$. The velocity is maximum there.

9.7 Exercises and Step-by-Step Solutions

Exercise 9.1 · ⋆ Starting a crate

A crate of mass 50.0 kg rests on a horizontal floor. The coefficients of friction are $\mu_s = 0.500$ and $\mu_k = 0.400$. Find the minimum horizontal force needed to start the crate moving, and the force needed to keep it moving at constant speed.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 9.1.

$$N = mg = 50.0(9.81) = 491 \text{ N}.$$

To start the motion:

$$F_s = \mu_s N = 0.500(491) = 245 \text{ N}.$$

To keep it moving at constant speed:

$$F_k = \mu_k N = 0.400(491) = 196 \text{ N}.$$

Exercise 9.2 · ⋆ **Rough horizontal surface**

A 10.0 kg block is pulled horizontally by a force of 50.0 N on a surface with $\mu_k = 0.300$. Find its acceleration.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 9.2.

$$N = mg = 10.0(9.81) = 98.1 \text{ N}, \qquad f_k = \mu_k N = 0.300(98.1) = 29.4 \text{ N}.$$

$$a = \frac{F - f_k}{m} = \frac{50.0 - 29.4}{10.0} = 2.06 \text{ m}\,\text{s}^{-2}.$$

Exercise 9.3 · ⋆ **Spring extension**

A mass of 2.00 kg hangs from a vertical spring of stiffness $k = 300 \text{ N}\,\text{m}^{-1}$. Find the extension at equilibrium and the force needed to stretch the spring by 5.00 cm.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 9.3. At equilibrium,

$$kx = mg \implies x = \frac{mg}{k} = \frac{2.00(9.81)}{300} = 0.0654 \text{ m}.$$

For a stretch $x = 0.0500$ m,

$$F = kx = 300(0.0500) = 15.0 \text{ N}.$$

Exercise 9.4 · Rough incline

A block of mass 5.00 kg slides down a plane inclined at 35.0° with $\mu_k = 0.200$. Find the normal reaction, the friction force, and the acceleration.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 9.4.

$$N = mg\cos 35.0^\circ = 5.00(9.81)(0.8192) = 40.2 \text{ N}.$$

$$f_k = \mu_k N = 0.200(40.2) = 8.04 \text{ N}.$$

$$a = g(\sin 35.0^\circ - \mu_k \cos 35.0^\circ) = 9.81(0.5736 - 0.1638) = 4.02 \text{ m s}^{-2}.$$

Exercise 9.5 · Angle of repose

The coefficient of static friction between a body and a plane is $\mu_s = 0.450$. Find the angle of repose. If the plane is inclined at 30.0°, does the body slide?

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 9.5.

$$\tan\alpha_c = \mu_s = 0.450 \implies \alpha_c = \arctan(0.450) = 24.2^\circ.$$

At $30.0^\circ > \alpha_c$, the body slides, because $\tan 30.0^\circ = 0.577 > 0.450$.

Exercise 9.6 · Table, pulley, and friction

A block $m_1 = 4.00$ kg on a rough horizontal table ($\mu_k = 0.250$) is connected by a string over a pulley to a hanging mass $m_2 = 2.00$ kg. Find the acceleration and the tension.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 9.6. For m_2 (downward positive):

$$m_2 g - T = m_2 a.$$

For m_1:

$$T - f_k = m_1 a, \qquad f_k = \mu_k m_1 g.$$

Adding,

$$m_2 g - \mu_k m_1 g = (m_1 + m_2)a,$$

$$a = \frac{2.00(9.81) - 0.250(4.00)(9.81)}{6.00} = \frac{19.62 - 9.81}{6.00} = 1.64\ \mathrm{m\,s^{-2}}.$$

Then

$$T = m_1(a + \mu_k g) = 4.00(1.64 + 2.45) = 16.4\ \mathrm{N}.$$

Exercise 9.7 · Double incline

Two masses $m_1 = 3.00$ kg and $m_2 = 5.00$ kg are connected by a string over a pulley at the top of a double incline with angles $\alpha = 30.0^\circ$ for m_1 and $\beta = 45.0^\circ$ for m_2. The surfaces are frictionless. Find the acceleration and the tension.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 9.7. Assume m_2 slides down its plane and m_1 moves up its plane. For m_1:

$$T - m_1 g \sin\alpha = m_1 a.$$

For m_2:

$$m_2 g \sin\beta - T = m_2 a.$$

Adding,

$$a = \frac{m_2 g \sin\beta - m_1 g \sin\alpha}{m_1 + m_2}.$$

$$a = \frac{5.00(9.81)\sin 45.0^\circ - 3.00(9.81)\sin 30.0^\circ}{8.00} = \frac{34.68 - 14.72}{8.00} = 2.50 \text{ m s}^{-2}.$$

$$T = m_1(g\sin\alpha + a) = 3.00(4.91 + 2.50) = 22.2 \text{ N}.$$

Exercise 9.8 · Terminal velocity

A body of mass 0.200 kg falls from rest subject to linear drag $f = kv$ with $k = 0.500$ kg s^{-1}. Find the terminal velocity, the time constant, the speed at $t = 1.20$ s, and the time to reach 99% of the terminal velocity.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 9.8.

$$v_L = \frac{mg}{k} = \frac{0.200(9.81)}{0.500} = 3.92\ \text{m s}^{-1},$$

$$\tau = \frac{m}{k} = \frac{0.200}{0.500} = 0.400\ \text{s}.$$

At $t = 1.20\ \text{s} = 3\tau$:

$$v = 3.92\left(1 - e^{-3}\right) = 3.92(0.950) = 3.73\ \text{m s}^{-1}.$$

For 99% of terminal velocity,

$$1 - e^{-t/\tau} = 0.99 \implies e^{-t/\tau} = 0.01 \implies t = \tau \ln 100 = 0.400(4.605) = 1.84\ \text{s}.$$

Exercise 9.9 · ⋆⋆⋆ **Springs in series and parallel**

Two springs of stiffness $k_1 = 100\ \text{N m}^{-1}$ and $k_2 = 200\ \text{N m}^{-1}$ are combined. Find the equivalent stiffness (a) in series and (b) in parallel. A mass of 2.00 kg hangs from each combination; find the equilibrium extension in each case.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 9.9. (a) Series:

$$\frac{1}{k_{\text{eq}}} = \frac{1}{100} + \frac{1}{200} = \frac{3}{200} \implies k_{\text{eq}} = 66.7\ \text{N m}^{-1}.$$

The extension under $m = 2.00$ kg:

$$x = \frac{mg}{k_{\text{eq}}} = \frac{19.62}{66.7} = 0.294 \text{ m}.$$

(b) Parallel:

$$k_{\text{eq}} = 100 + 200 = 300 \text{ N m}^{-1},$$

$$x = \frac{19.62}{300} = 0.0654 \text{ m}.$$

Exercise 9.10 · ⋆⋆⋆ **Optimal pulling angle**

A block of mass m is pulled on a rough horizontal floor by a force F making an angle θ above the horizontal. The coefficient of kinetic friction is μ_k. (a) Show that

$$a = \frac{F}{m}(\cos\theta + \mu_k \sin\theta) - \mu_k g.$$

(b) Show that the acceleration is maximum when $\tan\theta = \mu_k$. (c) For $m = 10.0$ kg, $F = 50.0$ N, and $\mu_k = 0.300$, compute the optimal angle and the maximum acceleration.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 9.10. (a) The normal reaction is reduced by the vertical component of the pull:

$$N = mg - F\sin\theta.$$

The friction is $f_k = \mu_k N$, and the horizontal equation is

$$ma = F\cos\theta - f_k = F\cos\theta - \mu_k(mg - F\sin\theta).$$

Dividing by m,

$$a = \frac{F}{m}(\cos\theta + \mu_k \sin\theta) - \mu_k g.$$

(b) Differentiate with respect to θ:

$$\frac{\mathrm{d}a}{\mathrm{d}\theta} = \frac{F}{m}(-\sin\theta + \mu_k \cos\theta).$$

Setting this to zero gives

$$\tan\theta = \mu_k.$$

The second derivative is negative at this point, so it is a maximum.
(c)

$$\theta = \arctan(0.300) = 16.7^\circ.$$

$$a_{\max} = \frac{50.0}{10.0}\left(\cos 16.7^\circ + 0.300 \sin 16.7^\circ\right) - 0.300(9.81).$$

$$a_{\max} = 5.00(0.9578 + 0.0862) - 2.943 = 5.22 - 2.94 = 2.28 \text{ m s}^{-2}.$$

CHAPTER 10

Dynamics in Non-Inertial Reference Frames

You will learn to: distinguish inertial from non-inertial frames; compute the fictitious forces of transport and Coriolis; formulate Newton's second law in a moving frame; analyse apparent weight and the shape of rotating fluids; and calculate the eastward deflection of falling bodies and the precession of the Foucault pendulum.

10.1 Inertial vs Non-Inertial Frames

Newton's laws of motion are valid only in **inertial (Galilean) reference frames**—frames that are either at rest or moving at constant velocity with respect to the distant stars.

When we observe motion from an **accelerating or rotating reference frame** $\mathcal{R}'(O', x', y', z')$ relative to an inertial frame $\mathcal{R}(O, x, y, z)$, the frame itself experiences acceleration. To apply Newton's second law in $\mathcal{R}'$, we must introduce **fictitious (inertial) forces**. These forces do not arise from physical interactions (no physical agent pulls on the body), but are purely consequences of the acceleration of the observer's frame.

10.2 Composition of Accelerations and Fictitious Forces

Recall from kinematics that the absolute acceleration $\vec{a}_a$ of a particle of mass m is related to its relative acceleration $\vec{a}_r$ in $\mathcal{R}'$ by:

$$\vec{a}_a = \vec{a}_r + \vec{a}_e + \vec{a}_c,$$

where:

- $\vec{a}_r = \left(\frac{\mathrm{d}^2\vec{r}'}{\mathrm{d}t^2}\right)_{\mathcal{R}'}$ is the **relative acceleration** observed in $\mathcal{R}'$;
- $\vec{a}_e = \vec{a}(O') + \left(\frac{\mathrm{d}\vec{\Omega}}{\mathrm{d}t}\right) \times \vec{r}' + \vec{\Omega} \times (\vec{\Omega} \times \vec{r}')$ is the **acceleration of transport** (acceleration of the point in $\mathcal{R}'$ coincident with M);
- $\vec{a}_c = 2\vec{\Omega} \times \vec{v}_r$ is the **Coriolis acceleration**.

Multiplying by mass m and using Newton's second law in the inertial frame, $m\vec{a}_a = \sum \vec{F}_{\text{real}}$, we obtain the **Fundamental Equation of Relative Dynamics**:

$$\boxed{m\vec{a}_r = \sum \vec{F}_{\text{real}} + \vec{F}_{ie} + \vec{F}_{ic}}$$

where the two inertial forces are:

$$\vec{F}_{ie} = -m\vec{a}_e \quad \textbf{(Inertial force of transport / Centrifugal force)}, \tag{10.1}$$

$$\vec{F}_{ic} = -m\vec{a}_c = -2m\,\vec{\Omega} \times \vec{v}_r \quad \textbf{(Coriolis force)}. \tag{10.2}$$

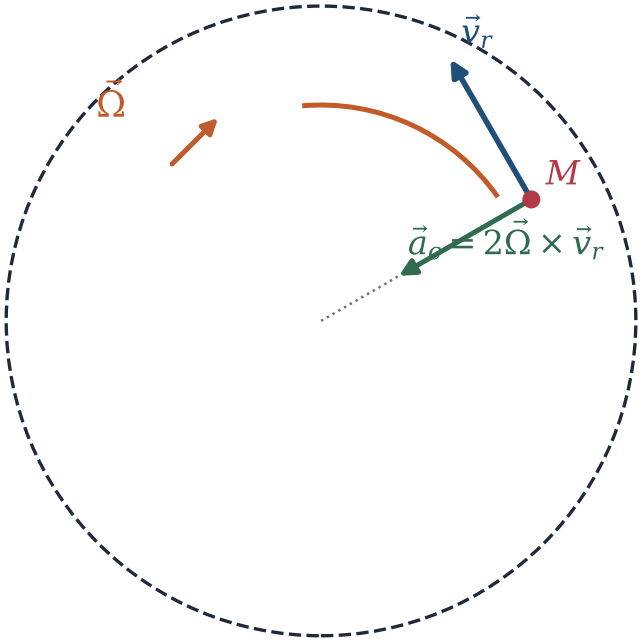


Figure 10.1: A particle M observed from both a fixed inertial frame $\mathcal{R}(O, \hat{\imath}, \hat{\jmath}, \hat{k})$ and a rotating frame $\mathcal{R}'(O', \vec{u}_r, \vec{u}_\theta, \hat{k})$ with angular velocity $\vec{\Omega}$.

Characteristics of Fictitious Forces

- $\vec{F}_{ie}$ exists whenever the frame translates with acceleration ($\vec{a}(O') \neq \vec{0}$) or rotates. In a purely rotating frame ($\vec{\Omega} = \text{const}$, $O' = O$), $\vec{F}_{ie} = -m\vec{\Omega} \times (\vec{\Omega} \times \vec{r}') = m\Omega^2 \vec{r}_\perp$ is directed *radially outward* from the axis of rotation: the **centrifugal force**.
- $\vec{F}_{ic}$ acts **only when the particle is moving** in the relative frame ($\vec{v}_r \neq \vec{0}$). It is always perpendicular to the relative velocity ($\vec{F}_{ic} \cdot \vec{v}_r = 0$), meaning **the Coriolis force does no mechanical work** in the relative frame!

Solving Dynamics in a Non-Inertial Frame

1. Identify the relative frame $\mathcal{R}'$ and its acceleration $\vec{a}(O')$ and angular velocity $\vec{\Omega}(t)$.
2. Draw the free-body diagram of the particle including all **real forces** ($\vec{P}, \vec{N}, \vec{T}, \vec{f}$, etc.).
3. Add the **fictitious forces**: $\vec{F}_{ie} = -m\vec{a}_e$ and $\vec{F}_{ic} = -2m\vec{\Omega} \times \vec{v}_r$.
4. Apply $m\vec{a}_r = \sum \vec{F}_{\text{real}} + \vec{F}_{ie} + \vec{F}_{ic}$ along chosen coordinate axes in $\mathcal{R}'$.

10.3 Worked Examples

Worked example 10.1 · Incline inside an accelerating truck

A block of mass $m = 2.0$ kg sits on a smooth ramp of angle $\alpha = 30°$ fixed inside a truck. The truck accelerates horizontally with acceleration a_0. Find the acceleration a_0 such that the block remains at rest relative to the ramp.

Solution 10.1. In the frame of the truck $\mathcal{R}'$ (translating horizontally with $\vec{a}_0 = a_0\hat{\imath}$):

- Real forces: weight $\vec{P} = -mg\hat{\jmath}$, normal force $\vec{N}$ perpendicular to the ramp ($N\sin\alpha\hat{\imath} + N\cos\alpha\hat{\jmath}$).
- Fictitious force: $\vec{F}_{ie} = -m\vec{a}_0 = -ma_0\hat{\imath}$. Since $\vec{\Omega} = \vec{0}$, $\vec{F}_{ic} = \vec{0}$.

Relative equilibrium requires $\vec{a}_r = \vec{0}$, hence:

$$\vec{P} + \vec{N} + \vec{F}_{ie} = \vec{0}.$$

Projecting along the horizontal and vertical axes:

$$\begin{aligned} \text{Horizontal:} \quad & N\sin\alpha - ma_0 = 0 \implies N\sin\alpha = ma_0, \\ \text{Vertical:} \quad & N\cos\alpha - mg = 0 \implies N\cos\alpha = mg. \end{aligned}$$

Dividing the two equations eliminates N:

$$\tan\alpha = \frac{a_0}{g} \implies \boxed{a_0 = g\tan\alpha.}$$

For $\alpha = 30°$, $a_0 = 9.81 \times \tan 30° = 9.81 \times 0.577 = 5.66\ \text{m s}^{-2}$.

Worked example 10.2 · Bead sliding on a rotating smooth rod

A thin smooth horizontal rod rotates about a vertical axis through one end O at constant angular velocity ω. A bead of mass m slides along the rod with distance $r(t)$ from O. Find the equation of motion for $r(t)$ and the horizontal reaction force exerted by the rod.

Solution 10.2. In the rotating frame $\mathcal{R}'$ attached to the rod with unit vector $\vec{u}_r$ along the rod and $\vec{u}_\theta$ perpendicular:

- The relative position is $\vec{r}' = r\vec{u}_r$, relative velocity $\vec{v}_r = \dot{r}\vec{u}_r$, relative acceleration $\vec{a}_r = \ddot{r}\vec{u}_r$.
- Real forces: weight $\vec{P} = -mg\hat{k}$, normal reaction $\vec{N} = N_\theta\vec{u}_\theta + N_z\hat{k}$ (since the rod is smooth, there is no component along $\vec{u}_r$).
- Centrifugal force: $\vec{F}_{\text{cf}} = -m\vec{\Omega} \times (\vec{\Omega} \times \vec{r}') = m\omega^2 r\vec{u}_r$.
- Coriolis force: $\vec{F}_{\text{Cor}} = -2m\vec{\Omega} \times \vec{v}_r = -2m(\omega\hat{k}) \times (\dot{r}\vec{u}_r) = -2m\omega\dot{r}\vec{u}_\theta$.

Applying $m\vec{a}_r = \vec{P} + \vec{N} + \vec{F}_{\text{cf}} + \vec{F}_{\text{Cor}}$:

$$\begin{aligned} \vec{u}_r: &\quad m\ddot{r} = m\omega^2 r \implies \boxed{\ddot{r} - \omega^2 r = 0,} \\ \vec{u}_\theta: &\quad 0 = N_\theta - 2m\omega\dot{r} \implies \boxed{N_\theta = 2m\omega\dot{r},} \\ \hat{k}: &\quad 0 = N_z - mg \implies N_z = mg. \end{aligned}$$

The radial equation $\ddot{r} - \omega^2 r = 0$ has solution $r(t) = A\cosh(\omega t) + B\sinh(\omega t)$. The bead accelerates exponentially outward! The rod must push horizontally with $N_\theta = 2m\omega\dot{r}$ to counteract the Coriolis force.

10.4 Exercises and Step-by-Step Solutions

Exercise 10.1 · ⋆ **Scale in an accelerating elevator**

A student of mass $m = 70.0$ kg stands on a spring scale inside an elevator. What does the scale read (in newtons and equivalent apparent mass in kg) when: (a) the elevator accelerates upward at $a = 2.00\ \text{m}\,\text{s}^{-2}$; (b) the elevator moves upward at constant speed $v = 4.00\ \text{m}\,\text{s}^{-1}$; (c) the elevator accelerates downward at $a = 2.00\ \text{m}\,\text{s}^{-2}$; (d) the elevator cable snaps and it falls freely?

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 10.1. In the reference frame of the elevator $\mathcal{R}'$, the elevator has vertical acceleration $\vec{a} = a\hat{\jmath}$ (with $\hat{\jmath}$ upward). The forces acting on the student in $\mathcal{R}'$ are:

- Real weight: $\vec{P} = -mg\hat{\jmath}$;
- Normal reaction from the scale: $\vec{N} = N\hat{\jmath}$ (the scale reads N);
- Fictitious inertial force: $\vec{F}_{ie} = -m\vec{a}_e = -ma\hat{\jmath}$.

Since the student is at rest in the elevator ($\vec{a}_r = \vec{0}$):

$$\vec{N} + \vec{P} + \vec{F}_{ie} = \vec{0} \implies N - mg - ma = 0 \implies N = m(g + a).$$

(a) Accelerating upward ($a = +2.00\ \mathrm{m\,s^{-2}}$):

$$N = 70.0(9.81 + 2.00) = 70.0(11.81) = 826.7\ \mathrm{N}.$$

Apparent mass reading: $m_{\text{app}} = N/g = 826.7/9.81 = 84.3$ kg.
(b) Constant speed ($a = 0$):

$$N = mg = 70.0 \times 9.81 = 686.7\ \mathrm{N} \implies m_{\text{app}} = 70.0\ \mathrm{kg}.$$

(c) Accelerating downward ($a = -2.00\ \mathrm{m\,s^{-2}}$):

$$N = m(g - 2.00) = 70.0(9.81 - 2.00) = 70.0(7.81) = 546.7\ \mathrm{N} \implies m_{\text{app}} = 55.7\ \mathrm{kg}.$$

(d) Free fall ($a = -g$):

$$N = m(g - g) = 0\ \mathrm{N} \implies m_{\text{app}} = 0\ \mathrm{kg} \quad \text{(weightlessness)}.$$

Exercise 10.2 · ⋆ **Direction and magnitude of the Coriolis force**

A bullet of mass $m = 20.0$ g is fired horizontally due north at speed $v = 800\ \mathrm{m\,s^{-1}}$ at a location in the Northern hemisphere with latitude $\lambda = 45.0°$. (a) Determine the direction of the Coriolis force $\vec{F}_{\text{Cor}} = -2m\vec{\Omega} \times \vec{v}_r$. (b) Calculate the magnitude of $\vec{F}_{\text{Cor}}$, given Earth's angular speed $\Omega \approx 7.292 \times 10^{-5}\ \mathrm{rad\,s^{-1}}$. (c) Compare this force with the bullet's weight mg.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 10.2. Set up a local coordinate frame with $\hat{\imath}$ pointing East, $\hat{\jmath}$ pointing North, and $\hat{k}$ pointing vertically upward (Zenith). Earth's angular velocity vector $\vec{\Omega}$ points parallel to Earth's polar axis (North):

$$\vec{\Omega} = \Omega\cos\lambda\,\hat{\jmath} + \Omega\sin\lambda\,\hat{k}.$$

(a) The bullet flies horizontally due North: $\vec{v}_r = v\hat{\jmath}$. The cross product is:

$$\vec{\Omega} \times \vec{v}_r = (\Omega\cos\lambda\,\hat{\jmath} + \Omega\sin\lambda\,\hat{k}) \times (v\hat{\jmath}) = -\Omega v\sin\lambda\,\hat{\imath}.$$

Therefore, the Coriolis force is:

$$\vec{F}_{\text{Cor}} = -2m(\vec{\Omega} \times \vec{v}_r) = +2m\Omega v\sin\lambda\,\hat{\imath}.$$

Since $\hat{\imath}$ is East, the Coriolis force is directed **due East** (deflecting the trajectory to the right).
(b) Magnitude:

$$F_{\text{Cor}} = 2m\Omega v \sin\lambda = 2(0.0200)(7.292\times10^{-5})(800)\sin 45^\circ = 2(0.0200)(7.292\times10^{-5})(800)(0.7071) \approx 1.65\times10^{-3}$$

(c) The weight is $P = mg = 0.0200 \times 9.81 = 0.1962$ N. The ratio is:

$$\frac{F_{\text{Cor}}}{mg} = \frac{1.65 \times 10^{-3}}{0.1962} \approx 8.4 \times 10^{-3} \approx 0.84\%.$$

Although small, over long trajectories (such as artillery shells or storm systems), this deflection is significant.

Exercise 10.3 · Plumb line deflection and apparent gravity

Because the Earth rotates with angular speed Ω about its North-South axis, a plumb line does not point exactly toward the Earth's geometric center O. (a) Show that the effective gravity $\vec{g}_{\text{eff}}$ at latitude λ is given by $\vec{g}_{\text{eff}} = \vec{g}_0 + \Omega^2 R_E \cos\lambda\, \vec{u}_\perp$, where $\vec{g}_0 = -(GM_E/R_E^2)\vec{u}_r$ and $\vec{u}_\perp$ is directed perpendicular to the rotation axis outward. (b) Find an expression for the deflection angle θ between the true vertical and the plumb line, assuming $\Omega^2 R_E \ll g_0$. (c) Calculate the maximum deflection angle $\theta_{\max}$ and the latitude at which it occurs (take $R_E = 6.37 \times 10^6$ m).

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 10.3. (a) A plumb bob of mass m hanging at rest relative to Earth ($\vec{v}_r = \vec{0}$) is in equilibrium under its true gravitational attraction $\vec{F}_g = m\vec{g}_0$, the string tension $\vec{T}$, and the centrifugal force $\vec{F}_{\text{cf}}$:

$$\vec{T} + m\vec{g}_0 + \vec{F}_{\text{cf}} = \vec{0}.$$

The apparent weight is defined as $\vec{P}_{\text{app}} = -\vec{T} = m\vec{g}_{\text{eff}}$. Thus:

$$m\vec{g}_{\text{eff}} = m\vec{g}_0 + \vec{F}_{\text{cf}}.$$

The distance from the bob to the rotation axis is $r_\perp = R_E \cos\lambda$. The centrifugal force is:

$$\vec{F}_{\text{cf}} = m\Omega^2 r_\perp \vec{u}_\perp = m\Omega^2 R_E \cos\lambda\, \vec{u}_\perp.$$

Dividing by m yields:

$$\vec{g}_{\text{eff}} = \vec{g}_0 + \Omega^2 R_E \cos\lambda\, \vec{u}_\perp.$$

(b) Resolve $\vec{u}_\perp$ into local radial ($-\vec{u}_r$, downward) and tangential ($\vec{u}_\theta$, Southward) components:

$$\vec{u}_\perp = \cos\lambda\, \vec{u}_r - \sin\lambda\, \vec{u}_\theta.$$

Thus:

$$g_{\text{eff},r} = -g_0 + \Omega^2 R_E \cos^2\lambda \approx -g_0,$$
$$g_{\text{eff},\theta} = -\Omega^2 R_E \cos\lambda \sin\lambda = -\frac{1}{2}\Omega^2 R_E \sin(2\lambda).$$

The deflection angle θ is small, so:

$$\tan\theta \approx \theta \approx \frac{|g_{\text{eff},\theta}|}{|g_{\text{eff},r}|} \approx \frac{\Omega^2 R_E \sin(2\lambda)}{2g_0}.$$

(c) The factor $\sin(2\lambda)$ reaches its maximum value of 1 when $2\lambda = 90^\circ \implies \lambda = 45^\circ$. At $\lambda = 45^\circ$:

$$\theta_{\max} = \frac{\Omega^2 R_E}{2g_0} = \frac{(7.292\times10^{-5})^2 \times (6.37\times10^6)}{2\times 9.81} = \frac{3.387\times10^{-2}}{19.62} \approx 1.73\times10^{-3}\ \text{rad} \approx 0.099^\circ \approx 5.9'.$$

The plumb line is deflected toward the equator by about 6 arcminutes at 45° latitude!

Exercise 10.4 · Liquid surface in an accelerating container

A rectangular beaker of water is placed on a cart that accelerates horizontally with constant acceleration a_0 along the $+x$ direction. (a) In the non-inertial frame of the beaker, determine the effective gravity $\vec{g}_{\text{eff}} = \vec{g} - \vec{a}_0$. (b) Explain why the free surface of the liquid must be perpendicular to $\vec{g}_{\text{eff}}$. (c) Deduce the angle of inclination θ of the liquid surface relative to the horizontal.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 10.4. (a) In the frame of the container $\mathcal{R}'$, each element of liquid experiences real gravity $\vec{g} = -g\hat{\jmath}$ and the inertial force per unit mass $\vec{f}_{ie} = -\vec{a}_0 = -a_0\hat{\imath}$. The effective gravitational acceleration in $\mathcal{R}'$ is:

$$\vec{g}_{\text{eff}} = \vec{g} - \vec{a}_0 = -a_0\hat{\imath} - g\hat{\jmath}.$$

(b) At hydrostatic equilibrium in the relative frame, the pressure gradient satisfies $\vec{\nabla}p = \rho\vec{g}_{\text{eff}}$. Along the free surface, the pressure is constant ($p = p_{\text{atm}}$), so $\mathrm{d}p = \vec{\nabla}p \cdot \mathrm{d}\vec{r} = 0$. Therefore:

$$\rho\vec{g}_{\text{eff}} \cdot \mathrm{d}\vec{r} = 0 \implies \vec{g}_{\text{eff}} \perp \mathrm{d}\vec{r}.$$

The free surface must everywhere be orthogonal to the effective gravity vector $\vec{g}_{\text{eff}}$.
(c) Let the surface profile be $y(x)$. A tangent vector to the surface is $\mathrm{d}\vec{r} = \mathrm{d}x\hat{\imath} + \mathrm{d}y\hat{\jmath}$. The

condition $\vec{g}_{\text{eff}} \cdot \mathrm{d}\vec{r} = 0$ gives:

$$(-a_0)\mathrm{d}x + (-g)\mathrm{d}y = 0 \implies \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{a_0}{g}.$$

The surface is a flat plane tilted at an angle θ relative to the horizontal, with:

$$\boxed{\tan\theta = \frac{a_0}{g}.}$$

The liquid rises at the rear of the container and drops at the front.

Exercise 10.5 · Parabolic mirror: fluid in a rotating cylinder

A cylindrical bucket containing liquid of density ρ rotates about its central vertical axis at constant angular velocity ω. (a) Set up cylindrical coordinates (r, θ, z) in the rotating frame and find the effective force per unit volume on a fluid element. (b) Integrate the hydrostatic equation $\vec{\nabla} p = \rho \vec{g}_{\text{eff}}$ to find the pressure distribution $p(r, z)$. (c) Show that the free surface of the liquid forms a paraboloid of revolution, $z(r) = z_0 + \frac{\omega^2 r^2}{2g}$.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 10.5. (a) In the rotating frame $\mathcal{R}'$, the fluid is at rest. The forces per unit mass are gravity $\vec{g} = -g\hat{k}$ and the centrifugal acceleration $\vec{a}_{\text{cf}} = \omega^2 r \vec{u}_r$. The effective field is:

$$\vec{g}_{\text{eff}} = \omega^2 r \vec{u}_r - g\hat{k}.$$

(Coriolis force is zero because $\vec{v}_r = \vec{0}$).
(b) Hydrostatic equilibrium in $\mathcal{R}'$:

$$\vec{\nabla} p = \rho \vec{g}_{\text{eff}} \implies \frac{\partial p}{\partial r}\vec{u}_r + \frac{1}{r}\frac{\partial p}{\partial \theta}\vec{u}_\theta + \frac{\partial p}{\partial z}\hat{k} = \rho\omega^2 r \vec{u}_r - \rho g \hat{k}.$$

Matching components:

$$\frac{\partial p}{\partial r} = \rho\omega^2 r, \qquad \frac{\partial p}{\partial \theta} = 0, \qquad \frac{\partial p}{\partial z} = -\rho g.$$

Integrating:

$$p(r, z) = p_0 + \frac{1}{2}\rho\omega^2 r^2 - \rho g z,$$

where $p_0 = p(0, 0)$.

(c) Along the free surface, the pressure is atmospheric pressure p_{atm}. Setting $p(r,z) = p_{\text{atm}}$:

$$p_{\text{atm}} = p_0 + \frac{1}{2}\rho\omega^2 r^2 - \rho g z \implies \rho g z = (p_0 - p_{\text{atm}}) + \frac{1}{2}\rho\omega^2 r^2.$$

Defining $z_0 = z(0) = \frac{p_0 - p_{\text{atm}}}{\rho g}$:

$$\boxed{z(r) = z_0 + \frac{\omega^2 r^2}{2g}.}$$

This is a paraboloid of revolution! This principle is used in liquid-mirror astronomical telescopes (using molten gallium or mercury).

Exercise 10.6 · Mass on a rotating turntable with friction

A small puck of mass $m = 0.30$ kg sits on a horizontal rotating turntable at a distance $r = 0.25$ m from the axis of rotation. The coefficient of static friction between puck and turntable is $\mu_s = 0.40$. (a) In the turntable frame, write the force balance condition for the puck to remain stationary. (b) Calculate the maximum angular velocity $\omega_{\max}$ (in $\text{rad}\,\text{s}^{-1}$ and in rpm) before the puck slips. (c) If the turntable rotates at $\omega = 5.0\ \text{rad}\,\text{s}^{-1}$ and the puck is pushed radially inward with speed $v_r = 0.50\ \text{m}\,\text{s}^{-1}$, compute the Coriolis force acting on it.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 10.6. (a) In the rotating frame of the turntable:

- Vertical: normal force $\vec{N} = N\hat{k}$ balances weight $\vec{P} = -mg\hat{k} \implies N = mg$.
- Horizontal: the centrifugal force $\vec{F}_{\text{cf}} = m\omega^2 r\vec{u}_r$ pulls radially outward. To remain stationary ($\vec{v}_r = \vec{0}, \vec{a}_r = \vec{0}$), static friction $\vec{f}_s$ must point radially inward: $\vec{f}_s = -m\omega^2 r\vec{u}_r$.

The condition for no slip is $f_s \le \mu_s N$:

$$m\omega^2 r \le \mu_s mg \implies \omega^2 r \le \mu_s g.$$

(b) Maximum angular velocity:

$$\omega_{\max} = \sqrt{\frac{\mu_s g}{r}} = \sqrt{\frac{0.40 \times 9.81}{0.25}} = \sqrt{15.696} \approx 3.96\ \text{rad}\,\text{s}^{-1}.$$

In revolutions per minute:

$$N_{\text{rpm}} = \frac{\omega_{\max} \times 60}{2\pi} = \frac{3.96 \times 60}{2\pi} \approx 37.8\ \text{rpm}.$$

(c) If the puck has relative velocity $\vec{v}_r = -v_r\vec{u}_r$ (radially inward): Earth's rotation vector is $\vec{\Omega} = \omega\hat{k}$. The Coriolis force is:

$$\vec{F}_{\text{Cor}} = -2m\vec{\Omega} \times \vec{v}_r = -2m(\omega\hat{k}) \times (-v_r\vec{u}_r) = +2m\omega v_r(\hat{k} \times \vec{u}_r) = 2m\omega v_r\vec{u}_\theta.$$

The Coriolis force pushes in the direction of rotation ($\vec{u}_\theta$, forward tangential). Magnitude:

$$F_{\text{Cor}} = 2(0.30)(5.0)(0.50) = 1.50 \text{ N}.$$

Exercise 10.7 · ⋆⋆⋆ **Eastward deflection of a falling body**

A stone is dropped from rest from the top of a tower of height $h = 100$ m at latitude $\lambda = 45°$ in the Northern hemisphere. (a) Using the coordinate system ($\hat{\imath}$ East, $\hat{\jmath}$ North, $\hat{k}$ Zenith), write the equations of motion for the stone in the Earth-bound frame including gravity and the Coriolis force (neglect the centrifugal term $\Omega^2 R_E$ and air resistance). (b) By successive approximations (first-order in Ω), show that the stone is deflected toward the East by an amount:

$$d_E = \frac{1}{3}\Omega\cos\lambda\sqrt{\frac{8h^3}{g}}.$$

(c) Calculate the numerical value of d_E.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 10.7. (a) In the local frame, $\vec{\Omega} = \Omega\cos\lambda\hat{\jmath} + \Omega\sin\lambda\hat{k}$. The velocity is $\vec{v}_r = \dot{x}\hat{\imath} + \dot{y}\hat{\jmath} + \dot{z}\hat{k}$. The Coriolis acceleration is:

$$\vec{a}_{\text{Cor}} = -2\vec{\Omega} \times \vec{v}_r = -2\begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 0 & \Omega\cos\lambda & \Omega\sin\lambda \\ \dot{x} & \dot{y} & \dot{z} \end{vmatrix}$$
$$= -2\left[(\dot{z}\Omega\cos\lambda - \dot{y}\Omega\sin\lambda)\hat{\imath} + (\dot{x}\Omega\sin\lambda)\hat{\jmath} - (\dot{x}\Omega\cos\lambda)\hat{k}\right].$$

Newton's second law $m\vec{a}_r = m\vec{g} + \vec{F}_{\text{Cor}}$ yields the three scalar differential equations:

$$\ddot{x} = 2\Omega(\dot{y}\sin\lambda - \dot{z}\cos\lambda),$$
$$\ddot{y} = -2\Omega\dot{x}\sin\lambda,$$
$$\ddot{z} = -g + 2\Omega\dot{x}\cos\lambda.$$

(b) At zeroth order in Ω (unperturbed vertical fall with $x(0) = y(0) = 0, z(0) = h$ and

$\vec{v}(0) = \vec{0}$):

$$\dot{x}^{(0)} = 0, \quad \dot{y}^{(0)} = 0, \quad \ddot{z}^{(0)} = -g \implies \dot{z}^{(0)}(t) = -gt, \quad z^{(0)}(t) = h - \frac{1}{2}gt^2.$$

The fall time to ground level ($z = 0$) is $t_f = \sqrt{2h/g}$.
Now substitute the zeroth-order solution into the $\ddot{x}$ equation to find the first-order correction:

$$\ddot{x}^{(1)} = 2\Omega(0 - (-gt)\cos\lambda) = 2\Omega g t \cos\lambda.$$

Integrating twice with initial conditions $x(0) = 0, \dot{x}(0) = 0$:

$$\dot{x}(t) = \Omega g t^2 \cos\lambda,$$
$$x(t) = \frac{1}{3}\Omega g t^3 \cos\lambda.$$

Since x is along the East axis, the deflection is strictly **Eastward**! Evaluating at $t = t_f = \sqrt{2h/g}$:

$$d_E = x(t_f) = \frac{1}{3}\Omega g \left(\frac{2h}{g}\right)^{3/2} \cos\lambda = \boxed{\frac{1}{3}\Omega\cos\lambda\sqrt{\frac{8h^3}{g}}.}$$

(c) Numerical calculation with $h = 100$ m, $g = 9.81\ \mathrm{m\,s^{-2}}$, $\Omega = 7.292 \times 10^{-5}\ \mathrm{rad\,s^{-1}}$, $\lambda = 45°$:

$$t_f = \sqrt{\frac{2 \times 100}{9.81}} = \sqrt{20.387} \approx 4.515 \text{ s}.$$

$$d_E = \frac{1}{3}(7.292 \times 10^{-5})(9.81)(4.515)^3 \cos 45° \approx 0.0155 \text{ m} = \mathbf{1.55} \text{ cm}.$$

The stone lands 1.55 cm East of the point directly below its release point.

Exercise 10.8 · ★★★ **Bead on a rotating vertical hoop**

A smooth circular wire hoop of radius R is placed in a vertical plane and rotates about its vertical diameter with constant angular speed ω. A bead of mass m slides without friction on the hoop. The position of the bead is specified by the angle θ from the lowest point. (a) In the rotating frame of the hoop, identify all real and fictitious forces acting on the bead. (b) Project the equation of motion along the tangent to the hoop to show:

$$R\ddot{\theta} = \left(\omega^2 R\cos\theta - g\right)\sin\theta.$$

(c) Find all equilibrium angles θ_e and determine their stability as a function of the rotation speed ω, identifying the critical threshold $\omega_c = \sqrt{g/R}$.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 10.8. (a) In the frame $\mathcal{R}'$ rotating with $\vec{\Omega} = \omega\hat{k}$:

- Real forces: weight $\vec{P} = -mg\hat{k}$, normal reaction from the wire $\vec{N}$ (no component along the tangent $\vec{u}_\theta$ because the wire is smooth).
- Centrifugal force: $\vec{F}_{\text{cf}} = m\omega^2 r_\perp \vec{u}_r = m\omega^2(R\sin\theta)\vec{u}_r$, directed horizontally outward from the vertical axis.
- Coriolis force: perpendicular to $\vec{u}_\theta$ and absorbed by the normal reaction $\vec{N}$.

(b) Choose the tangential unit vector $\vec{u}_\theta$ along the hoop pointing in the direction of increasing θ:

- Tangential component of weight: $P_\theta = -mg\sin\theta$.
- Tangential component of centrifugal force: $F_{\text{cf},\theta} = +m\omega^2(R\sin\theta)\cos\theta = m\omega^2 R\sin\theta\cos\theta$.

Applying $ma_\theta = mR\ddot{\theta} = P_\theta + F_{\text{cf},\theta}$:

$$mR\ddot{\theta} = -mg\sin\theta + m\omega^2 R\sin\theta\cos\theta \implies \boxed{R\ddot{\theta} = \left(\omega^2 R\cos\theta - g\right)\sin\theta.}$$

(c) Equilibria occur where $\ddot{\theta} = 0$:

$$\sin\theta\left(\omega^2 R\cos\theta - g\right) = 0.$$

This yields two possibilities:

1. $\sin\theta = 0 \implies \theta_1 = 0$ (lowest point) and $\theta_2 = \pi$ (highest point).
2. $\omega^2 R\cos\theta - g = 0 \implies \cos\theta_3 = \frac{g}{\omega^2 R}$. Since $|\cos\theta| \le 1$, this non-trivial equilibrium exists **if and only if** $\omega \ge \omega_c = \sqrt{g/R}$.

Stability analysis via effective potential $V_{\text{eff}}(\theta)$: Since $R\ddot{\theta} = -\frac{1}{mR}\frac{\mathrm{d}V_{\text{eff}}}{\mathrm{d}\theta}$:

$$V_{\text{eff}}(\theta) = -mgR\cos\theta - \frac{1}{2}m\omega^2 R^2\sin^2\theta.$$

Second derivative:

$$\frac{\mathrm{d}^2 V_{\text{eff}}}{\mathrm{d}\theta^2} = mgR\cos\theta - m\omega^2 R^2\cos(2\theta).$$

- At $\theta = 0$: $\left.\frac{\mathrm{d}^2 V_{\text{eff}}}{\mathrm{d}\theta^2}\right|_0 = mgR - m\omega^2 R^2 = mR(g - \omega^2 R)$.

  - If $\omega < \omega_c$: positive, so $\theta = 0$ is **stable**.
  - If $\omega > \omega_c$: negative, so $\theta = 0$ becomes **unstable**!
- For $\omega > \omega_c$, the new equilibrium $\cos\theta_3 = \frac{g}{\omega^2 R}$ appears:

$$\left.\frac{\mathrm{d}^2 V_{\text{eff}}}{\mathrm{d}\theta^2}\right|_{\theta_3} = m\omega^2 R^2 \sin^2\theta_3 > 0.$$

  Thus, the off-axis positions $\pm\theta_3$ are **stable**! This is a classic supercritical pitchfork bifurcation.

Exercise 10.9 · ★★★ **The Foucault Pendulum**

A simple pendulum of length L and bob mass m is set up at latitude λ. For small oscillations ($z \approx 0, \dot{z} \approx 0$), the tension in the string is $T \approx mg$. (a) In the horizontal plane (x, y) (x East, y North), show that the equations of motion in the rotating Earth frame are:

$$\ddot{x} - 2\omega_z \dot{y} + \omega_0^2 x = 0,$$
$$\ddot{y} + 2\omega_z \dot{x} + \omega_0^2 y = 0,$$

where $\omega_0 = \sqrt{g/L}$ and $\omega_z = \Omega \sin\lambda$. (b) Introduce the complex coordinate $\eta(t) = x(t) + \mathrm{i}y(t)$ to combine these equations into:

$$\ddot{\eta} + 2\mathrm{i}\omega_z \dot{\eta} + \omega_0^2 \eta = 0.$$

(c) Given $\omega_z \ll \omega_0$, solve for $\eta(t)$ with initial conditions $\eta(0) = x_0, \dot{\eta}(0) = 0$, and show that the plane of oscillation rotates with angular velocity $-\omega_z = -\Omega \sin\lambda$. What is the rotation period at the North Pole and at Paris ($\lambda = 48.8°$)?

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 10.9. (a) For small oscillations, the restoring force from the string tension has horizontal components:

$$F_{x,\text{rest}} \approx -T\frac{x}{L} \approx -mg\frac{x}{L} = -m\omega_0^2 x, \qquad F_{y,\text{rest}} \approx -m\omega_0^2 y.$$

The Coriolis force with $\vec{\Omega} = \Omega\cos\lambda\hat{\jmath} + \Omega\sin\lambda\hat{k}$ and $\vec{v}_r \approx \dot{x}\hat{\imath} + \dot{y}\hat{\jmath}$ gives:

$$\vec{F}_{\text{Cor}} = -2m\begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 0 & \Omega\cos\lambda & \Omega\sin\lambda \\ \dot{x} & \dot{y} & 0 \end{vmatrix} = 2m(\Omega\sin\lambda\dot{y})\hat{\imath} - 2m(\Omega\sin\lambda\dot{x})\hat{\jmath} + 2m(\Omega\cos\lambda\dot{x})\hat{k}.$$

Defining $\omega_z = \Omega\sin\lambda$, the horizontal equations of motion $m\ddot{x} = F_{x,\text{rest}} + F_{\text{Cor},x}$ and $m\ddot{y} = F_{y,\text{rest}} + F_{\text{Cor},y}$ become:

$$m\ddot{x} = -m\omega_0^2 x + 2m\omega_z\dot{y} \implies \boxed{\ddot{x} - 2\omega_z\dot{y} + \omega_0^2 x = 0,}$$
$$m\ddot{y} = -m\omega_0^2 y - 2m\omega_z\dot{x} \implies \boxed{\ddot{y} + 2\omega_z\dot{x} + \omega_0^2 y = 0.}$$

(b) Multiply the second equation by i and add it to the first:

$$(\ddot{x} + \mathrm{i}\ddot{y}) + 2\omega_z(\mathrm{i}\dot{x} - \dot{y}) + \omega_0^2(x + \mathrm{i}y) = 0.$$

Notice that $\mathrm{i}\dot{x} - \dot{y} = \mathrm{i}(\dot{x} + \mathrm{i}\dot{y}) = \mathrm{i}\dot{\eta}$. Thus:

$$\boxed{\ddot{\eta} + 2\mathrm{i}\omega_z\dot{\eta} + \omega_0^2\eta = 0.}$$

(c) Try a solution of the form $\eta(t) = e^{-\mathrm{i}\omega_z t}u(t)$. Substituting:

$$\dot{\eta} = e^{-\mathrm{i}\omega_z t}(\dot{u} - \mathrm{i}\omega_z u), \qquad \ddot{\eta} = e^{-\mathrm{i}\omega_z t}(\ddot{u} - 2\mathrm{i}\omega_z\dot{u} - \omega_z^2 u).$$

The middle cross-term cancels:

$$e^{-\mathrm{i}\omega_z t}\left[(\ddot{u} - 2\mathrm{i}\omega_z\dot{u} - \omega_z^2 u) + 2\mathrm{i}\omega_z(\dot{u} - \mathrm{i}\omega_z u) + \omega_0^2 u\right] = 0 \implies \ddot{u} + (\omega_0^2 + \omega_z^2)u = 0.$$

Since $\omega_z \approx 10^{-5}\ \text{rad}\,\text{s}^{-1} \ll \omega_0 \approx 1\ \text{rad}\,\text{s}^{-1}$, $\omega_0^2 + \omega_z^2 \approx \omega_0^2$. The solution for $u(t)$ with $u(0) = x_0, \dot{u}(0) = 0$ is:

$$u(t) = x_0\cos(\omega_0 t).$$

Therefore:

$$\boxed{\eta(t) = x_0\cos(\omega_0 t)\, e^{-\mathrm{i}\omega_z t}.}$$

The term $x_0\cos(\omega_0 t)$ represents simple harmonic oscillation along a line, while the factor $e^{-\mathrm{i}\omega_z t} = \cos(\omega_z t) - \mathrm{i}\sin(\omega_z t)$ rotates this line of oscillation clockwise with angular velocity:

$$\Omega_F = -\omega_z = -\Omega\sin\lambda.$$

The period of a complete 360° rotation is:

$$T_F = \frac{2\pi}{|\Omega_F|} = \frac{2\pi}{\Omega\sin\lambda} = \frac{24\ \text{h}}{\sin\lambda}.$$

- At the North Pole ($\lambda = 90°$): $T_F = 24.0$ h.
- At Paris ($\lambda = 48.8°$): $T_F = \frac{24}{\sin 48.8°} \approx \frac{24}{0.7524} \approx 31.9$ h ≈ 31 h 54 min.
- At the Equator ($\lambda = 0°$): $T_F \to \infty$ (no precession).

Exercise 10.10 · ⋆⋆⋆ **Centrifugal force potential and geoid flattening**

Because of the centrifugal force, the Earth is not a perfect sphere but an oblate spheroid flattened at the poles. (a) Show that the centrifugal force $\vec{F}_{\text{cf}} = m\Omega^2 r_\perp \vec{u}_r$ derives from a potential energy $E_{p,\text{cf}} = -\frac{1}{2}m\Omega^2 r^2 \cos^2\lambda$. (b) The effective potential energy of a unit mass at Earth's surface is $V_{\text{eff}} = -\frac{GM_E}{r} - \frac{1}{2}\Omega^2 r^2 \cos^2\lambda$. Since mean sea level is an equipotential surface of V_{eff}, equate V_{eff} at the pole ($r = R_p, \lambda = 90°$) and at the equator ($r = R_e, \lambda = 0°$) to estimate the equatorial bulge $\Delta R = R_e - R_p$ in terms of g, R_E, Ω. (c) Compute ΔR numerically.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 10.10. (a) The centrifugal force in cylindrical coordinates is $\vec{F}_{\text{cf}} = m\Omega^2 r_\perp \vec{u}_\perp$. We check that $\vec{\nabla} \times \vec{F}_{\text{cf}} = \vec{0}$, so it derives from a potential:

$$\vec{F}_{\text{cf}} = -\vec{\nabla} E_{p,\text{cf}} \implies \mathrm{d}E_{p,\text{cf}} = -\vec{F}_{\text{cf}} \cdot \mathrm{d}\vec{r} = -m\Omega^2 r_\perp \mathrm{d}r_\perp.$$

Integrating with zero at the axis:

$$E_{p,\text{cf}} = -\frac{1}{2}m\Omega^2 r_\perp^2 = -\frac{1}{2}m\Omega^2 r^2 \cos^2\lambda.$$

(b) The effective potential per unit mass is:

$$V_{\text{eff}}(r,\lambda) = -\frac{GM_E}{r} - \frac{1}{2}\Omega^2 r^2 \cos^2\lambda.$$

Equating potentials:

$$\text{At pole } (\lambda = 90°, r = R_p): \quad V_{\text{eff}} = -\frac{GM_E}{R_p},$$
$$\text{At equator } (\lambda = 0°, r = R_e): \quad V_{\text{eff}} = -\frac{GM_E}{R_e} - \frac{1}{2}\Omega^2 R_e^2.$$

Setting them equal:

$$-\frac{GM_E}{R_p} = -\frac{GM_E}{R_e} - \frac{1}{2}\Omega^2 R_e^2 \implies GM_E\left(\frac{1}{R_p} - \frac{1}{R_e}\right) = \frac{1}{2}\Omega^2 R_e^2.$$

Since $R_e - R_p = \Delta R \ll R_E$:

$$\frac{1}{R_p} - \frac{1}{R_e} = \frac{R_e - R_p}{R_p R_e} \approx \frac{\Delta R}{R_E^2}.$$

Using $g \approx \frac{GM_E}{R_E^2}$:

$$g\,\Delta R \approx \frac{1}{2}\Omega^2 R_E^2 \implies \boxed{\Delta R \approx \frac{\Omega^2 R_E^2}{2g}.}$$

(c) Numerical calculation:

$$\Delta R = \frac{(7.292\times10^{-5})^2 \times (6.37\times10^{6})^2}{2\times 9.81} \approx \frac{215770}{19.62} \approx 10997 \text{ m} \approx \mathbf{11.0} \text{ km}.$$

CHAPTER 11

Angular Momentum and Central Forces

You will learn to: define and compute angular momentum and torque; apply the angular momentum theorem; understand why central forces conserve angular momentum; use polar coordinates and the law of areas (Kepler's second law); analyse orbits with the effective potential $U_{\text{eff}}(r)$; and master Kepler's laws, orbital transfers, and Binet's formulas.

11.1 Angular Momentum and Torque

For a material point of mass m, position $\vec{r} = \vec{OM}$, and velocity $\vec{v}$ relative to a fixed origin O in an inertial frame:

- The **angular momentum** about O is:

$$\boxed{\vec{L}_O = \vec{r} \times \vec{p} = m\,\vec{r} \times \vec{v}.}$$

- The **moment of a force (torque)** $\vec{F}$ about O is:

$$\boxed{\vec{\tau}_O = \vec{r} \times \vec{F}.}$$

Differentiating $\vec{L}_O$ with respect to time:

$$\frac{\mathrm{d}\vec{L}_O}{\mathrm{d}t} = \frac{\mathrm{d}\vec{r}}{\mathrm{d}t} \times (m\vec{v}) + \vec{r} \times \left(m\frac{\mathrm{d}\vec{v}}{\mathrm{d}t}\right) = \vec{v} \times (m\vec{v}) + \vec{r} \times \vec{F}.$$

Since $\vec{v} \times \vec{v} = \vec{0}$, we obtain the **Angular Momentum Theorem**:

$$\boxed{\frac{\mathrm{d}\vec{L}_O}{\mathrm{d}t} = \vec{\tau}_O = \vec{r} \times \vec{F}.}$$

11.2 Central Forces and Conservation of Angular Momentum

A force is called a **central force** with center O if it is always directed along the line joining O to the particle:

$$\vec{F}(\vec{r}) = F(r)\,\vec{u}_r, \quad \text{where } \vec{u}_r = \frac{\vec{r}}{r}.$$

Key Properties of Motion under a Central Force

1. **Torque is identically zero**: $\vec{\tau}_O = \vec{r} \times (F(r)\vec{u}_r) = \vec{0}$.
2. **Angular momentum is strictly conserved**:

$$\frac{\mathrm{d}\vec{L}_O}{\mathrm{d}t} = \vec{0} \implies \vec{L}_O = \vec{L}_0 = \text{constant vector}.$$

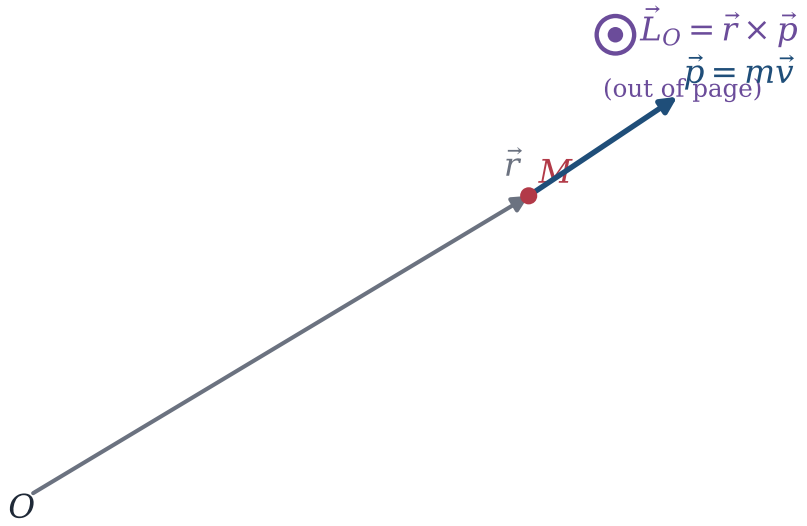


Figure 11.1: The position vector $\vec{r}$, linear momentum $\vec{p} = m\vec{v}$, and angular momentum vector $\vec{L}_O = \vec{r} \times \vec{p}$ perpendicular to the plane of motion.

3. **Planar trajectory**: Since $\vec{r} \cdot \vec{L}_0 = \vec{r} \cdot (\vec{r} \times \vec{p}) = 0$, the particle remains forever in the fixed plane passing through O orthogonal to $\vec{L}_0$.
4. **Constant areal velocity (Kepler's Second Law)**: In polar coordinates (r, θ) in the orbital plane:

$$\vec{L}_O = mr\vec{u}_r \times (\dot{r}\vec{u}_r + r\dot{\theta}\vec{u}_\theta) = mr^2\dot{\theta}\,\hat{k} \implies \boxed{r^2\dot{\theta} = C = \frac{L_z}{m} = \text{constant},}$$

where C is the *areal constant*. The rate at which area is swept out by the radius vector is:

$$\frac{\mathrm{d}A}{\mathrm{d}t} = \frac{1}{2}r^2\dot{\theta} = \frac{C}{2} = \text{constant}.$$

11.3 The Effective Potential

Since every central force $\vec{F} = F(r)\vec{u}_r$ is conservative ($\vec{\nabla} \times \vec{F} = \vec{0}$), it possesses a potential energy $U(r)$ such that $F(r) = -\frac{\mathrm{d}U}{\mathrm{d}r}$. The total mechanical energy in polar coordinates is:

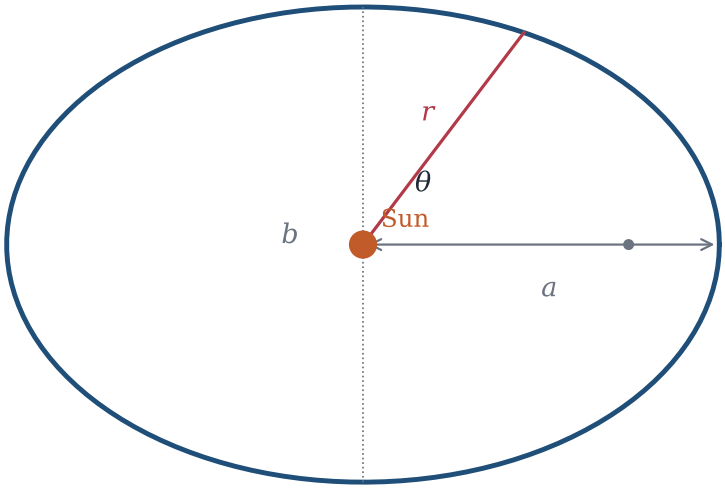
$$E_m = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2\right) + U(r).$$

Substituting $r\dot{\theta} = \frac{L}{mr}$ reduces the 2D problem to an equivalent **one-dimensional radial motion**:

$$\boxed{E_m = \frac{1}{2}m\dot{r}^2 + U_{\text{eff}}(r)}$$

where the **effective potential** is:

$$\boxed{U_{\text{eff}}(r) = U(r) + \frac{L^2}{2mr^2}.}$$



$$\frac{T^2}{a^3} = \frac{4\pi^2}{GM}$$

Figure 11.2: An elliptical Keplerian orbit with the central body at one focus. Equal areas are swept out in equal times ($\Delta t_1 = \Delta t_2 \implies \Delta A_1 = \Delta A_2$).

The extra term $\frac{L^2}{2mr^2}$ is the **centrifugal barrier**, which prevents a particle with non-zero angular momentum from falling into the origin.

11.4 Worked Examples

Worked example 11.1 · Angular momentum of a projectile

A projectile of mass m is launched from the origin with initial velocity $\vec{v}_0 = v_0\cos\alpha\,\hat{\imath} + v_0\sin\alpha\,\hat{\jmath}$ under gravity $\vec{g} = -g\hat{\jmath}$. (a) Find the angular momentum $\vec{L}_O(t)$ about the launch point O. (b) Verify that $\frac{\mathrm{d}\vec{L}_O}{\mathrm{d}t} = \vec{\tau}_O(t)$.

Solution 11.1. (a) The position and velocity as functions of time are:

$$\vec{r}(t) = (v_0\cos\alpha\,t)\hat{\imath} + \left(v_0\sin\alpha\,t - \frac{1}{2}gt^2\right)\hat{\jmath},$$

$$\vec{v}(t) = (v_0\cos\alpha)\hat{\imath} + (v_0\sin\alpha - gt)\hat{\jmath}.$$

The angular momentum about O is:

$$\begin{aligned}
\vec{L}_O(t) &= m\,\vec{r}\times\vec{v} = m\,[xv_y - yv_x]\,\hat{k} \\
&= m\left[(v_0\cos\alpha\,t)(v_0\sin\alpha - gt) - \left(v_0\sin\alpha\,t - \frac{1}{2}gt^2\right)(v_0\cos\alpha)\right]\hat{k} \\
&= m\left[v_0^2\sin\alpha\cos\alpha\,t - gv_0\cos\alpha\,t^2 - v_0^2\sin\alpha\cos\alpha\,t + \frac{1}{2}gv_0\cos\alpha\,t^2\right]\hat{k} \\
&= \boxed{-\frac{1}{2}mgv_0\cos\alpha\,t^2\,\hat{k}.}
\end{aligned}$$

(b) The torque about O due to gravity $\vec{P} = -mg\hat{\jmath}$ is:

$$\vec{\tau}_O = \vec{r} \times (-mg\hat{\jmath}) = (x\hat{\imath} + y\hat{\jmath}) \times (-mg\hat{\jmath}) = -mgx\,\hat{k} = -mg(v_0 \cos\alpha\, t)\,\hat{k}.$$

Differentiating $\vec{L}_O(t)$:

$$\frac{\mathrm{d}\vec{L}_O}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\left(-\frac{1}{2}mgv_0 \cos\alpha\, t^2\,\hat{k}\right) = -mgv_0 \cos\alpha\, t\,\hat{k} = \vec{\tau}_O.$$

The angular momentum theorem is verified!

Worked example 11.2 · Perihelion and aphelion velocities in an elliptical orbit

A satellite of mass m orbits Earth (mass M_E) in an elliptical orbit. Its closest approach (perigee) is at distance r_p and its farthest distance (apogee) is r_a. Express the speeds v_p and v_a in terms of G, M_E, r_p, r_a.

Solution 11.2. At both perigee and apogee, the velocity vector is perpendicular to the position vector ($\dot{r} = 0$).

- By conservation of angular momentum:

$$L = mr_p v_p = mr_a v_a \implies v_a = \frac{r_p}{r_a} v_p.$$

- By conservation of mechanical energy:

$$E_m = \frac{1}{2}mv_p^2 - \frac{GM_E m}{r_p} = \frac{1}{2}mv_a^2 - \frac{GM_E m}{r_a}.$$

Substituting $v_a = \frac{r_p}{r_a} v_p$:

$$\frac{1}{2}v_p^2\left(1 - \frac{r_p^2}{r_a^2}\right) = GM_E\left(\frac{1}{r_p} - \frac{1}{r_a}\right) = GM_E\frac{r_a - r_p}{r_p r_a}.$$

Notice that $1 - \frac{r_p^2}{r_a^2} = \frac{r_a^2 - r_p^2}{r_a^2} = \frac{(r_a - r_p)(r_a + r_p)}{r_a^2}$. Cancelling the factor $(r_a - r_p)$:

$$\frac{1}{2}v_p^2\frac{r_a + r_p}{r_a^2} = \frac{GM_E}{r_p r_a} \implies \boxed{v_p = \sqrt{\frac{2GM_E\, r_a}{r_p(r_p + r_a)}}}.$$

By symmetry:

$$\boxed{v_a = \sqrt{\frac{2GM_E\, r_p}{r_a(r_p + r_a)}}}.$$

Since the semi-major axis is $a = \frac{r_p + r_a}{2}$, this can also be written in the famous *vis-viva* form: $v^2 = GM_E\left(\frac{2}{r} - \frac{1}{a}\right)$.

11.5 Exercises and Step-by-Step Solutions

Exercise 11.1 · ⋆ **Angular momentum in uniform rectilinear motion**

A particle of mass $m = 0.50$ kg travels at constant velocity $\vec{v} = 12.0\,\hat{\imath}$ m s^{-1} along the line $y = 3.00$ m. (a) Compute its angular momentum $\vec{L}_O$ about the origin $O(0,0)$ as a function of its x-coordinate. (b) Does $\vec{L}_O$ depend on time or position? Explain why using the concept of torque.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 11.1. (a) The position vector is $\vec{r}(t) = x(t)\hat{\imath} + y_0\hat{\jmath}$ with $y_0 = 3.00$ m. The velocity is $\vec{v} = v_0\hat{\imath}$ with $v_0 = 12.0$ m s^{-1}. The angular momentum about O is:

$$\vec{L}_O = \vec{r} \times (m\vec{v}) = (x\hat{\imath} + y_0\hat{\jmath}) \times (mv_0\hat{\imath}) = -mv_0y_0\,\hat{k}.$$

Substituting the numerical values:

$$\vec{L}_O = -(0.50)(12.0)(3.00)\,\hat{k} = \mathbf{-18.0\,\hat{k}}\ \text{kg m}^2\,\text{s}^{-1}.$$

(b) $\vec{L}_O$ is strictly constant: it is completely independent of x and t. This is because the particle is in uniform rectilinear motion ($\vec{a} = \vec{0}$), so the net force is zero ($\vec{F} = \vec{0}$), meaning the net torque about the origin is $\vec{\tau}_O = \vec{r} \times \vec{F} = \vec{0}$. By the angular momentum theorem, $\frac{d\vec{L}_O}{dt} = \vec{0}$, so $\vec{L}_O = \text{const}$. Geometrically, $\left\|\vec{L}_O\right\| = mv_0d$, where $d = y_0$ is the perpendicular distance (impact parameter) from O to the line of motion.

Exercise 11.2 · ⋆ **Circular orbit speed and period of the ISS**

The International Space Station (ISS) orbits Earth in an approximately circular orbit at an altitude of $h = 415$ km above Earth's surface. (Earth's radius $R_E = 6371$ km, Earth's mass $M_E = 5.972 \times 10^{24}$ kg, $G = 6.674 \times 10^{-11}$ m^3 kg^{-1} s^{-2}). (a) Using Newton's second law, find the orbital speed v_c. (b) Compute the orbital period T in minutes. (c) How many revolutions does the ISS complete in 24 hours?

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 11.2. (a) The orbital radius is $r = R_E + h = 6371 + 415 = 6786$ km $= 6.786 \times 10^6$ m. For circular motion under gravity:

$$m\frac{v_c^2}{r} = \frac{GM_E m}{r^2} \implies \boxed{v_c = \sqrt{\frac{GM_E}{r}}.}$$

Computing:

$$v_c = \sqrt{\frac{(6.674 \times 10^{-11})(5.972 \times 10^{24})}{6.786 \times 10^6}} = \sqrt{\frac{3.986 \times 10^{14}}{6.786 \times 10^6}} = \sqrt{5.874 \times 10^7} \approx 7664 \text{ m s}^{-1} \approx \mathbf{7.66} \text{ km s}^{-1}.$$

(b) The orbital period is:

$$T = \frac{2\pi r}{v_c} = \frac{2\pi(6.786 \times 10^6)}{7664} \approx 5563 \text{ s} \approx \mathbf{92.7} \text{ min} \quad (1 \text{ h } 32.7 \text{ min}).$$

(c) Number of orbits per day (24 hours = 1440 min):

$$N = \frac{1440}{92.72} \approx \mathbf{15.5} \text{ orbits per day.}$$

Exercise 11.3 · ⋆ Escape velocity from celestial bodies

The escape velocity v_{esc} is the minimum speed needed for a projectile launched from the surface of a body to escape its gravitational pull completely ($r \to \infty$). (a) Using conservation of mechanical energy, show that $v_{\text{esc}} = \sqrt{\frac{2GM}{R}} = \sqrt{2gR}$. (b) Calculate v_{esc} for Earth ($R_E = 6371$ km, $g = 9.81$ m s^{-2}). (c) Calculate v_{esc} for the Moon ($M_M = 7.342 \times 10^{22}$ kg, $R_M = 1737$ km).

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 11.3. (a) Mechanical energy at launch ($r = R$, $v = v_{\text{esc}}$):

$$E_m = \frac{1}{2}mv_{\text{esc}}^2 - \frac{GMm}{R}.$$

At infinity ($r \to \infty$), the gravitational potential energy vanishes ($U \to 0$), and the condition for just escaping is that the speed reaches zero ($v \geq 0$), so $E_{m,\infty} \geq 0$. By conservation of energy:

$$\frac{1}{2}mv_{\text{esc}}^2 - \frac{GMm}{R} = 0 \implies \boxed{v_{\text{esc}} = \sqrt{\frac{2GM}{R}}.}$$

Since $g = \frac{GM}{R^2}$, we have $\frac{GM}{R} = gR$, so $v_{\text{esc}} = \sqrt{2gR}$.
(b) For Earth:

$$v_{\text{esc},E} = \sqrt{2(9.81)(6.371 \times 10^6)} = \sqrt{1.250 \times 10^8} \approx 11180 \text{ m s}^{-1} = \mathbf{11.2} \text{ km s}^{-1}.$$

(c) For the Moon:

$$v_{\text{esc},M} = \sqrt{\frac{2(6.674 \times 10^{-11})(7.342 \times 10^{22})}{1.737 \times 10^6}} = \sqrt{\frac{9.800 \times 10^{12}}{1.737 \times 10^6}} = \sqrt{5.642 \times 10^6} \approx 2375 \text{ m s}^{-1} = \mathbf{2.38} \text{ km s}^{-1}.$$

The Moon's escape velocity is nearly 5 times smaller than Earth's, which explains why the Moon could not retain an atmosphere.

Exercise 11.4 · Mass on a table with a hole: pulling the string

A small puck of mass $m = 0.20$ kg on a frictionless horizontal table is connected to a light string passing through a small hole at the center O. The puck initially moves in a circle of radius $r_1 = 0.50$ m with speed $v_1 = 2.0$ m s^{-1}. The string is then slowly pulled from below until the radius of the circular path is reduced to $r_2 = 0.25$ m. (a) Explain why angular momentum about O is conserved during the process. (b) Find the new speed v_2 of the puck. (c) Find the initial and final tension in the string. (d) Calculate the work done by the person pulling the string using the work-energy theorem.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 11.4. (a) The string tension $\vec{T} = -T\vec{u}_r$ is always directed toward the central hole O. The torque about O is:

$$\vec{\tau}_O = \vec{r} \times \vec{T} = r\vec{u}_r \times (-T\vec{u}_r) = \vec{0}.$$

Therefore, angular momentum about O is strictly conserved: $L = mr_1v_1 = mr_2v_2 = \text{const}$.
(b) Conserving angular momentum:

$$v_2 = \frac{r_1}{r_2}v_1 = \frac{0.50}{0.25} \times 2.0 = 2 \times 2.0 = \mathbf{4.0}\ \mathrm{m\,s^{-1}}.$$

(c) The tension provides the centripetal acceleration $T = \frac{mv^2}{r}$:

$$T_1 = \frac{mv_1^2}{r_1} = \frac{0.20 \times (2.0)^2}{0.50} = \frac{0.80}{0.50} = \mathbf{1.60}\ \mathrm{N},$$
$$T_2 = \frac{mv_2^2}{r_2} = \frac{0.20 \times (4.0)^2}{0.25} = \frac{3.20}{0.25} = \mathbf{12.8}\ \mathrm{N}.$$

The tension increases by a factor of $(r_1/r_2)^3 = 2^3 = 8$!
(d) By the work-energy theorem, since the normal force and gravity cancel and do no work, the work done by the tension force is the change in kinetic energy:

$$W = \Delta E_c = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 = \frac{1}{2}(0.20)\left[4.0^2 - 2.0^2\right] = 0.10(16-4) = 0.10 \times 12 = \mathbf{1.20}\ \mathrm{J}.$$

The person pulling the string performs $+1.20$ J of positive work against the centrifugal effect.

Exercise 11.5 · Effective potential and turning points

A particle of mass m moves under an attractive central force with potential $U(r) = -\frac{k}{r}$ ($k > 0$). Its angular momentum is L. (a) Write down the effective potential $U_{\text{eff}}(r)$ and sketch it as a function of r. (b) Find the radius r_0 of the circular orbit where U_{eff} has its minimum, and compute the minimum energy $E_{\min} = U_{\text{eff}}(r_0)$. (c) For total energy E_m such that $E_{\min} < E_m < 0$, find the turning points $r_{\min}$ (periapsis) and $r_{\max}$ (apoapsis) in terms of E_m, L, m, k.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 11.5. (a) The effective potential is:

$$\boxed{U_{\text{eff}}(r) = -\frac{k}{r} + \frac{L^2}{2mr^2}.}$$

As $r \to 0$, the centrifugal term $\frac{L^2}{2mr^2} \to +\infty$ dominates (repulsive barrier). As $r \to \infty$, $U_{\text{eff}}(r) \to 0^-$ (governed by $-\frac{k}{r}$). In between, it exhibits a unique negative potential well

minimum.
(b) The minimum occurs where $\frac{\mathrm{d}U_{\text{eff}}}{\mathrm{d}r} = 0$:

$$\frac{\mathrm{d}U_{\text{eff}}}{\mathrm{d}r} = \frac{k}{r^2} - \frac{L^2}{mr^3} = 0 \implies \frac{k}{r_0^2} = \frac{L^2}{mr_0^3} \implies \boxed{r_0 = \frac{L^2}{mk}.}$$

The second derivative is:

$$\frac{\mathrm{d}^2U_{\text{eff}}}{\mathrm{d}r^2} = -\frac{2k}{r^3} + \frac{3L^2}{mr^4} \implies \left.\frac{\mathrm{d}^2U_{\text{eff}}}{\mathrm{d}r^2}\right|_{r_0} = -\frac{2k}{(L^2/mk)^3} + \frac{3L^2}{m(L^2/mk)^4} = \frac{m^3k^4}{L^6} > 0,$$

confirming a stable circular orbit! The minimum energy is:

$$E_{\min} = U_{\text{eff}}(r_0) = -\frac{k}{L^2/(mk)} + \frac{L^2}{2m(L^2/mk)^2} = -\frac{mk^2}{L^2} + \frac{mk^2}{2L^2} = \boxed{-\frac{mk^2}{2L^2}.}$$

(c) Turning points are where the radial velocity vanishes: $\dot{r} = 0 \iff E_m = U_{\text{eff}}(r)$:

$$E_m = -\frac{k}{r} + \frac{L^2}{2mr^2} \iff E_m r^2 + kr - \frac{L^2}{2m} = 0.$$

Dividing by E_m (with $E_m < 0$, let $|E_m| = -E_m$):

$$r^2 - \frac{k}{|E_m|}r + \frac{L^2}{2m\,|E_m|} = 0.$$

Solving the quadratic equation:

$$\Delta = \frac{k^2}{E_m^2} - \frac{2L^2}{m\,|E_m|} = \frac{k^2}{E_m^2}\left(1 - \frac{2\,|E_m|\,L^2}{mk^2}\right) = \frac{k^2}{E_m^2}\left(1 - \frac{|E_m|}{|E_{\min}|}\right).$$

Since $E_{\min} < E_m < 0$, $\Delta > 0$. The two roots are:

$$\boxed{r_{\min,\max} = \frac{k}{2\,|E_m|}\left[1 \mp \sqrt{1 - \frac{2\,|E_m|\,L^2}{mk^2}}\right].}$$

Notice that the semi-major axis is $a = \frac{r_{\min}+r_{\max}}{2} = \frac{k}{2|E_m|} \implies E_m = -\frac{k}{2a}$, and the eccentricity is $e = \sqrt{1 - \frac{2|E_m|L^2}{mk^2}}$.

Exercise 11.6 · Hohmann Transfer Orbit to Mars

A spacecraft is in a circular parking orbit around the Sun at Earth's distance $r_1 = 1.00$ AU $= 1.496 \times 10^{11}$ m. It needs to transfer to Mars's orbit, assumed circular at $r_2 = 1.524$ AU $= 2.280 \times 10^{11}$ m. The Sun's gravitational parameter is $\mu = GM_\odot = 1.327 \times 10^{20}\ \mathrm{m^3\,s^{-2}}$. (a) Find the semi-major axis a_{trans} of the elliptical Hohmann transfer orbit. (b) Calculate the travel time (in days) to reach Mars. (c) Compute the initial velocity increment Δv_1 required to leave Earth's orbit and enter the transfer ellipse.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 11.6. (a) The Hohmann transfer orbit is an ellipse with perihelion at Earth's orbit ($r_p = r_1$) and aphelion at Mars's orbit ($r_a = r_2$). The major axis is $2a_{\text{trans}} = r_1 + r_2$, so:

$$a_{\text{trans}} = \frac{r_1 + r_2}{2} = \frac{1.000 + 1.524}{2} = \mathbf{1.262}\ \text{AU} = 1.262 \times (1.496 \times 10^{11}) = \mathbf{1.888 \times 10^{11}}\ \text{m}.$$

(b) By Kepler's third law, the period of the complete ellipse is:

$$T_{\text{trans}} = T_{\text{Earth}} \left(\frac{a_{\text{trans}}}{a_1}\right)^{3/2} = 1.00\ \text{year} \times (1.262)^{3/2} = 1.00 \times 1.418 = 1.418\ \text{years}.$$

The flight time to Mars is half of this orbital period (from perihelion to aphelion):

$$t_{\text{flight}} = \frac{1}{2} T_{\text{trans}} = \frac{1.418 \times 365.25}{2} \approx \mathbf{259}\ \text{days} \quad (\approx 8.5\ \text{months}).$$

(c) Earth's circular orbital speed:

$$v_{c1} = \sqrt{\frac{\mu}{r_1}} = \sqrt{\frac{1.327 \times 10^{20}}{1.496 \times 10^{11}}} \approx 29780\ \text{m s}^{-1} = 29.78\ \text{km s}^{-1}.$$

The required perihelion speed on the transfer ellipse (using the *vis-viva* equation $v^2 = \mu\left(\frac{2}{r} - \frac{1}{a}\right)$):

$$v_p = \sqrt{\mu\left(\frac{2}{r_1} - \frac{1}{a_{\text{trans}}}\right)} = \sqrt{(1.327 \times 10^{20})\left(\frac{2}{1.496 \times 10^{11}} - \frac{1}{1.888 \times 10^{11}}\right)} \approx 32730\ \text{m s}^{-1} = 32.73\ \text{km s}^{-1}.$$

The required impulse at departure is:

$$\Delta v_1 = v_p - v_{c1} = 32.73 - 29.78 = \mathbf{2.95}\ \text{km s}^{-1}.$$

Exercise 11.7 · ⋆⋆⋆ Binet's Formulas and Kepler's First Law

In polar coordinates (r, θ), define $u(\theta) = \frac{1}{r(\theta)}$. (a) Using $r^2\dot{\theta} = C$, show that:

$$\dot{r} = -C\frac{\mathrm{d}u}{\mathrm{d}\theta}, \qquad \ddot{r} - r\dot{\theta}^2 = -C^2u^2\left(\frac{\mathrm{d}^2u}{\mathrm{d}\theta^2} + u\right).$$

(b) Deduce **Binet's formula for acceleration**:

$$a_r = -C^2u^2\left(\frac{\mathrm{d}^2u}{\mathrm{d}\theta^2} + u\right).$$

(c) An observed orbit is an ellipse with the force center at one focus: $r(\theta) = \frac{p}{1+e\cos\theta}$. Prove that the central force must be an inverse-square law $F(r) \propto -\frac{1}{r^2}$.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 11.7. (a) Using the chain rule $\frac{\mathrm{d}}{\mathrm{d}t} = \dot\theta\frac{\mathrm{d}}{\mathrm{d}\theta} = \frac{C}{r^2}\frac{\mathrm{d}}{\mathrm{d}\theta} = Cu^2\frac{\mathrm{d}}{\mathrm{d}\theta}$:

$$\dot r = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{u}\right) = -\frac{1}{u^2}\dot u = -\frac{1}{u^2}\left(Cu^2\frac{\mathrm{d}u}{\mathrm{d}\theta}\right) = \boxed{-C\frac{\mathrm{d}u}{\mathrm{d}\theta}.}$$

Differentiating again with respect to t:

$$\ddot r = \frac{\mathrm{d}}{\mathrm{d}t}\left(-C\frac{\mathrm{d}u}{\mathrm{d}\theta}\right) = Cu^2\frac{\mathrm{d}}{\mathrm{d}\theta}\left(-C\frac{\mathrm{d}u}{\mathrm{d}\theta}\right) = -C^2u^2\frac{\mathrm{d}^2u}{\mathrm{d}\theta^2}.$$

The second centripetal term is:

$$r\dot\theta^2 = \frac{1}{u}\left(Cu^2\right)^2 = C^2u^3.$$

Combining the terms:

$$\ddot r - r\dot\theta^2 = -C^2u^2\frac{\mathrm{d}^2u}{\mathrm{d}\theta^2} - C^2u^3 = \boxed{-C^2u^2\left(\frac{\mathrm{d}^2u}{\mathrm{d}\theta^2} + u\right).}$$

(b) In polar coordinates, the radial acceleration is precisely $a_r = \ddot r - r\dot\theta^2$. Therefore:

$$\boxed{a_r = -C^2u^2\left(\frac{\mathrm{d}^2u}{\mathrm{d}\theta^2} + u\right).}$$

Newton's second law $ma_r = F(r)$ gives Binet's equation:

$$F(1/u) = -mC^2u^2\left(\frac{\mathrm{d}^2u}{\mathrm{d}\theta^2} + u\right).$$

(c) Given the focal ellipse equation $r(\theta) = \frac{p}{1+e\cos\theta}$:

$$u(\theta) = \frac{1}{r(\theta)} = \frac{1}{p} + \frac{e}{p}\cos\theta.$$

Differentiating twice with respect to θ:

$$\frac{\mathrm{d}u}{\mathrm{d}\theta} = -\frac{e}{p}\sin\theta, \qquad \frac{\mathrm{d}^2u}{\mathrm{d}\theta^2} = -\frac{e}{p}\cos\theta.$$

Adding u:

$$\frac{\mathrm{d}^2u}{\mathrm{d}\theta^2} + u = -\frac{e}{p}\cos\theta + \left(\frac{1}{p} + \frac{e}{p}\cos\theta\right) = \frac{1}{p} = \text{constant!}$$

Substituting into Binet's force formula:

$$F(r) = -mC^2u^2\left(\frac{1}{p}\right) = -\frac{mC^2}{p}u^2 = \boxed{-\frac{mC^2}{p}\frac{1}{r^2}.}$$

This completes Newton's celebrated deduction: an elliptical orbit with the focus at the center of force uniquely requires an **inverse-square gravitational force** $F(r) \propto -\frac{1}{r^2}$!

Exercise 11.8 · ⋆⋆⋆ Rutherford Scattering: Repulsive Central Force

An alpha particle of mass m and charge $q_1 = +2e$ is fired with initial speed v_0 and impact parameter b toward a stationary heavy gold nucleus of charge $q_2 = +Ze$. The Coulomb repulsive force is $\vec{F} = \frac{k}{r^2}\vec{u}_r$, where $k = \frac{q_1q_2}{4\pi\varepsilon_0}$. (a) Write the conservation of angular momentum L and total mechanical energy E_m in terms of m, v_0, b. (b) Using Binet's equation $\frac{\mathrm{d}^2u}{\mathrm{d}\theta^2}+u = -\frac{mk}{L^2}$, show that the trajectory is a hyperbola:

$$u(\theta) = \frac{1}{r(\theta)} = \frac{1}{p}\left(e\cos\theta - 1\right),$$

where $p = \frac{L^2}{mk}$. (c) Deduce Rutherford's scattering formula relating the deflection angle θ_s to the impact parameter b:

$$\tan\left(\frac{\theta_s}{2}\right) = \frac{k}{mv_0^2b}.$$

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 11.8. (a) At $r \to \infty$, the particle moves with speed v_0 along a line at distance b from the nucleus:

$$L = mv_0 b = \text{const}, \qquad E_m = \frac{1}{2}mv_0^2 = \text{const}.$$

(b) Binet's equation for a repulsive force $F(r) = +\frac{k}{r^2} = +ku^2$:

$$-mC^2u^2\left(\frac{\mathrm{d}^2u}{\mathrm{d}\theta^2} + u\right) = ku^2 \implies \frac{\mathrm{d}^2u}{\mathrm{d}\theta^2} + u = -\frac{k}{mC^2} = -\frac{mk}{L^2}.$$

Defining $p = \frac{L^2}{mk}$, this is a harmonic oscillator with constant drive:

$$\frac{\mathrm{d}^2u}{\mathrm{d}\theta^2} + u = -\frac{1}{p}.$$

The general solution is:

$$u(\theta) = A\cos(\theta - \theta_0) - \frac{1}{p}.$$

Aligning the symmetry axis (periapsis at $\theta = 0$):

$$u(\theta) = \frac{1}{p}\left(e\cos\theta - 1\right).$$

Since $r > 0$, this represents the repulsive branch of a hyperbola ($e > 1$).
(c) The asymptotes of the hyperbola correspond to $r \to \infty \implies u(\theta_\infty) = 0$:

$$e\cos\theta_\infty - 1 = 0 \implies \cos\theta_\infty = \frac{1}{e}.$$

From the geometry of hyperbolic scattering, the incoming direction is $-\theta_\infty$ and the outgoing direction is $+\theta_\infty$. The total deflection (scattering angle) is $\theta_s = \pi - 2\theta_\infty \implies \theta_\infty = \frac{\pi - \theta_s}{2}$. Thus:

$$\cos\theta_\infty = \cos\left(\frac{\pi}{2} - \frac{\theta_s}{2}\right) = \sin\left(\frac{\theta_s}{2}\right) = \frac{1}{e}.$$

Consequently:

$$\cot\left(\frac{\theta_s}{2}\right) = \sqrt{e^2 - 1}.$$

To relate e to b, note that at periapsis $r_0 = \frac{p}{e-1}$, and using energy conservation $E_m = \frac{k(e^2-1)}{2p} = \frac{1}{2}mv_0^2$:

$$e^2 - 1 = \frac{2E_m p}{k} = \frac{2(\frac{1}{2}mv_0^2)(m^2v_0^2b^2/mk)}{k} = \left(\frac{mv_0^2 b}{k}\right)^2.$$

Taking the square root:

$$\cot\left(\frac{\theta_s}{2}\right) = \frac{mv_0^2 b}{k} \implies \boxed{\tan\left(\frac{\theta_s}{2}\right) = \frac{k}{mv_0^2 b}.}$$

This is the fundamental Rutherford scattering relation that revealed the atomic nucleus in 1911!

Exercise 11.9 · ⋆⋆⋆ **Distance of Closest Approach for an Alpha Particle**

In a head-on collision ($b = 0$), an alpha particle of mass $m = 6.64 \times 10^{-27}$ kg and kinetic energy $K = 5.00$ MeV is fired directly at a gold nucleus ($Z = 79$). (1 eV $= 1.602 \times 10^{-19}$ J, $\frac{1}{4\pi\varepsilon_0} = 8.988 \times 10^9$ N m^2 C^{-2}, $e = 1.602 \times 10^{-19}$ C). (a) Express the distance of closest approach $d_{\min}$ using conservation of energy. (b) Calculate $d_{\min}$ in femtometres (fm $= 10^{-15}$ m). (c) What upper bound does this place on the radius of the gold nucleus?

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 11.9. (a) For a head-on collision ($b = 0$), the particle travels along a straight line. At closest approach, its speed drops momentarily to zero ($v = 0$). By conservation of mechanical energy:

$$E_m = K_\infty = U(d_{\min}) \implies K = \frac{1}{4\pi\varepsilon_0}\frac{(2e)(Ze)}{d_{\min}} = \frac{2Ze^2}{4\pi\varepsilon_0\, d_{\min}}.$$

Solving for $d_{\min}$:

$$\boxed{d_{\min} = \frac{2Ze^2}{4\pi\varepsilon_0 K}.}$$

(b) Convert $K = 5.00$ MeV to joules:

$$K = 5.00 \times 10^6 \times 1.602 \times 10^{-19}\ \text{J} = 8.01 \times 10^{-13}\ \text{J}.$$

Calculate $d_{\min}$ with $Z = 79$:

$$\begin{aligned} d_{\min} &= \frac{(8.988 \times 10^9) \times 2 \times 79 \times (1.602 \times 10^{-19})^2}{8.01 \times 10^{-13}} \\ &= \frac{(8.988 \times 10^9) \times 158 \times (2.566 \times 10^{-38})}{8.01 \times 10^{-13}} \\ &= \frac{3.644 \times 10^{-26}}{8.01 \times 10^{-13}} \approx 4.55 \times 10^{-14}\ \text{m} = \mathbf{45.5}\ \text{fm}. \end{aligned}$$

(c) Because the alpha particle turns around purely under electrostatic repulsion without hitting the nuclear surface, the radius of the gold nucleus must be strictly less than $d_{\min}$:

$$R_{\text{gold}} < 45.5\ \text{fm}.$$

(The actual measured nuclear radius of gold is $R \approx 1.2A^{1/3} \approx 1.2(197)^{1/3} \approx 7.0$ fm).

Exercise 11.10 · ⋆⋆⋆ **Precession of the Perihelion under a Perturbed Central Force**

Consider a nearly circular planetary orbit under a modified gravitational potential $U(r) = -\frac{k}{r} - \frac{\epsilon}{r^2}$, where ϵ is a tiny relativistic correction. (a) Find the effective potential $U_{\text{eff}}(r)$ and the radius r_0 of the circular orbit in terms of L, m, k, ϵ. (b) Perform a Taylor expansion of $U_{\text{eff}}(r)$ around r_0 to second order: $U_{\text{eff}}(r) \approx U_{\text{eff}}(r_0) + \frac{1}{2}k_{\text{eff}}(r - r_0)^2$. Find the radial oscillation frequency $\omega_r = \sqrt{k_{\text{eff}}/m}$. (c) Comparing ω_r with the orbital angular frequency $\omega_\theta = \dot{\theta} = \frac{L}{mr_0^2}$, show that the perihelion precesses by an angle $\Delta\theta_p \approx \frac{2\pi\epsilon}{kr_0}$ per orbit.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 11.10. (a) The effective potential is:

$$U_{\text{eff}}(r) = -\frac{k}{r} - \frac{\epsilon}{r^2} + \frac{L^2}{2mr^2} = -\frac{k}{r} + \frac{L^2 - 2m\epsilon}{2mr^2}.$$

Define the modified angular momentum parameter $L'^2 = L^2 - 2m\epsilon$. The circular orbit occurs at the minimum:

$$\frac{\mathrm{d}U_{\text{eff}}}{\mathrm{d}r} = \frac{k}{r^2} - \frac{L'^2}{mr^3} = 0 \implies \boxed{r_0 = \frac{L'^2}{mk} \approx \frac{L^2}{mk}\left(1 - \frac{2m\epsilon}{L^2}\right).}$$

(b) Second derivative at r_0:

$$\frac{\mathrm{d}^2U_{\text{eff}}}{\mathrm{d}r^2} = -\frac{2k}{r^3} + \frac{3L'^2}{mr^4}.$$

Evaluating at $r_0 = \frac{L'^2}{mk}$:

$$k_{\text{eff}} = \left.\frac{\mathrm{d}^2U_{\text{eff}}}{\mathrm{d}r^2}\right|_{r_0} = -\frac{2k}{r_0^3} + \frac{3(mkr_0)}{mr_0^4} = \frac{k}{r_0^3}.$$

The radial oscillation frequency is:

$$\omega_r = \sqrt{\frac{k_{\text{eff}}}{m}} = \sqrt{\frac{k}{mr_0^3}}.$$

(c) The orbital frequency for circular motion is $\omega_\theta = \frac{L}{mr_0^2}$. Using $\frac{L^2}{mr_0^3} = \frac{L'^2 + 2m\epsilon}{mr_0^3} = \frac{mkr_0 + 2m\epsilon}{mr_0^3} = \frac{k}{r_0^2} + \frac{2\epsilon}{r_0^3}$:

$$\omega_\theta = \sqrt{\frac{k}{mr_0^3} + \frac{2\epsilon}{mr_0^4}} = \omega_r\sqrt{1 + \frac{2\epsilon}{kr_0}} \approx \omega_r\left(1 + \frac{\epsilon}{kr_0}\right).$$

The time between two consecutive perihelion passages is the radial period $T_r = \frac{2\pi}{\omega_r}$. During this time, the polar angle increases by:

$$\Delta\theta = \omega_\theta T_r = \omega_\theta \frac{2\pi}{\omega_r} \approx 2\pi \left(1 + \frac{\epsilon}{kr_0}\right) = 2\pi + \Delta\theta_p.$$

Therefore, the perihelion advances each orbit by:

$$\boxed{\Delta\theta_p \approx \frac{2\pi\epsilon}{kr_0}.}$$

This orbital precession was precisely what Einstein used in 1915 to verify General Relativity through the 43 arcseconds per century anomalous precession of Mercury's perihelion!

PART IV

Work and Energy

CHAPTER 12

Work, Power, and the Work-Energy Theorem

You will learn to: define work as a line integral; calculate instantaneous and average power; apply the Work-Energy Theorem for particles; determine the work done by gravity, springs, and friction; and solve classic problems including vertical loops, detachment from surfaces, and resistive forces.

12.1 Definition of Work

When a force $\vec{F}$ acts on a material point undergoing an infinitesimal displacement $\mathrm{d}\vec{r}$, the **elementary mechanical work** done by $\vec{F}$ is the scalar product:

$$\boxed{\mathrm{d}W = \vec{F} \cdot \mathrm{d}\vec{r} = F_x\,\mathrm{d}x + F_y\,\mathrm{d}y + F_z\,\mathrm{d}z = F\,\mathrm{d}s\cos\theta,}$$

where θ is the angle between $\vec{F}$ and the displacement tangent $\mathrm{d}\vec{r}$, and $\mathrm{d}s = \|\mathrm{d}\vec{r}\|$ is the arc length.

For a finite displacement along a curve $\mathcal{C}$ from initial point A to final point B, the total work is the **line integral**:

$$\boxed{W_{A\to B}(\vec{F}) = \int_{\mathcal{C}} \vec{F} \cdot \mathrm{d}\vec{r}.}$$

Work is measured in **joules** ($1\ \mathrm{J} = 1\ \mathrm{N\,m} = 1\ \mathrm{kg\,m^2\,s^{-2}}$).

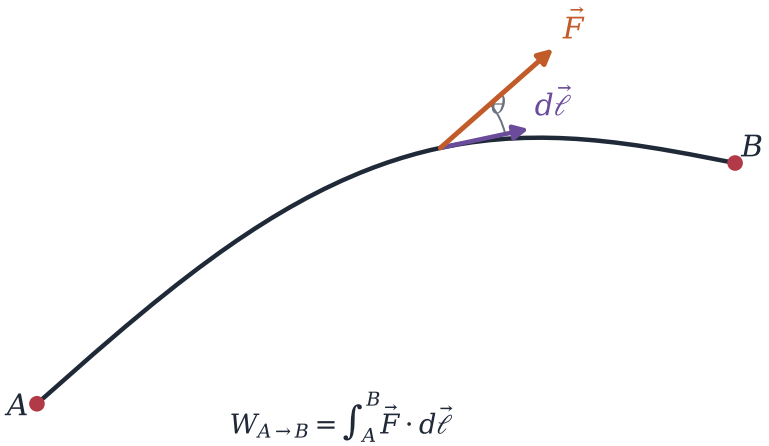


Figure 12.1: The line integral of force $\vec{F}$ along a curved trajectory from point A to point B. The tangential projection $F_t = \vec{F} \cdot \vec{u}_t$ contributes to the work.

Sign of Mechanical Work

- **Motive work** ($W > 0$): $\theta < 90°$ ($\vec{F} \cdot \vec{v} > 0$). The force accelerates the particle, adding energy.
- **Resistive work** ($W < 0$): $\theta > 90°$ ($\vec{F} \cdot \vec{v} < 0$). The force opposes motion, removing energy (e.g. friction).
- **Zero work** ($W = 0$): $\theta = 90°$ ($\vec{F} \perp \vec{v}$). Forces perpendicular to velocity do **no work!** Examples: normal reactions from smooth rigid surfaces, magnetic forces ($\vec{F} = q\vec{v} \times \vec{B}$), and the Coriolis force ($\vec{F}_{ic} = -2m\vec{\Omega} \times \vec{v}_r$).

12.2 Instantaneous Power

The rate at which work is performed with respect to time is the **instantaneous power**:

$$\boxed{P(t) = \frac{\mathrm{d}W}{\mathrm{d}t} = \vec{F} \cdot \frac{\mathrm{d}\vec{r}}{\mathrm{d}t} = \vec{F} \cdot \vec{v}.}$$

Power is measured in **watts** ($1\ \mathrm{W} = 1\ \mathrm{J\,s^{-1}} = 1\ \mathrm{kg\,m^2\,s^{-3}}$).

12.3 The Work-Energy Theorem

The **kinetic energy** of a material point of mass m moving at speed v is:

$$\boxed{E_c = \frac{1}{2}mv^2 = \frac{p^2}{2m}.}$$

Taking the time derivative of E_c:

$$\frac{\mathrm{d}E_c}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}m\vec{v} \cdot \vec{v}\right) = m\vec{v} \cdot \frac{\mathrm{d}\vec{v}}{\mathrm{d}t} = \vec{v} \cdot (m\vec{a}) = \vec{v} \cdot \sum \vec{F} = P_{\text{net}}(t).$$

Multiplying by $\mathrm{d}t$:

$$\mathrm{d}E_c = \mathrm{d}W_{\text{net}} = \sum \vec{F} \cdot \mathrm{d}\vec{r}.$$

Integrating between points A and B gives the **Work-Energy Theorem** (*Théorème de l'énergie cinétique*):

$$\boxed{\Delta E_c = E_c(B) - E_c(A) = \frac{1}{2}mv_B^2 - \frac{1}{2}mv_A^2 = W_{A\to B}\left(\sum \vec{F}\right).}$$

Power of the Work-Energy Theorem

The work-energy theorem allows us to find speeds directly from displacements **without needing to integrate Newton's second law with respect to time!**

12.4 Work Done by Common Forces

Force	**Expression**	**Work $W_{A\to B}$**
Uniform gravity	$\vec{P} = -mg\hat{k}$	$-mg(z_B - z_A)$
Ideal spring	$\vec{F}_s = -kx\hat{\imath}$	$-\frac{1}{2}k(x_B^2 - x_A^2)$
Kinetic friction	$\vec{f}_k = -\mu_k N\,\vec{u}_t$	$-\int_A^B \mu_k N\,\mathrm{d}s = -\mu_k N\,d$ (along straight line)
Normal reaction	$\vec{N} \perp \vec{v}$	0

12.5 Worked Examples

Worked example 12.1 · Braking distance with kinetic friction

A car travels at $v_0 = 90.0\ \text{km}\,\text{h}^{-1} = 25.0\ \text{m}\,\text{s}^{-1}$ on a horizontal road. The driver slams on the brakes, locking all four wheels. The coefficient of kinetic friction between tires and asphalt is $\mu_k = 0.700$. Using the work-energy theorem, find the stopping distance d.

Solution 12.1. The forces acting on the car during braking are:

- Vertical: gravity $\vec{P} = -mg\hat{\jmath}$ and normal force $\vec{N} = N\hat{\jmath}$. Since $a_y = 0$, $N = mg$.
- Horizontal: kinetic friction $\vec{f_k} = -\mu_k N\hat{\imath} = -\mu_k mg\hat{\imath}$.

Work done:

- $W(\vec{P}) = 0$ and $W(\vec{N}) = 0$ (perpendicular to displacement).
- $W(\vec{f_k}) = -f_k d = -\mu_k mgd$.

By the work-energy theorem between the initial state ($v_A = v_0$) and final stopped state ($v_B = 0$):

$$\Delta E_c = 0 - \frac{1}{2}mv_0^2 = W_{\text{net}} = -\mu_k mgd.$$

The mass m cancels:

$$\frac{1}{2}v_0^2 = \mu_k gd \implies \boxed{d = \frac{v_0^2}{2\mu_k g}.}$$

Computing:

$$d = \frac{(25.0)^2}{2(0.700)(9.81)} = \frac{625}{13.734} \approx \mathbf{45.5}\ \text{m}.$$

Worked example 12.2 · The vertical loop-the-loop

A small roller coaster car of mass m starts from rest at height h and slides down a frictionless track ending in a vertical circular loop of radius R. (a) Find the speed of the car at the highest point of the loop (point T, at height $2R$). (b) Find the normal force N_T exerted by the track at the top. (c) Deduce the minimum release height $h_{\min}$ to complete the loop without leaving the track.

Solution 12.2. (a) Applying the work-energy theorem between the release point A (height h, $v_A = 0$) and the top of the loop T (height $2R$, speed v_T): Since the normal force $\vec{N} \perp \vec{v}$ does no work ($W(\vec{N}) = 0$), only gravity performs work:

$$W_{A\to T}(\vec{P}) = -mg(z_T - z_A) = -mg(2R - h) = mg(h - 2R).$$

Thus:

$$\Delta E_c = \frac{1}{2}mv_T^2 - 0 = mg(h - 2R) \implies \boxed{v_T^2 = 2g(h - 2R).}$$

(b) At point T, both gravity and the track normal force $\vec{N}_T$ point vertically downward toward the center of the circle:

$$\sum F_{\text{rad}} = N_T + mg = m\frac{v_T^2}{R}.$$

Solving for N_T:

$$N_T = m\frac{v_T^2}{R} - mg = \frac{m}{R}\left[2g(h - 2R)\right] - mg = mg\left(\frac{2h}{R} - 5\right).$$

(c) The car remains on the track as long as the normal force is non-negative ($N_T \geq 0$):

$$mg\left(\frac{2h}{R} - 5\right) \geq 0 \implies \frac{2h}{R} \geq 5 \implies \boxed{h_{\min} = \frac{5}{2}R = 2.5R.}$$

At this minimum height, the speed at the top is $v_T = \sqrt{gR}$, and the car is momentarily in free fall ($N_T = 0$).

12.6 Exercises and Step-by-Step Solutions

Exercise 12.1 · ⋆ **Pulling a crate at an angle**

A worker pulls a crate of mass $m = 40.0$ kg across a smooth horizontal floor over a distance $d = 15.0$ m by applying a force of magnitude $F = 120$ N via a rope angled at $\alpha = 30.0^\circ$ above the horizontal. (a) Compute the work done by each force: the pulling force $\vec{F}$, gravity $\vec{P}$, and normal force $\vec{N}$. (b) If the crate starts from rest, find its final speed v_f using the work-energy theorem.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 12.1. (a) Work calculations:

- Pulling force: $W(\vec{F}) = Fd\cos\alpha = (120)(15.0)\cos 30.0^\circ = 1800 \times 0.8660 \approx \mathbf{1559}$ J $= 1.56$ kJ.
- Gravity: $\vec{P} = -mg\hat{\jmath}$ is perpendicular to displacement $\mathrm{d}\vec{r} = \mathrm{d}x\hat{\imath} \implies W(\vec{P}) = \mathbf{0}$ J.
- Normal force: $\vec{N} = N\hat{\jmath} \perp \mathrm{d}\vec{r} \implies W(\vec{N}) = \mathbf{0}$ J.

(b) Total net work: $W_{\text{net}} = 1559$ J. By the work-energy theorem:

$$\Delta E_c = \frac{1}{2}mv_f^2 - 0 = W_{\text{net}} \implies v_f = \sqrt{\frac{2W_{\text{net}}}{m}} = \sqrt{\frac{2 \times 1559}{40.0}} = \sqrt{77.94} \approx \mathbf{8.83}\ \mathrm{m\,s^{-1}}.$$

Exercise 12.2 · ⋆ **Cruising power of an automobile**

A car of mass $m = 1400$ kg cruises at a constant speed of $v = 108\ \mathrm{km\,h^{-1}} = 30.0\ \mathrm{m\,s^{-1}}$ on a flat horizontal highway. The total resistance force opposing its motion (aerodynamic drag plus rolling resistance) is $F_{\text{res}} = 650$ N. (a) What driving force F_{engine} must the engine deliver at constant speed? (b) Calculate the mechanical power output P delivered to the wheels in kilowatts and horsepower (1 hp $= 746$ W). (c) If the car now climbs an uphill slope of grade 5.0% ($\sin\alpha \approx 0.050$), what power is needed to maintain the same speed?

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 12.2. (a) At constant speed, $\vec{a} = \vec{0}$, so $\sum \vec{F} = \vec{0}$:

$$F_{\text{engine}} - F_{\text{res}} = 0 \implies F_{\text{engine}} = F_{\text{res}} = \mathbf{650}\ \text{N}.$$

(b) Mechanical power:

$$P = F_{\text{engine}} v = (650\ \text{N})(30.0\ \text{m}\,\text{s}^{-1}) = 19500\ \text{W} = \mathbf{19.5}\ \text{kW}.$$

In horsepower:

$$P_{\text{hp}} = \frac{19500}{746} \approx \mathbf{26.1}\ \text{hp}.$$

(c) On the 5% incline, gravity adds a component opposing motion: $P_{\parallel} = mg\sin\alpha$.

$$P_{\parallel} = (1400)(9.81)(0.050) = 686.7\ \text{N}.$$

The new required engine thrust is:

$$F_{\text{new}} = F_{\text{res}} + mg\sin\alpha = 650 + 686.7 = 1336.7\ \text{N}.$$

The required power is:

$$P_{\text{climb}} = F_{\text{new}} v = (1336.7)(30.0) = 40101\ \text{W} \approx \mathbf{40.1}\ \text{kW} \approx \mathbf{53.8}\ \text{hp}.$$

The required power more than doubles to climb the incline!

Exercise 12.3 · Mass sliding down a rough ramp

A block of mass $m = 3.00$ kg is released from rest at the top of a ramp inclined at $\theta = 37.0°$ of length $L = 5.00$ m. The coefficient of kinetic friction is $\mu_k = 0.250$. (a) Calculate the work done by gravity W_g as the block slides down the entire ramp. (b) Calculate the work done by friction W_f. (c) Using the work-energy theorem, find the speed of the block at the bottom of the ramp.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 12.3. (a) The vertical height difference is $h = L\sin\theta = 5.00\sin 37.0^\circ = 5.00(0.6018) = 3.009$ m. Work done by gravity:

$$W_g = +mgh = (3.00)(9.81)(3.009) \approx \mathbf{88.56}\ \text{J}.$$

(b) The normal force perpendicular to the ramp is $N = mg\cos\theta$. The kinetic friction force is $f_k = \mu_k N = \mu_k mg\cos\theta$. Since friction opposes displacement over length L:

$$W_f = -f_k L = -\mu_k mg\cos\theta\, L = -(0.250)(3.00)(9.81)(\cos 37.0^\circ)(5.00) = -(0.250)(3.00)(9.81)(0.7986)(5.00) \approx$$

(c) Net work:

$$W_{\text{net}} = W_g + W_f = 88.56 - 29.38 = 59.18\ \text{J}.$$

By the work-energy theorem:

$$\frac{1}{2}mv_f^2 - 0 = W_{\text{net}} \implies v_f = \sqrt{\frac{2W_{\text{net}}}{m}} = \sqrt{\frac{2\times 59.18}{3.00}} = \sqrt{39.45} \approx \mathbf{6.28}\ \text{m}\,\text{s}^{-1}.$$

Exercise 12.4 · Vertical spring: maximum stretch

A mass $m = 1.50$ kg is attached to an unstretched vertical spring of stiffness $k = 120\ \text{N}\,\text{m}^{-1}$ and released from rest. (a) State all forces acting on the mass and express their work when the spring stretches by a distance x. (b) Use the work-energy theorem to find the maximum extension $x_{\max}$ of the spring. (c) Compare $x_{\max}$ with the static equilibrium extension x_{eq}. Explain why $x_{\max} = 2x_{\text{eq}}$.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 12.4. (a) Two forces act during the vertical descent by distance x:

- Gravity: downward displacement by $x \implies W_g = +mgx$.
- Spring restoring force: $F_s = -kx$ opposing extension $\implies W_s = -\int_0^x kx'\mathrm{d}x' = -\frac{1}{2}kx^2$.

(b) At maximum extension $x_{\max}$, the mass momentarily comes to rest ($v = 0$):

$$\Delta E_c = 0 - 0 = 0.$$

By the work-energy theorem:

$$W_{\text{net}} = W_g + W_s = mgx_{\max} - \frac{1}{2}kx_{\max}^2 = 0.$$

Since $x_{\max} \neq 0$:

$$mg - \frac{1}{2}kx_{\max} = 0 \implies \boxed{x_{\max} = \frac{2mg}{k}.}$$

Computing:

$$x_{\max} = \frac{2(1.50)(9.81)}{120} = \frac{29.43}{120} \approx \mathbf{0.245}\ \text{m} = 24.5\ \text{cm}.$$

(c) At static equilibrium, the net force is zero:

$$\sum F = mg - kx_{\text{eq}} = 0 \implies x_{\text{eq}} = \frac{mg}{k} = \frac{24.5}{2} = 12.25\ \text{cm}.$$

Therefore, $x_{\max} = 2x_{\text{eq}}$. Because the system has no friction, the mass does not stop at equilibrium; when passing through x_{eq}, it has maximum kinetic energy! It overshoots equilibrium by exactly an equal distance x_{eq}, oscillating symmetrically between $x = 0$ and $x = 2x_{\text{eq}}$.

Exercise 12.5 · **Path-dependent work: conservative vs non-conservative forces**

Consider the force field $\vec{F}(x, y) = y^2\,\hat{\imath} + 2xy\,\hat{\jmath}$ and a second force field $\vec{G}(x, y) = y\,\hat{\imath} + 2x\,\hat{\jmath}$. (a) Compute the work done by each force along Path 1: straight line from $(0, 0)$ to $(1, 1)$ ($y = x$). (b) Compute the work done by each force along Path 2: from $(0, 0)$ along the x-axis to $(1, 0)$, then vertically to $(1, 1)$. (c) State which force is conservative and compute $\vec{\nabla} \times \vec{F}$ and $\vec{\nabla} \times \vec{G}$.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 12.5. (a) Path 1 ($y = x \implies \mathrm{d}y = \mathrm{d}x$ from $x = 0$ to $x = 1$):

$$W_1(\vec{F}) = \int_0^1 (x^2\,\mathrm{d}x + 2x^2\,\mathrm{d}x) = \int_0^1 3x^2\,\mathrm{d}x = \Big[x^3\Big]_0^1 = \mathbf{1.0}\ \mathrm{J}.$$
$$W_1(\vec{G}) = \int_0^1 (x\,\mathrm{d}x + 2x\,\mathrm{d}x) = \int_0^1 3x\,\mathrm{d}x = \left[\frac{3}{2}x^2\right]_0^1 = \mathbf{1.5}\ \mathrm{J}.$$

(b) Path 2: Segment A: $(0,0) \to (1,0)$ ($y = 0, \mathrm{d}y = 0, x : 0 \to 1$); Segment B: $(1,0) \to (1,1)$ ($x = 1, \mathrm{d}x = 0, y : 0 \to 1$): For $\vec{F}$:

$$W_{2A}(\vec{F}) = \int_0^1 0^2\,\mathrm{d}x = 0,$$
$$W_{2B}(\vec{F}) = \int_0^1 2(1)y\,\mathrm{d}y = \Big[y^2\Big]_0^1 = 1.0\ \mathrm{J}.$$
$$W_2(\vec{F}) = 0 + 1.0 = \mathbf{1.0}\ \mathrm{J}.$$

For $\vec{G}$:

$$W_{2A}(\vec{G}) = \int_0^1 0\,\mathrm{d}x = 0,$$
$$W_{2B}(\vec{G}) = \int_0^1 2(1)\,\mathrm{d}y = [2y]_0^1 = 2.0\ \mathrm{J}.$$
$$W_2(\vec{G}) = 0 + 2.0 = \mathbf{2.0}\ \mathrm{J}.$$

(c) Evaluating curls:

$$(\vec{\nabla} \times \vec{F})_z = \frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} = \frac{\partial(2xy)}{\partial x} - \frac{\partial(y^2)}{\partial y} = 2y - 2y = 0 \implies \vec{F} \text{ is } \mathbf{conservative!}$$

Hence $W_1(\vec{F}) = W_2(\vec{F}) = 1.0$ J (independent of path; $E_p(x,y) = -xy^2$). For $\vec{G}$:

$$(\vec{\nabla} \times \vec{G})_z = \frac{\partial G_y}{\partial x} - \frac{\partial G_x}{\partial y} = \frac{\partial(2x)}{\partial x} - \frac{\partial(y)}{\partial y} = 2 - 1 = 1 \neq 0 \implies \vec{G} \text{ is } \mathbf{non\text{-}conservative!}$$

Hence $W_1(\vec{G}) = 1.5\ \mathrm{J} \neq W_2(\vec{G}) = 2.0\ \mathrm{J}$ (path-dependent).

Exercise 12.6 · Bead sliding off a spherical dome: detachment angle

A small ice cube of mass m is placed at the top of a smooth frictionless spherical igloo of radius R. Given a negligible nudge, it begins to slide down the surface. (a) Using the work-energy theorem, find the speed $v(\theta)$ as a function of the polar angle θ measured from the top. (b) Write Newton's second law along the radial direction to find the normal force $N(\theta)$. (c) Find the angle θ_{det} at which the cube leaves the surface of the sphere, and the height above the ground at detachment.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 12.6. (a) The initial height at $\theta = 0$ is $z_0 = R$. At angle θ, the height is $z(\theta) = R\cos\theta$. The normal force is perpendicular to velocity, so $W(\vec{N}) = 0$. By the work-energy theorem:

$$\Delta E_c = \frac{1}{2}mv^2 - 0 = W_g = mg(R - R\cos\theta) \implies \boxed{v^2(\theta) = 2gR(1 - \cos\theta).}$$

(b) Along the inward radial direction toward the center of the sphere:

$$mg\cos\theta - N = m\frac{v^2}{R}.$$

Substituting $v^2(\theta)$:

$$N(\theta) = mg\cos\theta - m\frac{2gR(1-\cos\theta)}{R} = mg\cos\theta - 2mg(1-\cos\theta) = \boxed{mg(3\cos\theta - 2).}$$

(c) The cube loses contact with the surface when the normal reaction drops to zero ($N(\theta_{\text{det}}) = 0$):

$$mg(3\cos\theta_{\text{det}} - 2) = 0 \implies \boxed{\cos\theta_{\text{det}} = \frac{2}{3}.}$$

The angle of detachment is:

$$\theta_{\text{det}} = \arccos(2/3) \approx \mathbf{48.2}^\circ.$$

The height above ground at detachment is:

$$z_{\text{det}} = R\cos\theta_{\text{det}} = \frac{\mathbf{2}}{\mathbf{3}}\mathbf{R}.$$

The cube flies off into parabolic trajectory after dropping through one-third of the radius!

Exercise 12.7 · ⋆⋆⋆ **Velocity-dependent drag: distance to halve speed**

A motorboat of mass $m = 600$ kg cuts its engine when traveling at speed $v_0 = 20.0\ \text{m}\,\text{s}^{-1}$. Water resistance exerts a quadratic drag force $F_d = -bv^2$ with drag constant $b = 15.0\ \text{kg}\,\text{m}^{-1}$. (a) Write the work-energy theorem in differential form $\text{d}E_c = \text{d}W$ and deduce a differential equation relating speed v and position x. (b) Integrate the equation to find the speed $v(x)$ as a function of the distance x traveled since the engine was cut. (c) Calculate the distance $d_{1/2}$ the boat travels before its speed is reduced to $v_0/2 = 10.0\ \text{m}\,\text{s}^{-1}$.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 12.7. (a) The differential work-energy relation is:

$$\mathrm{d}E_c = \mathrm{d}\left(\frac{1}{2}mv^2\right) = mv\,\mathrm{d}v.$$

The work done over distance $\mathrm{d}x$ is:

$$\mathrm{d}W = F_d\,\mathrm{d}x = -bv^2\,\mathrm{d}x.$$

Equating $\mathrm{d}E_c = \mathrm{d}W$:

$$mv\,\mathrm{d}v = -bv^2\,\mathrm{d}x \implies m\frac{\mathrm{d}v}{v} = -b\,\mathrm{d}x.$$

(b) Separating variables and integrating from $(x = 0, v = v_0)$ to (x, v):

$$\int_{v_0}^{v} \frac{\mathrm{d}v'}{v'} = -\frac{b}{m}\int_0^x \mathrm{d}x' \implies \ln\left(\frac{v}{v_0}\right) = -\frac{b}{m}x.$$

Exponentiating:

$$\boxed{v(x) = v_0\,e^{-\frac{b}{m}x}.}$$

The speed decreases exponentially with distance traveled!
(c) Setting $v = v_0/2$:

$$\frac{1}{2} = e^{-\frac{b}{m}d_{1/2}} \implies \frac{b}{m}d_{1/2} = \ln 2 \implies \boxed{d_{1/2} = \frac{m\ln 2}{b}.}$$

Computing:

$$d_{1/2} = \frac{(600)(0.69315)}{15.0} = 40.0 \times 0.69315 \approx \mathbf{27.7}\text{ m}.$$

Exercise 12.8 · ⋆⋆⋆ **Power delivered in lifting a heavy chain**

A heavy flexible uniform chain of total length L and total mass M (linear mass density $\lambda = M/L$) rests coiled on the floor. A crane lifts one end vertically at a constant speed v_0. (a) At the instant when a length y of the chain is off the ground, find the total upward force $F(t)$ that the crane must exert. (Hint: consider both the weight of the suspended portion and the rate of change of momentum of each incoming link $\frac{\mathrm{d}p}{\mathrm{d}t} = \lambda v_0^2$). (b) Compute the instantaneous power $P(t) = F(t)v_0$ delivered by the crane. (c) Integrate $P(t)$ to find the total work W_{crane} done to lift the entire chain ($y = L$). (d) Compare W_{crane} with the final mechanical energy $E_m = E_c + E_p$ of the suspended chain. Account for the difference.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 12.8. (a) Let a length y be in the air moving at speed v_0. In an infinitesimal time interval $\mathrm{d}t$, a mass element $\mathrm{d}m = \lambda\,\mathrm{d}y = \lambda v_0 \mathrm{d}t$ at rest on the floor is abruptly jerked into motion at speed v_0. The momentum acquired is $\mathrm{d}p = (\mathrm{d}m)v_0 = \lambda v_0^2 \mathrm{d}t$. The required impulse force is $\frac{\mathrm{d}p}{\mathrm{d}t} = \lambda v_0^2$. In addition, the crane must support the weight of the suspended segment, $P(y) = (\lambda y)g$. Thus, the total lifting force is:

$$\boxed{F(t) = \lambda g y + \lambda v_0^2 = \lambda\left(gy + v_0^2\right).}$$

(b) The instantaneous power delivered by the crane is:

$$\boxed{P(t) = F(t)v_0 = \lambda v_0\left(gy + v_0^2\right) = \lambda g y v_0 + \lambda v_0^3.}$$

(c) The work done by the crane as y goes from 0 to L (with $\mathrm{d}y = v_0\mathrm{d}t$):

$$W_{\text{crane}} = \int_0^{L/v_0} P(t)\mathrm{d}t = \int_0^L F(y)\mathrm{d}y = \int_0^L (\lambda g y + \lambda v_0^2)\mathrm{d}y = \frac{1}{2}\lambda g L^2 + \lambda v_0^2 L.$$

Using $\lambda L = M$:

$$\boxed{W_{\text{crane}} = \frac{1}{2}MgL + Mv_0^2.}$$

(d) The mechanical energy of the fully lifted chain is:

- Kinetic energy: $E_c = \frac{1}{2}Mv_0^2$;
- Potential energy (center of mass is at $h_{CM} = L/2$): $E_p = Mg\frac{L}{2} = \frac{1}{2}MgL$.

Total final mechanical energy:

$$E_m = \frac{1}{2}MgL + \frac{1}{2}Mv_0^2.$$

The difference between the work supplied and the final mechanical energy is:

$$W_{\text{crane}} - E_m = \left(\frac{1}{2}MgL + Mv_0^2\right) - \left(\frac{1}{2}MgL + \frac{1}{2}Mv_0^2\right) = \mathbf{\frac{1}{2}Mv_0^2}.$$

This remaining half of the kinetic energy supplied ($\frac{1}{2}Mv_0^2$) is **dissipated as internal heat and sound** during the inelastic shocks when each coiled link is jerked off the floor!

CHAPTER 13

Potential Energy and Conservative Forces

You will learn to: test whether a force field is conservative using the curl operator; derive potential energy functions; apply the conservation of mechanical energy; account for energy dissipation by non-conservative forces; and analyse systems with gravity, springs, and electrostatic potentials.

13.1 Conservative Forces and Potential Energy

A force $\vec{F}(\vec{r})$ is said to be **conservative** if the work it performs on a particle moving between two points A and B depends **only on the endpoints** A and B, and not on the path taken:

$$W_{A\to B}(\vec{F}) = \int_{\mathcal{C}_1} \vec{F}\cdot \mathrm{d}\vec{r} = \int_{\mathcal{C}_2} \vec{F}\cdot \mathrm{d}\vec{r}.$$

Three Equivalent Conditions for a Conservative Force

1. **Path Independence**: The circulation along any closed loop is zero:
$$\oint \vec{F}\cdot \mathrm{d}\vec{r} = 0.$$
2. **Irrotational (Curl-Free)**: In a simply-connected domain:
$$\boxed{\vec{\nabla}\times\vec{F} = \vec{0} \iff \frac{\partial F_z}{\partial y} = \frac{\partial F_y}{\partial z},\quad \frac{\partial F_x}{\partial z} = \frac{\partial F_z}{\partial x},\quad \frac{\partial F_y}{\partial x} = \frac{\partial F_x}{\partial y}.}$$
3. **Derives from a Scalar Potential Energy**: There exists a scalar function $E_p(x,y,z)$ such that:
$$\boxed{\vec{F} = -\vec{\nabla}E_p = -\left(\frac{\partial E_p}{\partial x}\hat{\imath} + \frac{\partial E_p}{\partial y}\hat{\jmath} + \frac{\partial E_p}{\partial z}\hat{k}\right).}$$

The elementary work is the exact differential of the potential energy:

$$\mathrm{d}W = \vec{F}\cdot\mathrm{d}\vec{r} = -\vec{\nabla}E_p\cdot\mathrm{d}\vec{r} = -\mathrm{d}E_p.$$

Integrating between A and B:

$$\boxed{W_{A\to B}(\vec{F}) = -\Delta E_p = E_p(A) - E_p(B).}$$

The Minus Sign

Remember the crucial minus sign in $\vec{F} = -\vec{\nabla}E_p$! Physical systems naturally accelerate **toward lower potential energy** (downhill).

13.2 Conservation of Mechanical Energy

The **total mechanical energy** of a material point is the sum of its kinetic and potential energies:

$$\boxed{E_m = E_c + E_p = \frac{1}{2}mv^2 + E_p(\vec{r}).}$$

Taking the time derivative:

$$\frac{\mathrm{d}E_m}{\mathrm{d}t} = \frac{\mathrm{d}E_c}{\mathrm{d}t} + \frac{\mathrm{d}E_p}{\mathrm{d}t} = \vec{v}\cdot\sum\vec{F} + \vec{\nabla}E_p\cdot\frac{\mathrm{d}\vec{r}}{\mathrm{d}t} = \vec{v}\cdot\left(\sum\vec{F}_{\text{cons}} + \sum\vec{F}_{\text{nc}}\right) - \vec{F}_{\text{cons}}\cdot\vec{v} = \vec{F}_{\text{nc}}\cdot\vec{v}.$$

Thus:

$$\boxed{\frac{\mathrm{d}E_m}{\mathrm{d}t} = P_{\text{nc}}(t) = \vec{F}_{\text{nc}}\cdot\vec{v}.}$$

- If **all forces are conservative** ($\vec{F}_{\text{nc}} = \vec{0}$):

$$\boxed{\Delta E_m = 0 \iff E_c(B) + E_p(B) = E_c(A) + E_p(A) = \text{constant}.}$$

- If **non-conservative forces** (such as friction or drag) act:

$$\boxed{\Delta E_m = E_m(B) - E_m(A) = W_{A\to B}\left(\vec{F}_{\text{nc}}\right).}$$

13.3 Worked Examples

Worked example 13.1 · Deriving potential energy from a 3D force field

Consider the force field $\vec{F}(x,y,z) = (2xy+z^3)\hat{\imath}+x^2\hat{\jmath}+3xz^2\hat{k}$. (a) Show that $\vec{F}$ is conservative by calculating its curl. (b) Find the potential energy function $E_p(x,y,z)$, taking $E_p(0,0,0) = 0$. (c) Find the work done by $\vec{F}$ from $(0,0,0)$ to $(1,2,1)$.

Solution 13.1. (a) Compute $\vec{\nabla}\times\vec{F}$:

$$(\vec{\nabla}\times\vec{F})_x = \frac{\partial(3xz^2)}{\partial y} - \frac{\partial(x^2)}{\partial z} = 0 - 0 = 0,$$
$$(\vec{\nabla}\times\vec{F})_y = \frac{\partial(2xy+z^3)}{\partial z} - \frac{\partial(3xz^2)}{\partial x} = 3z^2 - 3z^2 = 0,$$
$$(\vec{\nabla}\times\vec{F})_z = \frac{\partial(x^2)}{\partial x} - \frac{\partial(2xy+z^3)}{\partial y} = 2x - 2x = 0.$$

Since $\vec{\nabla}\times\vec{F} = \vec{0}$ everywhere in $\mathbb{R}^3$, the force is **conservative**.
(b) Find $E_p(x,y,z)$ such that $\vec{F} = -\vec{\nabla}E_p$:

$$-\frac{\partial E_p}{\partial x} = 2xy + z^3 \implies E_p(x,y,z) = -x^2y - xz^3 + g(y,z),$$
$$-\frac{\partial E_p}{\partial y} = x^2 \implies x^2 - \frac{\partial g}{\partial y} = x^2 \implies \frac{\partial g}{\partial y} = 0 \implies g(y,z) = h(z),$$
$$-\frac{\partial E_p}{\partial z} = 3xz^2 \implies 3xz^2 - h'(z) = 3xz^2 \implies h'(z) = 0 \implies h(z) = C.$$

With $E_p(0,0,0) = 0$, $C = 0$. Thus:

$$\boxed{E_p(x,y,z) = -x^2y - xz^3.}$$

(c) The work from $A(0,0,0)$ to $B(1,2,1)$ is simply:

$$W_{A\to B} = E_p(A) - E_p(B) = 0 - \left[-(1)^2(2) - (1)(1)^3\right] = 0 - (-2-1) = +\mathbf{3.0}\ \text{J}.$$

Worked example 13.2 · The spring gun

A spring gun uses a spring of stiffness $k = 400\ \text{N m}^{-1}$ compressed by $\Delta x = 0.150$ m to fire a ball of mass $m = 50.0$ g $= 0.0500$ kg vertically upward. Find the maximum height $h_{\max}$ reached by the ball above the compressed release point.

Solution 13.2. Set the origin of vertical coordinates $y = 0$ at the compressed release position of the ball. The system is conservative (ignoring air resistance):

- Initial state (compressed spring at rest): $y_1 = 0$, $v_1 = 0$, spring compression $\Delta x = 0.150$ m.

$$E_{m1} = E_{c1} + E_{p,\text{grav},1} + E_{p,\text{spring},1} = 0 + 0 + \frac{1}{2}k(\Delta x)^2.$$

- Final state at maximum height: $y_2 = h_{\max}$, $v_2 = 0$, spring fully released (uncompressed).

$$E_{m2} = E_{c2} + E_{p,\text{grav},2} + E_{p,\text{spring},2} = 0 + mgh_{\max} + 0.$$

By conservation of mechanical energy $E_{m1} = E_{m2}$:

$$\frac{1}{2}k(\Delta x)^2 = mgh_{\max} \implies \boxed{h_{\max} = \frac{k(\Delta x)^2}{2mg}}.$$

Computing:

$$h_{\max} = \frac{400 \times (0.150)^2}{2(0.0500)(9.81)} = \frac{400 \times 0.0225}{0.981} = \frac{9.00}{0.981} \approx \mathbf{9.17}\ \text{m}.$$

13.4 Exercises and Step-by-Step Solutions

Exercise 13.1 · ⋆ Finding force from a polynomial potential

A particle moves along the x-axis subject to a potential energy:

$$E_p(x) = ax^4 - bx^2,$$

where $a = 2.00\ \text{J m}^{-4}$ and $b = 8.00\ \text{J m}^{-2}$. (a) Derive the expression for the force $F_x(x) = -\frac{\mathrm{d}E_p}{\mathrm{d}x}$. (b) Find all points where the force is zero (equilibrium points). (c) Calculate the potential energy at each equilibrium point.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 13.1. (a) The 1D conservative force is the negative derivative of $E_p(x)$:

$$F_x(x) = -\frac{\mathrm{d}E_p}{\mathrm{d}x} = -\frac{\mathrm{d}}{\mathrm{d}x}\left(ax^4 - bx^2\right) = -(4ax^3 - 2bx) = \boxed{2bx - 4ax^3.}$$

With numerical values: $F_x(x) = 16.0x - 8.00x^3 = 8.00x(2 - x^2)$ N.
(b) Equilibria occur where $F_x(x) = 0$:

$$8.00x(2 - x^2) = 0 \implies x_1 = \mathbf{0}\ \text{m}, \quad x_{2,3} = \pm\sqrt{2} \approx \pm\mathbf{1.414}\ \text{m}.$$

(c) Potential values:

- At $x = 0$: $E_p(0) = \mathbf{0}$ J.
- At $x = \pm\sqrt{2}$:

$$E_p(\pm\sqrt{2}) = 2.00(\sqrt{2})^4 - 8.00(\sqrt{2})^2 = 2.00(4) - 8.00(2) = 8.00 - 16.00 = -\mathbf{8.00}\ \text{J}.$$

The potential has the classic symmetric double-well shape with minima at $\pm\sqrt{2}$ m and a local barrier at $x = 0$.

Exercise 13.2 · ⋆ **The simple pendulum: maximum speed and string tension**

A simple pendulum consists of a bob of mass $m = 0.400$ kg suspended by a light string of length $L = 1.20$ m. The bob is pulled aside until the string makes an angle $\theta_0 = 60.0°$ with the vertical and released from rest. (a) Using conservation of mechanical energy, find the speed $v_{\max}$ of the bob as it passes through the lowest point ($\theta = 0$). (b) Determine the string tension $T_{\max}$ at the lowest point. Compare it with the static weight mg.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 13.2. (a) Set the zero of gravitational potential energy at the lowest point of the swing ($z = 0$). The initial height at angle θ_0 is:

$$z_0 = L(1 - \cos\theta_0) = 1.20(1 - \cos 60^\circ) = 1.20(1 - 0.50) = 0.600 \text{ m}.$$

Conservation of mechanical energy between the release point ($v_0 = 0$) and the bottom ($z = 0$, speed v):

$$E_{m1} = mgz_0 = E_{m2} = \frac{1}{2}mv_{\max}^2 \implies v_{\max} = \sqrt{2gz_0} = \sqrt{2gL(1 - \cos\theta_0)}.$$

Computing:

$$v_{\max} = \sqrt{2(9.81)(0.600)} = \sqrt{11.772} \approx \mathbf{3.43} \text{ m s}^{-1}.$$

(b) At the lowest point, the bob undergoes circular motion with centripetal acceleration $a_c = \frac{v_{\max}^2}{L}$ directed upward:

$$\sum F_z = T_{\max} - mg = m\frac{v_{\max}^2}{L} \implies T_{\max} = mg + m\frac{2gL(1 - \cos\theta_0)}{L} = mg(3 - 2\cos\theta_0).$$

For $\theta_0 = 60^\circ$ ($\cos 60^\circ = 0.5$):

$$T_{\max} = mg(3 - 2(0.5)) = \mathbf{2mg}.$$

Numerical value:

$$T_{\max} = 2(0.400)(9.81) = \mathbf{7.85} \text{ N}.$$

The dynamic tension at the bottom is exactly twice the static weight!

Exercise 13.3 · Curl test and 3D potential function

Consider the two force fields:

$$\vec{F}_1 = (2xy + z)\hat{\imath} + (x^2 + 2yz)\hat{\jmath} + (y^2 + x)\hat{k},$$

$$\vec{F}_2 = yz\,\hat{\imath} + xz\,\hat{\jmath} + 2xy\,\hat{k}.$$

(a) Compute the curl of both fields and determine which one is conservative. (b) For the conservative field, find the potential energy $E_p(x, y, z)$ such that $E_p(0, 0, 0) = 0$. (c) Evaluate the work done by the conservative force along any curve connecting $(1, 0, 0)$ to $(1, 2, 3)$.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 13.3. (a) For $\vec{F}_1$:

$$(\vec{\nabla}\times\vec{F}_1)_x = \frac{\partial(y^2+x)}{\partial y} - \frac{\partial(x^2+2yz)}{\partial z} = 2y - 2y = 0,$$
$$(\vec{\nabla}\times\vec{F}_1)_y = \frac{\partial(2xy+z)}{\partial z} - \frac{\partial(y^2+x)}{\partial x} = 1 - 1 = 0,$$
$$(\vec{\nabla}\times\vec{F}_1)_z = \frac{\partial(x^2+2yz)}{\partial x} - \frac{\partial(2xy+z)}{\partial y} = 2x - 2x = 0.$$

Thus $\vec{\nabla}\times\vec{F}_1 = \vec{0}$: $\vec{F}_1$ is **conservative**.
For $\vec{F}_2$:

$$(\vec{\nabla}\times\vec{F}_2)_x = \frac{\partial(2xy)}{\partial y} - \frac{\partial(xz)}{\partial z} = 2x - x = x \neq 0.$$

Thus $\vec{F}_2$ is **non-conservative**.
(b) Integrate $\vec{F}_1 = -\vec{\nabla}E_p$:

$$-\frac{\partial E_p}{\partial x} = 2xy + z \implies E_p = -x^2y - xz + g(y,z),$$
$$-\frac{\partial E_p}{\partial y} = x^2 + 2yz \implies x^2 - \frac{\partial g}{\partial y} = x^2 + 2yz \implies \frac{\partial g}{\partial y} = -2yz \implies g(y,z) = -y^2z + h(z),$$
$$-\frac{\partial E_p}{\partial z} = y^2 + x \implies x + y^2 - h'(z) = y^2 + x \implies h'(z) = 0 \implies h(z) = C.$$

With $E_p(0,0,0) = 0$, $C = 0$:

$$\boxed{E_p(x,y,z) = -x^2y - y^2z - xz.}$$

(c) Work is path-independent:

$$W = E_p(1,0,0) - E_p(1,2,3).$$

Evaluating:

$$E_p(1,0,0) = -(1)^2(0) - (0)^2(3) - (1)(0) = 0,$$
$$E_p(1,2,3) = -(1)^2(2) - (2)^2(3) - (1)(3) = -2 - 12 - 3 = -17 \text{ J}.$$

Thus:

$$W = 0 - (-17) = +\mathbf{17.0} \text{ J}.$$

Exercise 13.4 · High-altitude vertical launch: exact vs approximate energy

A rocket of mass $m = 500$ kg is launched vertically upward from Earth's surface with speed $v_0 = 3.00 \text{ km s}^{-1} = 3000 \text{ m s}^{-1}$. (a) Using the exact gravitational potential energy $E_p(r) = -G\frac{M_E m}{r}$, find the maximum altitude $h_{\max}$ reached above Earth's surface. ($R_E = 6371$ km, $M_E = 5.972 \times 10^{24}$ kg). (b) Calculate h_{approx} using the naive flat-Earth formula $v_0^2 = 2gh$. (c) Find the percentage error made by the flat-Earth approximation.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 13.4. (a) Let $r_1 = R_E$ and $r_2 = R_E + h_{\max}$. At maximum height, $v = 0$. By conservation of mechanical energy:

$$\frac{1}{2}mv_0^2 - \frac{GM_E m}{R_E} = 0 - \frac{GM_E m}{R_E + h_{\max}}.$$

Divide by m and define $\mu = GM_E$:

$$\frac{1}{R_E + h_{\max}} = \frac{1}{R_E} - \frac{v_0^2}{2\mu} = \frac{2\mu - R_E v_0^2}{2\mu R_E}.$$

Inverting:

$$R_E + h_{\max} = \frac{R_E}{1 - \frac{R_E v_0^2}{2\mu}} \implies \boxed{h_{\max} = \frac{v_0^2 R_E^2}{2\mu - R_E v_0^2}}.$$

Computing: $\mu = (6.674 \times 10^{-11})(5.972 \times 10^{24}) = 3.986 \times 10^{14}\ \text{m}^3\,\text{s}^{-2}$. $R_E = 6.371 \times 10^6$ m, $v_0^2 = 9.00 \times 10^6\ \text{m}^2\,\text{s}^{-2}$.

$$R_E v_0^2 = (6.371 \times 10^6)(9.00 \times 10^6) = 5.734 \times 10^{13}\ \text{m}^3\,\text{s}^{-2}.$$

$$2\mu - R_E v_0^2 = 7.972 \times 10^{14} - 0.5734 \times 10^{14} = 7.399 \times 10^{14}\ \text{m}^3\,\text{s}^{-2}.$$

$$h_{\max} = \frac{(9.00 \times 10^6) \times (6.371 \times 10^6)^2}{7.399 \times 10^{14}} = \frac{3.653 \times 10^{20}}{7.399 \times 10^{14}} \approx 4.937 \times 10^5\ \text{m} = \mathbf{493.7}\ \text{km}.$$

(b) Flat-Earth approximation ($g = 9.81\ \text{m}\,\text{s}^{-2}$):

$$h_{\text{approx}} = \frac{v_0^2}{2g} = \frac{(3000)^2}{2(9.81)} = \frac{9.00 \times 10^6}{19.62} \approx 4.587 \times 10^5\ \text{m} = \mathbf{458.7}\ \text{km}.$$

(c) Percentage error:

$$\text{Error} = \frac{h_{\max} - h_{\text{approx}}}{h_{\max}} \times 100\% = \frac{493.7 - 458.7}{493.7} \times 100\% \approx \mathbf{7.1}\%.$$

Gravity weakens significantly at this altitude, allowing the rocket to climb 35 km higher than the constant-g approximation predicts.

Exercise 13.5 · Spring compression on a rough surface

A block of mass $m = 2.00$ kg slides on a horizontal floor with initial speed $v_0 = 4.00\ \mathrm{m\,s^{-1}}$ toward an unstretched horizontal spring of stiffness $k = 500\ \mathrm{N\,m^{-1}}$. The coefficient of kinetic friction between the block and floor is $\mu_k = 0.200$. (a) From the moment the block contacts the spring until it momentarily stops at maximum compression x_c, identify the conservative and non-conservative forces. (b) Write the energy balance $\Delta E_m = W_{\mathrm{nc}}$. (c) Solve the resulting quadratic equation to find the maximum compression x_c.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 13.5. (a) Forces acting on the block:

- Conservative: spring restoring force $\vec{F}_s = -kx\hat{\imath}$ (potential energy $E_p = \frac{1}{2}kx^2$).
- Non-conservative: kinetic friction $\vec{f}_k = -\mu_k mg\hat{\imath}$.
- Normal force and gravity cancel vertically and do no work.

(b) Initial state (contact with unstretched spring): $E_m(0) = \frac{1}{2}mv_0^2 + 0$. Final state (maximum compression x_c, $v = 0$): $E_m(x_c) = 0 + \frac{1}{2}kx_c^2$. Work of friction during compression: $W_{\mathrm{nc}} = -f_k x_c = -\mu_k mgx_c$. Energy relation:

$$\Delta E_m = E_m(x_c) - E_m(0) = W_{\mathrm{nc}} \implies \frac{1}{2}kx_c^2 - \frac{1}{2}mv_0^2 = -\mu_k mgx_c.$$

Rearranging into standard quadratic form:

$$\frac{1}{2}kx_c^2 + \mu_k mgx_c - \frac{1}{2}mv_0^2 = 0 \iff kx_c^2 + 2\mu_k mgx_c - mv_0^2 = 0.$$

(c) Numerical coefficients: $k = 500$, $2\mu_k mg = 2(0.200)(2.00)(9.81) = 7.848$, $mv_0^2 = (2.00)(4.00)^2 = 32.0$.

$$500x_c^2 + 7.848x_c - 32.0 = 0.$$

Solving for positive root:

$$\Delta = (7.848)^2 - 4(500)(-32.0) = 61.59 + 64000 = 64061.59 \implies \sqrt{\Delta} \approx 253.10.$$

$$x_c = \frac{-7.848 + 253.10}{2(500)} = \frac{245.25}{1000} \approx \mathbf{0.245}\ \mathrm{m} = 24.5\ \mathrm{cm}.$$

Exercise 13.6 · ⋆⋆⋆ **The Yukawa Potential**

In nuclear and particle physics, the interaction between two nucleons is modeled by the **Yukawa potential**:

$$U(r) = -g_0^2 \frac{e^{-\alpha r}}{r},$$

where g_0 is the coupling constant and $\alpha = 1/r_0$ is the inverse screening radius. (a) Show that the central force derived from this potential is:

$$\vec{F}(r) = -g_0^2 \frac{e^{-\alpha r}}{r^2}(1 + \alpha r)\,\vec{u}_r.$$

(b) Examine the two limiting cases $\alpha r \ll 1$ and $\alpha r \gg 1$. (c) For a particle of mass m with angular momentum L, write the effective potential $U_{\text{eff}}(r)$. Show that circular orbits are possible only if the screening length r_0 is sufficiently large.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 13.6. (a) Since the potential is spherically symmetric, $\vec{F} = -\frac{\mathrm{d}U}{\mathrm{d}r}\vec{u}_r$:

$$\begin{aligned}\frac{\mathrm{d}U}{\mathrm{d}r} &= -g_0^2 \frac{\mathrm{d}}{\mathrm{d}r}\left(\frac{e^{-\alpha r}}{r}\right) = -g_0^2\left[\frac{-\alpha e^{-\alpha r}\cdot r - e^{-\alpha r}\cdot 1}{r^2}\right] \\ &= +g_0^2 \frac{e^{-\alpha r}}{r^2}(1+\alpha r).\end{aligned}$$

Therefore:

$$\boxed{\vec{F}(r) = -g_0^2 \frac{e^{-\alpha r}}{r^2}(1 + \alpha r)\,\vec{u}_r.}$$

The force is strictly attractive and points radially inward.
(b) Limiting behaviors:

- Short range ($\alpha r \ll 1$): $e^{-\alpha r} \approx 1$ and $(1 + \alpha r) \approx 1$, so:

  $$\vec{F}(r) \approx -\frac{g_0^2}{r^2}\vec{u}_r.$$

  It recovers the standard Coulomb / inverse-square force law!

- Long range ($\alpha r \gg 1$): The exponential decay $e^{-\alpha r}$ completely suppresses the interaction. Unlike Coulomb forces, the Yukawa force is **short-ranged** (decaying exponentially past $r_0 = 1/\alpha$).

(c) The effective potential is:

$$U_{\text{eff}}(r) = -g_0^2 \frac{e^{-\alpha r}}{r} + \frac{L^2}{2mr^2}.$$

A circular orbit at radius r_c requires $\left.\frac{\mathrm{d}U_{\text{eff}}}{\mathrm{d}r}\right|_{r_c} = 0$:

$$g_0^2 \frac{e^{-\alpha r_c}}{r_c^2}(1+\alpha r_c) = \frac{L^2}{m r_c^3} \implies g_0^2 r_c e^{-\alpha r_c}(1+\alpha r_c) = \frac{L^2}{m}.$$

The function $f(r) = r(1+\alpha r)e^{-\alpha r}$ has a maximum at $\alpha r = \frac{\sqrt{5}-1}{2} \approx 0.618$. If L^2/m exceeds this peak value, no circular orbit can exist! Thus, screening limits the existence of bound states.

Exercise 13.7 · ★★★ **Rate of mechanical energy dissipation by air drag**

A projectile of mass m is in flight under gravity $\vec{g} = -g\hat{\jmath}$ and a non-linear aerodynamic drag force $\vec{f} = -\gamma v^n \vec{u}_t = -\gamma v^{n-1}\vec{v}$, where $\gamma > 0$ and $n \geq 1$. (a) Show that the time rate of change of total mechanical energy is:

$$\frac{\mathrm{d}E_m}{\mathrm{d}t} = -\gamma v^{n+1}.$$

(b) Explain why $\frac{\mathrm{d}E_m}{\mathrm{d}t} \leq 0$ at all instants, and state what becomes of the lost mechanical energy. (c) For linear drag ($n = 1$, $\vec{f} = -b\vec{v}$), integrate $\frac{\mathrm{d}E_m}{\mathrm{d}t}$ along a horizontal trajectory where $v(t) = v_0 e^{-bt/m}$ to find the total mechanical energy dissipated from $t = 0$ to $t \to \infty$. Compare with the initial kinetic energy.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 13.7. (a) The mechanical energy is $E_m = E_c + E_p$. The only non-conservative force is the drag force $\vec{f}$. The rate of change of mechanical energy is the power of the non-conservative forces:

$$\frac{\mathrm{d}E_m}{\mathrm{d}t} = \vec{f}\cdot\vec{v} = \left(-\gamma v^{n-1}\vec{v}\right)\cdot\vec{v} = -\gamma v^{n-1}(\vec{v}\cdot\vec{v}) = \boxed{-\gamma v^{n+1}.}$$

(b) Since $\gamma > 0$, $v \geq 0$, and $n \geq 1$, the term $\gamma v^{n+1} \geq 0$ for all physical states. Therefore:

$$\frac{\mathrm{d}E_m}{\mathrm{d}t} \leq 0.$$

Mechanical energy is strictly non-increasing (monotonically dissipated). By the First Law of Thermodynamics, the lost mechanical energy is converted into **internal thermal energy** (heating both the projectile and the surrounding air) and acoustic energy (sound waves).

(c) For linear drag ($n = 1, \gamma = b$), on a horizontal path ($E_p = \text{const}$):

$$\frac{\mathrm{d}E_m}{\mathrm{d}t} = -bv^2(t).$$

Substituting $v(t) = v_0 e^{-bt/m}$:

$$v^2(t) = v_0^2 e^{-2bt/m}.$$

Integrating the dissipated power from $t = 0$ to ∞:

$$\Delta E_{\text{diss}} = \int_0^\infty bv^2(t)\mathrm{d}t = bv_0^2 \int_0^\infty e^{-2bt/m}\mathrm{d}t = bv_0^2 \left[-\frac{m}{2b}e^{-2bt/m}\right]_0^\infty = bv_0^2 \left(0 - \left(-\frac{m}{2b}\right)\right) = \boxed{\frac{1}{2}\mathbf{m}\mathbf{v}_\mathbf{0}^\mathbf{2}}.$$

The total mechanical energy dissipated is exactly equal to the initial kinetic energy $\frac{1}{2}mv_0^2$: 100% of the initial kinetic energy is converted into heat.

CHAPTER 14

1D Potential Wells, Stability, and Small Oscillations

You will learn to: read and interpret 1D potential energy curves $E_p(x)$; determine allowed motion zones and turning points; classify equilibrium states as stable, unstable, or neutral; expand potentials in Taylor series; compute the effective stiffness k_{eff}; and derive the frequency of small harmonic oscillations around stable minima.

14.1 Analysis of 1D Potential Energy Curves

In one-dimensional motion, the conservative force is related to the potential energy $E_p(x)$ by:

$$\boxed{F(x) = -\frac{\mathrm{d}E_p}{\mathrm{d}x}.}$$

The force is the **negative slope** of the potential curve:

- If $\frac{\mathrm{d}E_p}{\mathrm{d}x} > 0$ (uphill slope), then $F(x) < 0$ (force pushes to the left).
- If $\frac{\mathrm{d}E_p}{\mathrm{d}x} < 0$ (downhill slope), then $F(x) > 0$ (force pushes to the right).
- If $\frac{\mathrm{d}E_p}{\mathrm{d}x} = 0$ (flat slope), then $F(x) = 0$ (**equilibrium position**).

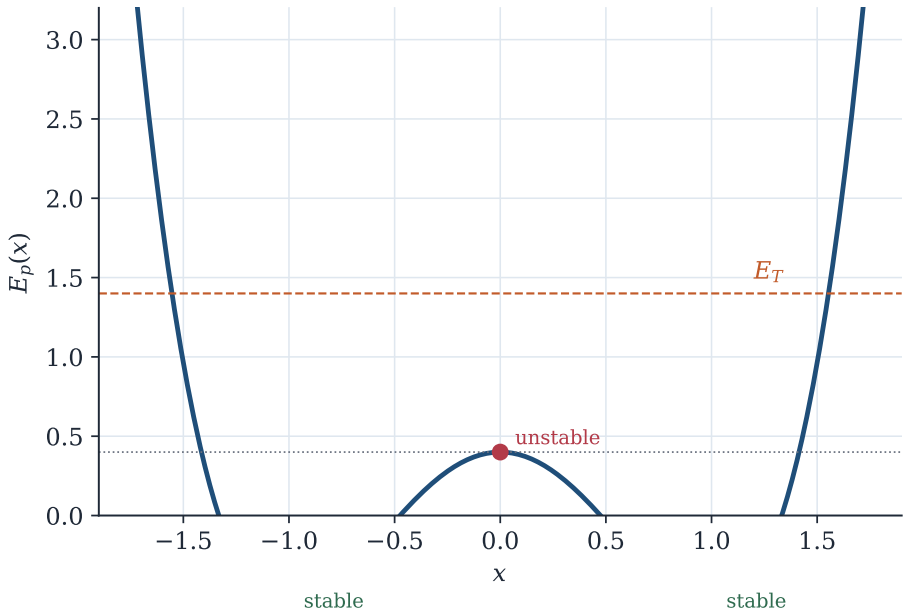


Figure 14.1: A one-dimensional potential energy curve $E_p(x)$ with total mechanical energy E_m. The particle oscillates between the classical turning points x_1 and x_2 where $E_m = E_p(x)$.

14.2 Permitted Regions and Turning Points

Since kinetic energy cannot be negative:

$$E_c = \frac{1}{2}mv^2 = E_m - E_p(x) \geq 0 \implies \boxed{E_m \geq E_p(x).}$$

- **Permitted region**: The set of positions where $E_p(x) \leq E_m$.
- **Forbidden region**: Positions where $E_p(x) > E_m$ cannot be reached classically.
- **Turning points**: Positions x_t where $E_p(x_t) = E_m$. At these points, $E_c = 0 \implies v = 0$. The particle stops and reverses direction.
- **Bound motion**: If the permitted region is bounded by two turning points $[x_1, x_2]$, the particle is trapped in a **potential well** and undergoes periodic oscillations.

14.3 Equilibrium and Stability Criteria

An equilibrium position x_0 satisfies $\left.\frac{\mathrm{d}E_p}{\mathrm{d}x}\right|_{x_0} = 0$. Its stability is determined by the second derivative:

Second Derivative	Geometry of $E_p(x)$	Stability	Restoring Force
$\left.\frac{\mathrm{d}^2E_p}{\mathrm{d}x^2}\right\|_{x_0} > 0$	Local minimum (well)	**Stable**	$F \approx -k_{\text{eff}}(x - x_0)$ directed towa
$\left.\frac{\mathrm{d}^2E_p}{\mathrm{d}x^2}\right\|_{x_0} < 0$	Local maximum (crest)	**Unstable**	$F \approx +k(x - x_0)$ pushes away fr
$\left.\frac{\mathrm{d}^2E_p}{\mathrm{d}x^2}\right\|_{x_0} = 0$	Flat inflection / plateau	**Neutral / Higher order**	Examine third and fourth deriva

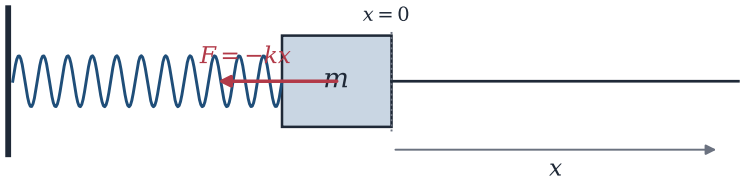


Figure 14.2: The classic harmonic oscillator: a mass on a spring around its stable equilibrium position.

14.4 Small Oscillations around a Stable Minimum

Let x_0 be a stable equilibrium point. Expand $E_p(x)$ in a Taylor series about x_0 for small displacements $\xi = x - x_0$:

$$E_p(x) = E_p(x_0) + \underbrace{\left.\frac{\mathrm{d}E_p}{\mathrm{d}x}\right|_{x_0}}_{=0}\xi + \frac{1}{2}\underbrace{\left.\frac{\mathrm{d}^2E_p}{\mathrm{d}x^2}\right|_{x_0}}_{\equiv k_{\text{eff}}}\xi^2 + \mathcal{O}(\xi^3).$$

Dropping higher-order terms, the potential near the minimum is **parabolic**:

$$\boxed{E_p(x) \approx E_p(x_0) + \frac{1}{2}k_{\text{eff}}\,\xi^2,} \quad \text{where} \quad \boxed{k_{\text{eff}} = \left.\frac{\mathrm{d}^2E_p}{\mathrm{d}x^2}\right|_{x_0} > 0.}$$

The restoring force is linear in ξ:

$$F(\xi) = -\frac{\mathrm{d}E_p}{\mathrm{d}\xi} \approx -k_{\text{eff}}\,\xi.$$

Newton's second law $m\ddot{\xi} = -k_{\text{eff}}\xi$ becomes the equation of a **simple harmonic oscillator**:

$$\boxed{\ddot{\xi} + \omega_0^2\,\xi = 0,} \quad \text{where} \quad \boxed{\omega_0 = \sqrt{\frac{k_{\text{eff}}}{m}} = \sqrt{\frac{1}{m}\frac{\mathrm{d}^2E_p}{\mathrm{d}x^2}\bigg|_{x_0}}.}$$

The period of small oscillations is:

$$\boxed{T_0 = \frac{2\pi}{\omega_0} = 2\pi\sqrt{\frac{m}{k_{\text{eff}}}}.}$$

14.5 Worked Examples

Worked example 14.1 · The Lennard-Jones (6-12) molecular potential

The interaction between two noble gas atoms is modeled by the Lennard-Jones potential:

$$U(r) = 4\epsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right],$$

where ϵ is the depth of the potential well and σ is the finite distance at which $U(\sigma) = 0$. (a) Find the equilibrium bond length r_0 and the minimum potential energy $U(r_0)$. (b) Calculate the effective spring constant k_{eff} for small vibrations about r_0. (c) Find the angular frequency ω_0 of small vibrations for a reduced mass μ.

Solution 14.1. (a) Set $\frac{\mathrm{d}U}{\mathrm{d}r} = 0$:

$$\frac{\mathrm{d}U}{\mathrm{d}r} = 4\epsilon\left[-12\frac{\sigma^{12}}{r^{13}} + 6\frac{\sigma^6}{r^7}\right] = \frac{24\epsilon}{r}\left[\left(\frac{\sigma}{r}\right)^6 - 2\left(\frac{\sigma}{r}\right)^{12}\right].$$

Setting this to zero:

$$\left(\frac{\sigma}{r_0}\right)^6 = 2\left(\frac{\sigma}{r_0}\right)^{12} \implies \left(\frac{r_0}{\sigma}\right)^6 = 2 \implies \boxed{r_0 = 2^{1/6}\sigma \approx 1.1225\,\sigma.}$$

The minimum potential energy is:

$$U(r_0) = 4\epsilon\left[\left(\frac{1}{2}\right)^2 - \frac{1}{2}\right] = 4\epsilon\left[\frac{1}{4} - \frac{1}{2}\right] = \boxed{-\epsilon.}$$

The dissociation energy required to separate the molecules to infinity is ϵ.
(b) Compute the second derivative:

$$\frac{\mathrm{d}^2U}{\mathrm{d}r^2} = 4\epsilon\left[12\times 13\frac{\sigma^{12}}{r^{14}} - 6\times 7\frac{\sigma^6}{r^8}\right] = \frac{4\epsilon}{r^2}\left[156\left(\frac{\sigma}{r}\right)^{12} - 42\left(\frac{\sigma}{r}\right)^6\right].$$

Evaluating at $r = r_0$ where $(\sigma/r_0)^6 = 1/2$ and $(\sigma/r_0)^{12} = 1/4$:

$$k_{\text{eff}} = \frac{\mathrm{d}^2U}{\mathrm{d}r^2}\bigg|_{r_0} = \frac{4\epsilon}{r_0^2}\left[156\left(\frac{1}{4}\right) - 42\left(\frac{1}{2}\right)\right] = \frac{4\epsilon}{r_0^2}\left[39 - 21\right] = \frac{4\epsilon}{r_0^2}(18) = \boxed{\frac{72\epsilon}{r_0^2}.}$$

Since $r_0^2 = 2^{1/3}\sigma^2$, this is $k_{\text{eff}} \approx \frac{57.14\epsilon}{\sigma^2} > 0$ (stable equilibrium!).
(c) The angular frequency of small vibrations is:

$$\boxed{\omega_0 = \sqrt{\frac{k_{\text{eff}}}{\mu}} = \frac{1}{r_0}\sqrt{\frac{72\epsilon}{\mu}} = \frac{6}{r_0}\sqrt{\frac{2\epsilon}{\mu}}.}$$

Worked example 14.2 · Equilibrium and stability of a bead on an inverted parabola

A bead of mass m slides without friction on a rigid wire bent into the shape of a parabola $y = -\frac{1}{2}cx^2$ $(c > 0)$ in a vertical plane, under gravity $\vec{g} = -g\hat{\jmath}$. Show that $x = 0$ is an unstable equilibrium, and find the characteristic time τ for the bead to fall away.

Solution 14.2. The potential energy of the bead in gravity is:

$$E_p(x) = mgy = -mg\left(\frac{1}{2}cx^2\right) = -\frac{1}{2}mgc\,x^2.$$

The first and second derivatives are:

$$\frac{\mathrm{d}E_p}{\mathrm{d}x} = -mgc\,x, \qquad \frac{\mathrm{d}^2E_p}{\mathrm{d}x^2} = -mgc.$$

At $x = 0$, $\frac{\mathrm{d}E_p}{\mathrm{d}x} = 0$, so $x_0 = 0$ is an equilibrium point. However:

$$\left.\frac{\mathrm{d}^2E_p}{\mathrm{d}x^2}\right|_0 = -mgc < 0.$$

Because the second derivative is strictly negative, the origin is a local maximum: it is an **unstable equilibrium**. The equation of motion for small displacements ($\dot{x}^2 \ll 1 \implies \mathrm{d}s \approx \mathrm{d}x$) is:

$$m\ddot{x} = -\frac{\mathrm{d}E_p}{\mathrm{d}x} = +mgc\,x \implies \ddot{x} - (gc)x = 0.$$

The general solution is $x(t) = Ae^{t/\tau} + Be^{-t/\tau}$ with characteristic time constant:

$$\boxed{\tau = \frac{1}{\sqrt{gc}}.}$$

Any tiny perturbation causes the displacement to grow exponentially with time constant τ.

14.6 Exercises and Step-by-Step Solutions

Exercise 14.1 · ⋆ Classifying equilibria of a polynomial potential

Consider a particle of mass $m = 0.25$ kg moving in the potential:

$$E_p(x) = x^3 - 3x \quad \text{(with } x \text{ in metres, } E_p \text{ in joules).}$$

(a) Find the positions of all equilibrium points. (b) Determine the stability of each equilibrium point using the second derivative test. (c) For the stable equilibrium point, calculate the effective spring constant k_{eff} and the frequency f_0 of small oscillations.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 14.1. (a) Equilibria occur where $\frac{\mathrm{d}E_p}{\mathrm{d}x} = 0$:

$$\frac{\mathrm{d}E_p}{\mathrm{d}x} = 3x^2 - 3 = 0 \implies x^2 = 1 \implies \mathbf{x_1 = -1.0}\ \mathrm{m}, \quad \mathbf{x_2 = +1.0}\ \mathrm{m}.$$

(b) Second derivative:

$$\frac{\mathrm{d}^2 E_p}{\mathrm{d}x^2} = 6x.$$

- At $x_1 = -1.0$ m: $\left.\frac{\mathrm{d}^2 E_p}{\mathrm{d}x^2}\right|_{-1} = -6.0\ \mathrm{J\,m^{-2}} < 0 \implies$ **Unstable equilibrium** (local peak).
- At $x_2 = +1.0$ m: $\left.\frac{\mathrm{d}^2 E_p}{\mathrm{d}x^2}\right|_{+1} = +6.0\ \mathrm{J\,m^{-2}} > 0 \implies$ **Stable equilibrium** (local well).

(c) At the stable minimum $x_2 = 1.0$ m:

$$k_{\text{eff}} = \left.\frac{\mathrm{d}^2 E_p}{\mathrm{d}x^2}\right|_{+1} = \mathbf{6.0}\ \mathrm{N\,m^{-1}}.$$

The angular frequency is:

$$\omega_0 = \sqrt{\frac{k_{\text{eff}}}{m}} = \sqrt{\frac{6.0}{0.25}} = \sqrt{24} \approx 4.90\ \mathrm{rad\,s^{-1}}.$$

The oscillation frequency is:

$$f_0 = \frac{\omega_0}{2\pi} = \frac{4.90}{2\pi} \approx \mathbf{0.78}\ \mathrm{Hz}.$$

Exercise 14.2 · ⋆ **Permitted regions and turning points from energy**

A particle of mass $m = 1.0$ kg moves along the x-axis in the potential $E_p(x) = \frac{1}{2}x^4 - 2x^2$ (in joules). Its total mechanical energy is $E_m = -1.0$ J. (a) Find the positions of the two potential minima and the potential barrier at $x = 0$. (b) Find the turning points x_t where $E_p(x) = E_m$. (c) Identify the permitted intervals of motion for $E_m = -1.0$ J. Can a particle in the left well reach the right well?

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 14.2. (a) Finding extrema:

$$\frac{\mathrm{d}E_p}{\mathrm{d}x} = 2x^3 - 4x = 2x(x^2-2) = 0 \implies x = 0, \quad x = \pm\sqrt{2}.$$

- At $x = 0$: $E_p(0) = 0$ J (local maximum / potential barrier).
- At $x = \pm\sqrt{2}$: $E_p(\pm\sqrt{2}) = \frac{1}{2}(4) - 2(2) = 2 - 4 = -2.0$ J (two symmetric minima).

(b) Turning points satisfy $E_p(x) = E_m = -1.0$ J:

$$\frac{1}{2}x^4 - 2x^2 = -1.0 \iff x^4 - 4x^2 + 2 = 0.$$

Let $u = x^2$:

$$u^2 - 4u + 2 = 0 \implies u = \frac{4 \pm \sqrt{16-8}}{2} = 2 \pm \sqrt{2}.$$

Since $2 - \sqrt{2} \approx 0.5858 > 0$ and $2 + \sqrt{2} \approx 3.4142$:

$$x = \pm\sqrt{2-\sqrt{2}} \approx \pm 0.765 \text{ m}, \qquad x = \pm\sqrt{2+\sqrt{2}} \approx \pm 1.848 \text{ m}.$$

(c) The condition $E_p(x) \le E_m = -1.0$ J requires:

$$\text{Left well:} \quad x \in [-1.848 \text{ m}, \ -0.765 \text{ m}], \qquad \text{Right well:} \quad x \in [+0.765 \text{ m}, \ +1.848 \text{ m}].$$

The interval $(-0.765 \text{ m}, +0.765 \text{ m})$ is classically **forbidden** because $E_p(x) > -1.0$ J there (with a peak of 0 J at $x = 0$). Therefore, a classical particle starting in the left well is trapped forever and **cannot reach the right well** (in quantum mechanics, it could tunnel through).

Exercise 14.3 · The Morse potential for molecular bonds

The potential energy between two atoms in a diatomic molecule (such as HCl) is well approximated by the Morse potential:

$$V(r) = D_e\left[1 - e^{-a(r-r_e)}\right]^2,$$

where D_e is the well depth, r_e is the equilibrium separation, and a is the width parameter. (a) Verify that $r = r_e$ is the equilibrium separation and find $V(r_e)$. (b) Compute the effective spring constant $k_{\text{eff}} = \left.\frac{\mathrm{d}^2V}{\mathrm{d}r^2}\right|_{r_e}$. (c) For HCl, $D_e = 4.43$ eV $= 7.10\times10^{-19}$ J, $a = 1.90\times10^{10}$ m^{-1}, and reduced mass $\mu = 1.63\times10^{-27}$ kg. Calculate k_{eff} and the vibrational frequency $\nu = \omega_0/2\pi$.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 14.3. (a) First derivative:

$$\frac{\mathrm{d}V}{\mathrm{d}r} = 2D_e\left[1-e^{-a(r-r_e)}\right]\left[ae^{-a(r-r_e)}\right] = 2aD_e e^{-a(r-r_e)}\left[1-e^{-a(r-r_e)}\right].$$

At $r = r_e$:

$$\left.\frac{\mathrm{d}V}{\mathrm{d}r}\right|_{r_e} = 2aD_e(1)(1-1) = 0.$$

Thus r_e is indeed an equilibrium position. The potential energy at equilibrium is $V(r_e) = D_e(1-1)^2 = \mathbf{0}$ J.
(b) Second derivative:

$$\begin{aligned}\frac{\mathrm{d}^2V}{\mathrm{d}r^2} &= 2aD_e\frac{\mathrm{d}}{\mathrm{d}r}\left[e^{-a(r-r_e)}-e^{-2a(r-r_e)}\right]\\ &= 2aD_e\left[-ae^{-a(r-r_e)}+2ae^{-2a(r-r_e)}\right] = 2a^2D_e\left[2e^{-2a(r-r_e)}-e^{-a(r-r_e)}\right].\end{aligned}$$

Evaluating at $r = r_e$:

$$k_{\text{eff}} = \left.\frac{\mathrm{d}^2V}{\mathrm{d}r^2}\right|_{r_e} = 2a^2D_e(2-1) = \boxed{2a^2D_e.}$$

(c) Numerical values:

$$k_{\text{eff}} = 2(1.90\times10^{10})^2\times(7.10\times10^{-19}) = 2(3.61\times10^{20})\times(7.10\times10^{-19}) = \mathbf{512.6}\ \mathrm{N\,m^{-1}}.$$

Vibrational angular frequency:

$$\omega_0 = \sqrt{\frac{k_{\text{eff}}}{\mu}} = \sqrt{\frac{512.6}{1.63\times10^{-27}}} = \sqrt{3.145\times10^{29}} \approx 5.61\times10^{14}\ \mathrm{rad\,s^{-1}}.$$

Vibrational frequency:

$$\nu = \frac{\omega_0}{2\pi} = \frac{5.61\times10^{14}}{2\pi} \approx \mathbf{8.93\times10^{13}}\ \mathrm{Hz} = 89.3\ \mathrm{THz}.$$

(This corresponds to an infrared absorption wavelength $\lambda = c/\nu \approx 3.36\ \mu$m, typical of molecular vibration).

Exercise 14.4 · **Oscillations of a floating cylinder**

A uniform solid vertical cylinder of mass m, radius R, and length H floats upright in a liquid of density ρ ($\rho > m/(\pi R^2 H)$). Let z be the downward vertical displacement from its static equilibrium floating position. (a) Determine the equilibrium submerged depth h_0. (b) Show that the net restoring force for a small displacement z is $F_z = -(\rho g \pi R^2) z$. (c) Derive the effective potential energy $E_p(z)$ and show that the vertical bobbing frequency is $\omega_0 = \sqrt{\frac{g}{h_0}}$.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 14.4. (a) At static equilibrium, Archimedes' buoyant force balances gravity:

$$F_B = \rho g V_{\text{sub}} = \rho g(\pi R^2 h_0) = mg \implies \boxed{h_0 = \frac{m}{\rho \pi R^2}.}$$

(b) When the cylinder is displaced downward by z, the submerged depth increases to $h_0 + z$. The buoyant force becomes $F_B(z) = \rho g \pi R^2 (h_0 + z)$. The net vertical force (with downward positive) is:

$$F_{\text{net}} = mg - F_B(z) = mg - \rho g \pi R^2 h_0 - \rho g \pi R^2 z.$$

Using $mg = \rho g \pi R^2 h_0$, the static terms cancel:

$$\boxed{F_{\text{net}}(z) = -(\rho g \pi R^2)\, z.}$$

The restoring force is purely linear (Hookean)!

(c) Potential energy:

$$E_p(z) = -\int_0^z F_{\text{net}}(z')\mathrm{d}z' = \frac{1}{2}(\rho g \pi R^2) z^2.$$

Since $\frac{\mathrm{d}^2 E_p}{\mathrm{d}z^2} = \rho g \pi R^2 = k_{\text{eff}} > 0$, the equilibrium is unconditionally stable. Newton's second law gives:

$$m\ddot{z} + (\rho g \pi R^2) z = 0 \implies \ddot{z} + \omega_0^2 z = 0,$$

with angular frequency:

$$\omega_0 = \sqrt{\frac{\rho g \pi R^2}{m}}.$$

Substituting $m = \rho \pi R^2 h_0$:

$$\boxed{\omega_0 = \sqrt{\frac{g}{h_0}}.}$$

Remarkably, the bobbing frequency of any floating cylinder depends solely on its submerged depth h_0, identical to a simple pendulum of length $L = h_0$!

Exercise 14.5 · ⋆⋆⋆ **Mass between two pre-tensioned transverse springs**

A particle of mass m can slide along a smooth horizontal x-axis. It is held between two identical horizontal springs perpendicular to the x-axis, each of spring constant k and natural length ℓ_0, attached to fixed posts at $(0, +d)$ and $(0, -d)$ with $d > \ell_0$ (pre-tensioned). (a) Express the total potential energy $E_p(x)$ of the two springs as a function of the transverse displacement x. (b) Expand $E_p(x)$ in powers of x up to order x^4. (c) Find the effective spring constant k_{eff} at $x = 0$ and the frequency ω_0 of small oscillations. (d) What happens if the posts are moved to $d = \ell_0$ (unstretched at $x = 0$) or $d < \ell_0$ (compressed)?

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 14.5. (a) The distance from the mass at $(x, 0)$ to either post at $(0, \pm d)$ is $\ell(x) = \sqrt{x^2 + d^2}$. The extension of each spring is $\Delta\ell = \ell(x) - \ell_0 = \sqrt{x^2 + d^2} - \ell_0$. The total potential energy of both identical springs is:

$$\boxed{E_p(x) = 2 \times \frac{1}{2} k \left(\sqrt{x^2 + d^2} - \ell_0\right)^2 = k \left(\sqrt{x^2 + d^2} - \ell_0\right)^2 .}$$

(b) For small displacements $x \ll d$:

$$\sqrt{x^2 + d^2} = d\sqrt{1 + \frac{x^2}{d^2}} \approx d \left(1 + \frac{1}{2}\frac{x^2}{d^2} - \frac{1}{8}\frac{x^4}{d^4}\right).$$

The extension is:

$$\sqrt{x^2 + d^2} - \ell_0 \approx (d - \ell_0) + \frac{x^2}{2d} - \frac{x^4}{8d^3}.$$

Squaring this:

$$\begin{aligned} E_p(x) &\approx k \left[(d - \ell_0)^2 + 2(d - \ell_0)\frac{x^2}{2d} + \frac{x^4}{4d^2} - 2(d - \ell_0)\frac{x^4}{8d^3}\right] \\ &= k(d - \ell_0)^2 + k\left(1 - \frac{\ell_0}{d}\right)x^2 + \frac{k\ell_0}{4d^3}x^4. \end{aligned}$$

(c) The linear restoring term corresponds to the coefficient of $\frac{1}{2}x^2$:

$$\frac{1}{2}k_{\text{eff}} = k\left(1 - \frac{\ell_0}{d}\right) \implies \boxed{k_{\text{eff}} = 2k\left(1 - \frac{\ell_0}{d}\right).}$$

Since $d > \ell_0$, $k_{\text{eff}} > 0$: the equilibrium at $x = 0$ is **stable**. The angular frequency of small transverse oscillations is:

$$\boxed{\omega_0 = \sqrt{\frac{2k}{m}\left(1 - \frac{\ell_0}{d}\right)}.}$$

(d) Physical discussion of regimes:

- If $d = \ell_0$ (unstretched at $x = 0$): $k_{\text{eff}} = 0$. The harmonic term vanishes! The potential becomes quartic: $E_p(x) \approx \frac{k}{4d^2}x^4$. The restoring force is non-linear ($F \propto -x^3$), and the oscillation frequency depends on amplitude!
- If $d < \ell_0$ (springs compressed at $x = 0$): $k_{\text{eff}} < 0$. The origin $x = 0$ becomes **unstable**! The compressed springs buckle to the left or right, creating two new symmetric stable equilibrium points at $x_e = \pm\sqrt{\ell_0^2 - d^2}$.

CHAPTER 15

Harmonic, Damped, and Driven Oscillations

You will learn to: solve the differential equations of free, damped, and forced harmonic oscillators; analyze underdamped, critically damped, and overdamped regimes; determine the quality factor Q and logarithmic decrement; calculate resonance frequency, amplitude, and phase lag; and study energy dissipation and resonance bandwidth.

15.1 The Simple Harmonic Oscillator

The ideal undamped harmonic oscillator obeys the linear differential equation:

$$\boxed{\ddot{x} + \omega_0^2\, x = 0,}$$

where $\omega_0 = \sqrt{k/m}$ is the **natural angular frequency**. The general solution can be written in equivalent forms:

$$\boxed{x(t) = A\cos(\omega_0 t + \phi) = C_1\cos(\omega_0 t) + C_2\sin(\omega_0 t) = \mathrm{Re}\left(\tilde{X}e^{\mathrm{i}\omega_0 t}\right),}$$

where A is the amplitude and ϕ is the initial phase determined by initial conditions:

$$A = \sqrt{x_0^2 + \left(\frac{v_0}{\omega_0}\right)^2}, \qquad \tan\phi = -\frac{v_0}{\omega_0 x_0}.$$

The period and frequency are:

$$T_0 = \frac{2\pi}{\omega_0} = 2\pi\sqrt{\frac{m}{k}}, \qquad f_0 = \frac{1}{T_0} = \frac{\omega_0}{2\pi}.$$

The total mechanical energy is conserved:

$$\boxed{E_m = \frac{1}{2}mv^2 + \frac{1}{2}kx^2 = \frac{1}{2}kA^2 = \frac{1}{2}m\omega_0^2 A^2 = \text{constant}.}$$

Over one complete cycle, kinetic and potential energies share the total energy equally on average:

$$\langle E_c\rangle = \langle E_p\rangle = \frac{1}{2}E_m.$$

15.2 Damped Harmonic Oscillations

When a viscous damping force $\vec{f} = -b\vec{v} = -b\dot{x}\hat{\imath}$ ($b > 0$) opposes motion, Newton's second law gives:

$$m\ddot{x} + b\dot{x} + kx = 0 \implies \boxed{\ddot{x} + 2\lambda\dot{x} + \omega_0^2\, x = 0,}$$

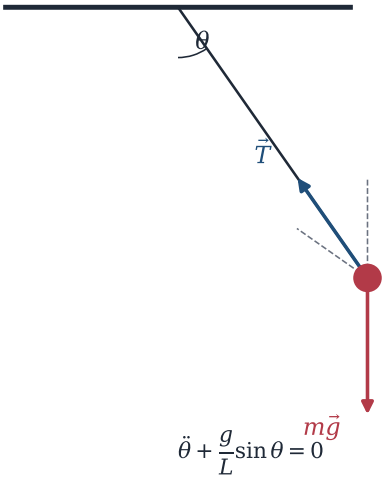


Figure 15.1: The simple pendulum of length L oscillating at angle θ from the vertical. For small angles $\theta \ll 1$, it executes simple harmonic motion with natural frequency $\omega_0 = \sqrt{g/L}$.

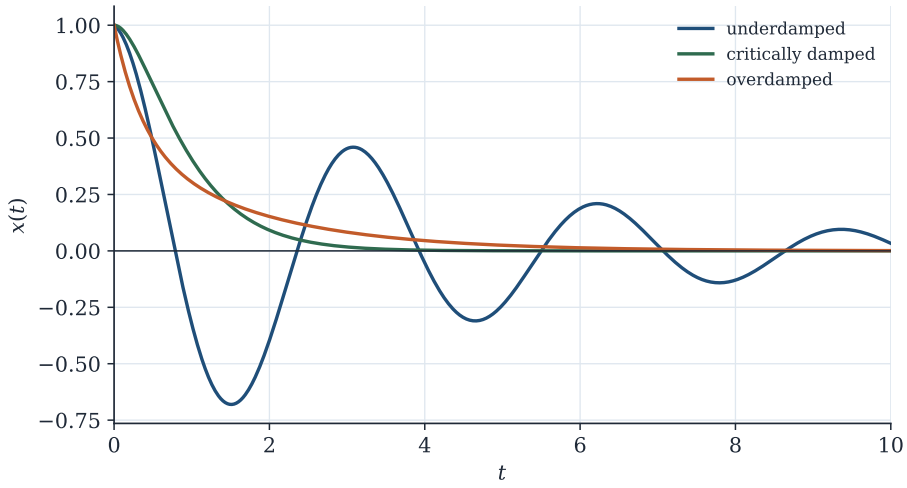


Figure 15.2: Free damped oscillations in the underdamped regime ($\lambda < \omega_0$). The displacement decays within an exponential envelope $\pm A_0 e^{-\lambda t}$.

where $\lambda = \frac{b}{2m}$ is the **damping parameter**. The characteristic equation is $r^2 + 2\lambda r + \omega_0^2 = 0$, with roots $r_{1,2} = -\lambda \pm \sqrt{\lambda^2 - \omega_0^2}$.

Three distinct physical regimes arise depending on the discriminant $\Delta = \lambda^2 - \omega_0^2$:

Regime	Condition	General Solution $x(t)$	Physical Behavior
Underdamped	$\lambda < \omega_0$	$A_0 e^{-\lambda t}\cos(\omega_d t + \phi)$	Oscillations with decaying amplitude; $\omega_d = \sqrt{\omega$
Critically Damped	$\lambda = \omega_0$	$(C_1 + C_2 t)e^{-\lambda t}$	Fastest return to equilibrium without oscillati
Overdamped	$\lambda > \omega_0$	$C_1 e^{r_1 t} + C_2 e^{r_2 t}$	Sluggish non-oscillatory exponential return

Key Parameters of Damped Systems

- **Pseudo-period**: $T_d = \frac{2\pi}{\omega_d} = \frac{2\pi}{\sqrt{\omega_0^2 - \lambda^2}} > T_0$.
- **Logarithmic decrement**: The natural logarithm of the ratio of two successive peak displacements separated by one period T_d:

$$\boxed{\delta = \ln\left(\frac{x(t)}{x(t+T_d)}\right) = \lambda T_d = \frac{2\pi\lambda}{\omega_d}.}$$

- **Quality factor** Q: Dimensionless figure of merit measuring sharpness of resonance and low dissipation:

$$\boxed{Q = \frac{\omega_0}{2\lambda} = \frac{m\omega_0}{b} \approx \frac{\pi}{\delta}.}$$

  In one cycle, the fractional energy lost is $\frac{\Delta E_m}{E_m} \approx \frac{2\pi}{Q}$.

15.3 Driven (Forced) Oscillations and Resonance

Apply an external sinusoidal driving force $F(t) = F_0\cos(\omega t)$:

$$\boxed{\ddot{x} + 2\lambda\dot{x} + \omega_0^2\, x = \frac{F_0}{m}\cos(\omega t).}$$

After transients die out, the **steady-state response** is sinusoidal at the driving frequency ω:

$$\boxed{x_p(t) = X(\omega)\cos(\omega t - \phi),}$$

where the steady-state amplitude $X(\omega)$ and phase lag $\phi(\omega)$ are:

$$\boxed{X(\omega) = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\lambda^2\omega^2}},} \qquad \boxed{\tan\phi(\omega) = \frac{2\lambda\omega}{\omega_0^2 - \omega^2}.}$$

Resonance Phenomena

- **Amplitude Resonance**: The amplitude $X(\omega)$ reaches its maximum at the **resonance**

frequency:

$$\boxed{\omega_{\text{res}} = \sqrt{\omega_0^2 - 2\lambda^2} = \omega_0\sqrt{1 - \frac{1}{2Q^2}}.}$$

(Resonance occurs only if $\lambda < \omega_0/\sqrt{2}$, i.e., $Q > 1/\sqrt{2}$).

- **Peak Amplitude**: At $\omega = \omega_{\text{res}}$:

$$X_{\max} = \frac{F_0/m}{2\lambda\sqrt{\omega_0^2 - \lambda^2}} \approx Q\left(\frac{F_0}{k}\right) \quad (\text{for } Q \gg 1).$$

  The static deflection $x_{\text{static}} = F_0/k$ is magnified by a factor of Q!

- **Velocity Resonance**: The velocity amplitude is maximized exactly at $\omega = \omega_0$, where the response is in **quadrature** ($\phi = \pi/2$), meaning the driving force is always in phase with velocity ($P(t) \geq 0$).

- **Resonance Bandwidth**: The width between half-power points ($\Delta\omega = \omega_2 - \omega_1$) satisfies:

$$\boxed{\Delta\omega = 2\lambda = \frac{\omega_0}{Q}.}$$

15.4 Worked Examples

Worked example 15.1 · Determining damping and Q from an oscilloscope trace

An underdamped mechanical oscillator of mass $m = 0.100$ kg has a natural frequency $f_0 = 5.00$ Hz. Its recorded amplitude drops from $x_0 = 4.00$ cm to $x_{10} = 1.00$ cm after $N = 10$ full oscillations. Find the logarithmic decrement δ, the damping constant λ, the quality factor Q, and the damping coefficient b.

Solution 15.1. The amplitude after N cycles decays according to $x_N = x_0 e^{-N\lambda T_d}$:

$$\frac{x_0}{x_N} = e^{N\lambda T_d} = e^{N\delta} \implies N\delta = \ln\left(\frac{x_0}{x_N}\right).$$

For $N = 10$ and $x_0/x_{10} = 4.00/1.00 = 4.00$:

$$10\delta = \ln(4.00) = 1.3863 \implies \boxed{\delta = \frac{1.3863}{10} \approx \mathbf{0.1386}.}$$

Since $\delta \ll 1$, damping is weak, so $\omega_d \approx \omega_0$:

$$T_d \approx T_0 = \frac{1}{f_0} = \frac{1}{5.00} = 0.200 \text{ s}.$$

The damping parameter is:

$$\lambda = \frac{\delta}{T_d} = \frac{0.1386}{0.200} = \mathbf{0.693} \text{ s}^{-1}.$$

The quality factor is:

$$Q = \frac{\pi}{\delta} = \frac{\pi}{0.1386} \approx \mathbf{22.7}.$$

The viscous damping coefficient is:

$$b = 2m\lambda = 2(0.100 \text{ kg})(0.693 \text{ s}^{-1}) = \mathbf{0.139} \text{ kg s}^{-1}.$$

Worked example 15.2 · Driven oscillator at resonance

A spring-mass system with $m = 0.500$ kg, $k = 200\ \text{N m}^{-1}$, and $b = 0.800\ \text{kg s}^{-1}$ is driven by an external force $F(t) = F_0 \cos(\omega t)$ with $F_0 = 4.00$ N. (a) Find the natural angular frequency ω_0, damping factor λ, and quality factor Q. (b) Find the resonance frequency ω_{res} and the amplitude at resonance $X_{\max}$. (c) Compute the average power supplied by the driver at velocity resonance ($\omega = \omega_0$).

Solution 15.2. (a) Fundamental parameters:

$$\omega_0 = \sqrt{\frac{k}{m}} = \sqrt{\frac{200}{0.500}} = \mathbf{20.0}\ \text{rad s}^{-1}.$$

$$\lambda = \frac{b}{2m} = \frac{0.800}{2(0.500)} = \mathbf{0.800}\ \text{s}^{-1}.$$

$$Q = \frac{\omega_0}{2\lambda} = \frac{20.0}{2(0.800)} = \mathbf{12.5}.$$

(b) Resonance frequency:

$$\omega_{\text{res}} = \sqrt{\omega_0^2 - 2\lambda^2} = \sqrt{20.0^2 - 2(0.800)^2} = \sqrt{400 - 1.28} = \sqrt{398.72} \approx \mathbf{19.97}\ \text{rad s}^{-1}.$$

The amplitude at resonance is:

$$X_{\max} = \frac{F_0/m}{2\lambda\sqrt{\omega_0^2 - \lambda^2}} = \frac{4.00/0.500}{2(0.800)\sqrt{400 - 0.64}} = \frac{8.00}{1.60 \times 19.98} = \frac{8.00}{31.97} \approx \mathbf{0.250}\ \text{m} = 25.0\ \text{cm}.$$

Notice that static deflection is $x_{\text{st}} = F_0/k = 4.00/200 = 0.020$ m $= 2.0$ cm. The resonance magnifies the motion by a factor of $X_{\max}/x_{\text{st}} = 25.0/2.0 = 12.5 = Q$!
(c) At $\omega = \omega_0 = 20.0\ \text{rad s}^{-1}$, the force and velocity are in phase ($\phi = \pi/2$). The velocity amplitude is $V_0 = \omega_0 X(\omega_0) = \frac{F_0}{b}$:

$$V_0 = \frac{4.00}{0.800} = 5.00\ \text{m s}^{-1}.$$

The average power supplied by the driver (which equals the power dissipated by friction) is:

$$\langle P \rangle = \frac{1}{2} F_0 V_0 = \frac{1}{2}(4.00\ \text{N})(5.00\ \text{m s}^{-1}) = \mathbf{10.0}\ \text{W}.$$

15.5 Exercises and Step-by-Step Solutions

Exercise 15.1 · ⋆ General solution from initial conditions

A horizontal spring-mass oscillator has mass $m = 0.200$ kg and spring stiffness $k = 80.0\ \text{N m}^{-1}$. At $t = 0$, the mass is displaced to $x_0 = +5.00$ cm $= 0.0500$ m and given an initial velocity $v_0 = -0.600\ \text{m s}^{-1}$ directed toward the origin. (a) Find the natural angular frequency ω_0 and the period T_0. (b) Write the position $x(t) = A\cos(\omega_0 t + \phi)$, finding A and ϕ. (c) Calculate the total mechanical energy E_m.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 15.1. (a) Angular frequency:

$$\omega_0 = \sqrt{\frac{k}{m}} = \sqrt{\frac{80.0}{0.200}} = \sqrt{400} = \mathbf{20.0}\ \text{rad}\,\text{s}^{-1}.$$

Period:

$$T_0 = \frac{2\pi}{\omega_0} = \frac{2\pi}{20.0} = \frac{\pi}{10} \approx \mathbf{0.314}\ \text{s}.$$

(b) Amplitude:

$$A = \sqrt{x_0^2 + \left(\frac{v_0}{\omega_0}\right)^2} = \sqrt{(0.0500)^2 + \left(\frac{-0.600}{20.0}\right)^2} = \sqrt{0.00250 + (-0.0300)^2} = \sqrt{0.00250 + 0.00090} = \sqrt{0.0034}$$

Phase angle ϕ:

$$\tan\phi = -\frac{v_0}{\omega_0 x_0} = -\frac{-0.600}{20.0 \times 0.0500} = +\frac{0.600}{1.000} = +0.600.$$

Since $x_0 > 0$ and $\cos\phi = x_0/A > 0$, ϕ is in the first quadrant:

$$\phi = \arctan(0.600) \approx 0.540\ \text{rad} \approx 31.0^\circ.$$

The full trajectory is:

$$\boxed{x(t) = 0.0583\cos(20.0t + 0.540)\ \text{m}.}$$

(c) Mechanical energy:

$$E_m = \frac{1}{2}kA^2 = \frac{1}{2}(80.0)(0.0583)^2 = 40.0 \times 0.00340 = \mathbf{0.136}\ \text{J}.$$

Check with initial values: $E_m = \frac{1}{2}kx_0^2 + \frac{1}{2}mv_0^2 = \frac{1}{2}(80.0)(0.05)^2 + \frac{1}{2}(0.20)(-0.6)^2 = 0.100 + 0.036 = 0.136$ J (exact agreement!).

Exercise 15.2 · ⋆ **The Virial Theorem for harmonic oscillations**

For an undamped harmonic oscillator $x(t) = A\cos(\omega_0 t + \phi)$: (a) Write the instantaneous kinetic energy $E_c(t)$ and potential energy $E_p(t)$. (b) Calculate the time-average of both energies over one complete period T_0:

$$\langle E_c\rangle = \frac{1}{T_0}\int_0^{T_0} E_c(t)\mathrm{d}t, \qquad \langle E_p\rangle = \frac{1}{T_0}\int_0^{T_0} E_p(t)\mathrm{d}t.$$

(c) Verify that $\langle E_c\rangle = \langle E_p\rangle = \frac{1}{2}E_m$.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 15.2. (a) The position and velocity are:

$$x(t) = A\cos(\omega_0 t + \phi), \qquad v(t) = \dot{x}(t) = -\omega_0 A\sin(\omega_0 t + \phi).$$

Instantaneous energies:

$$E_p(t) = \frac{1}{2}kx^2(t) = \frac{1}{2}kA^2\cos^2(\omega_0 t + \phi),$$
$$E_c(t) = \frac{1}{2}mv^2(t) = \frac{1}{2}m\omega_0^2 A^2\sin^2(\omega_0 t + \phi) = \frac{1}{2}kA^2\sin^2(\omega_0 t + \phi).$$

(b) Using the standard trigonometric integral averages:

$$\frac{1}{T_0}\int_0^{T_0}\cos^2(\omega_0 t + \phi)\mathrm{d}t = \frac{1}{2}, \qquad \frac{1}{T_0}\int_0^{T_0}\sin^2(\omega_0 t + \phi)\mathrm{d}t = \frac{1}{2}.$$

Therefore:

$$\langle E_p\rangle = \frac{1}{2}kA^2\left(\frac{1}{2}\right) = \mathbf{\frac{1}{4}kA^2},$$
$$\langle E_c\rangle = \frac{1}{2}kA^2\left(\frac{1}{2}\right) = \mathbf{\frac{1}{4}kA^2}.$$

(c) The total energy is $E_m = \frac{1}{2}kA^2$. Comparing:

$$\boxed{\langle E_c\rangle = \langle E_p\rangle = \frac{1}{2}E_m.}$$

This equipartition of energy on average between kinetic and potential forms is a fundamental instance of the Virial Theorem for quadratic potentials ($V \propto r^2 \implies 2\langle E_c\rangle = 2\langle E_p\rangle$).

Exercise 15.3 · Critical damping for an automobile suspension

A car wheel and suspension can be modeled as a quarter-car of mass $m = 350$ kg supported by a spring of stiffness $k = 35.0\ \mathrm{kN\,m^{-1}} = 35000\ \mathrm{N\,m^{-1}}$ and a shock absorber (dashpot) of damping coefficient b. (a) Find the natural angular frequency ω_0 and the undamped oscillation frequency f_0. (b) Calculate the critical damping coefficient b_{crit} required for the suspension to return to equilibrium as quickly as possible without bouncing. (c) If the shock absorber wears out and its damping drops to $b = 0.25\,b_{\mathrm{crit}}$, find the pseudo-period T_d and the fraction of amplitude remaining after one bounce.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 15.3. (a) Natural angular frequency:

$$\omega_0 = \sqrt{\frac{k}{m}} = \sqrt{\frac{35000}{350}} = \sqrt{100} = \mathbf{10.0}\ \text{rad}\,\text{s}^{-1}.$$

Frequency:

$$f_0 = \frac{\omega_0}{2\pi} = \frac{10.0}{2\pi} \approx \mathbf{1.59}\ \text{Hz}.$$

(b) Critical damping occurs when $\lambda = \omega_0$:

$$\frac{b_{\text{crit}}}{2m} = \omega_0 \implies \boxed{b_{\text{crit}} = 2m\omega_0 = 2\sqrt{km}.}$$

Computing:

$$b_{\text{crit}} = 2(350)(10.0) = \mathbf{7000}\ \text{N}\,\text{s}\,\text{m}^{-1} = 7.00 \times 10^3\ \text{kg}\,\text{s}^{-1}.$$

(c) Worn shock absorber: $b = 0.25\,b_{\text{crit}} \implies \lambda = 0.25\,\omega_0 = 2.50\ \text{s}^{-1}$. The pseudo-frequency is:

$$\omega_d = \sqrt{\omega_0^2 - \lambda^2} = \sqrt{10.0^2 - 2.5^2} = \sqrt{100 - 6.25} = \sqrt{93.75} \approx 9.68\ \text{rad}\,\text{s}^{-1}.$$

The pseudo-period is:

$$T_d = \frac{2\pi}{\omega_d} = \frac{2\pi}{9.68} \approx \mathbf{0.649}\ \text{s}.$$

The decay per period is given by the factor $e^{-\lambda T_d}$:

$$\lambda T_d = (2.50)(0.649) \approx 1.623 \implies \frac{A(t+T_d)}{A(t)} = e^{-1.623} \approx \mathbf{0.197} \approx 19.7\%.$$

The car will noticeably bounce after hitting a pothole, retaining nearly 20% of its displacement on the rebound!

Exercise 15.4 · Oscillations in a U-tube manometer

A uniform glass U-tube of internal cross-sectional area S contains a column of liquid of density ρ and total column length L. The liquid is displaced by a distance z_0 in one arm and released. (a) Neglecting viscosity, show that the column executes simple harmonic motion with equation:

$$\ddot{z} + \frac{2g}{L} z = 0.$$

(b) Find the period of oscillation T. Compare it with a simple pendulum. (c) Liquid viscosity introduces a Poiseuille laminar drag force $F_d = -8\pi\eta L\dot{z}$, where η is the dynamic viscosity.

Write the damped differential equation and determine the condition on η for the motion to remain underdamped.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 15.4. (a) The total mass of the liquid column is $m = \rho SL$. When the liquid level rises by z in the right arm, it drops by z in the left arm. The height difference between the two columns is $2z$. The unbalanced weight of this excess column provides the restoring force:

$$F_{\text{rest}} = -(\text{mass of excess column})g = -(\rho S \cdot 2z)g = -2\rho g S z.$$

Applying Newton's second law $m\ddot{z} = F_{\text{rest}}$:

$$(\rho SL)\ddot{z} = -2\rho g S z \implies \rho SL\ddot{z} + 2\rho g S z = 0.$$

Dividing by ρSL:

$$\boxed{\ddot{z} + \frac{2g}{L} z = 0.}$$

(b) The natural angular frequency is $\omega_0 = \sqrt{\frac{2g}{L}}$. The period of oscillation is:

$$\boxed{T = 2\pi\sqrt{\frac{L}{2g}}.}$$

This is identical to the period of a simple pendulum of length $L_{\text{eff}} = \frac{L}{2}$!
(c) Adding the laminar viscous friction force $F_d = -b\dot{z}$ with $b = 8\pi\eta L$:

$$m\ddot{z} + b\dot{z} + (2\rho g S)z = 0 \implies \ddot{z} + \frac{8\pi\eta}{\rho S}\dot{z} + \frac{2g}{L} z = 0.$$

The damping parameter is $\lambda = \frac{b}{2m} = \frac{8\pi\eta L}{2\rho SL} = \frac{4\pi\eta}{\rho S}$. For the motion to remain **underdamped** ($\lambda < \omega_0$):

$$\frac{4\pi\eta}{\rho S} < \sqrt{\frac{2g}{L}} \implies \boxed{\eta < \frac{\rho S}{4\pi}\sqrt{\frac{2g}{L}}.}$$

Exercise 15.5 · ★★★ Two coupled oscillators and normal modes

Two identical carts, each of mass m, can move along a frictionless horizontal track. Each cart is connected to a rigid wall by a spring of stiffness k, and the two carts are coupled to each other

by a central spring of stiffness k_c. (a) Write Newton's second law for displacements $x_1(t)$ and $x_2(t)$ from equilibrium. (b) Define the **normal coordinates** $u(t) = x_1 + x_2$ (symmetric mode) and $v(t) = x_1 - x_2$ (antisymmetric mode). Show that they decouple into two independent harmonic oscillators. (c) Find the two normal frequencies ω_1 and ω_2, and describe the physical motion in each mode.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 15.5. (a) The forces on each cart are:

$$m\ddot{x}_1 = -kx_1 + k_c(x_2 - x_1) = -(k + k_c)x_1 + k_c x_2,$$
$$m\ddot{x}_2 = -kx_2 - k_c(x_2 - x_1) = k_c x_1 - (k + k_c)x_2.$$

(b) Adding the two equations:

$$m(\ddot{x}_1 + \ddot{x}_2) = -k(x_1 + x_2) \implies m\ddot{u} + ku = 0 \implies \boxed{\ddot{u} + \omega_1^2 u = 0,}$$

where $\omega_1 = \sqrt{\frac{k}{m}}$.
Subtracting the two equations:

$$m(\ddot{x}_1 - \ddot{x}_2) = -(k + 2k_c)(x_1 - x_2) \implies m\ddot{v} + (k + 2k_c)v = 0 \implies \boxed{\ddot{v} + \omega_2^2 v = 0,}$$

where $\omega_2 = \sqrt{\frac{k+2k_c}{m}}$. The normal coordinates $u(t)$ and $v(t)$ are completely decoupled!
(c) Physical description of the normal modes:

- **Mode 1 (Symmetric / In-phase)**: $v = 0 \implies x_1(t) = x_2(t)$. Frequency: $\boxed{\omega_1 = \sqrt{\frac{k}{m}}}$.
  Both carts move synchronously in the same direction with identical amplitude. The coupling spring maintains constant length ($x_2 - x_1 = 0$) and exerts no force; the carts oscillate as if the central spring did not exist!
- **Mode 2 (Antisymmetric / Out-of-phase)**: $u = 0 \implies x_1(t) = -x_2(t)$. Frequency: $\boxed{\omega_2 = \sqrt{\frac{k + 2k_c}{m}}}$. The carts move in opposite directions, alternately compressing and expanding the central spring. The restoring force is stronger, so $\omega_2 > \omega_1$.

Any general motion is a linear superposition $x_1(t) = \frac{1}{2}(u(t) + v(t))$ and $x_2(t) = \frac{1}{2}(u(t) - v(t))$, giving rise to the phenomenon of **beats** when $k_c \ll k$.

PART V

Mastery

CHAPTER 16

Systems of Particles, Collisions, and Synoptic Mastery

You will learn to: locate the center of mass of discrete systems; apply the momentum conservation theorem to isolated systems; solve 1D and 2D elastic and inelastic collisions using the coefficient of restitution; transform between laboratory and center-of-mass frames using Koenig's theorems and reduced mass μ; derive the Tsiolkovsky rocket equation; and tackle comprehensive multi-stage exam synthesis problems.

16.1 Center of Mass and Total Momentum

For a system of N material points with masses m_i and positions $\vec{r}_i$:

- The **total mass** is $M = \sum_{i=1}^{N} m_i$.
- The **center of mass** (CM) position G is defined by:

$$\boxed{\vec{r}_G = \frac{1}{M}\sum_{i=1}^{N} m_i \vec{r}_i.}$$

- The **total linear momentum** is:

$$\boxed{\vec{P}_{\text{tot}} = \sum_{i=1}^{N} m_i \vec{v}_i = M\vec{v}_G.}$$

Differentiating with respect to time and separating forces into external forces $\vec{F}_{i,\text{ext}}$ and internal mutual forces $\vec{F}_{j\to i}$:

$$\frac{\mathrm{d}\vec{P}_{\text{tot}}}{\mathrm{d}t} = \sum_{i=1}^{N} m_i \vec{a}_i = \sum_{i=1}^{N} \vec{F}_{i,\text{ext}} + \sum_{i=1}^{N}\sum_{j\neq i}^{N} \vec{F}_{j\to i}.$$

By Newton's third law (action-reaction), internal forces cancel in pairs ($\vec{F}_{j\to i} + \vec{F}_{i\to j} = \vec{0}$). Hence, we obtain the **Fundamental Theorem of Linear Momentum for a System**:

$$\boxed{\frac{\mathrm{d}\vec{P}_{\text{tot}}}{\mathrm{d}t} = M\vec{a}_G = \sum \vec{F}_{\text{ext}}.}$$

Conservation of Total Momentum

If a system is isolated or subject to zero net external force ($\sum \vec{F}_{\text{ext}} = \vec{0}$):

$$\boxed{\vec{P}_{\text{tot}} = \text{constant vector} \iff \vec{v}_G = \text{constant velocity.}}$$

The center of mass moves in uniform rectilinear motion, completely unaffected by any internal

explosions, collisions, or interactions!

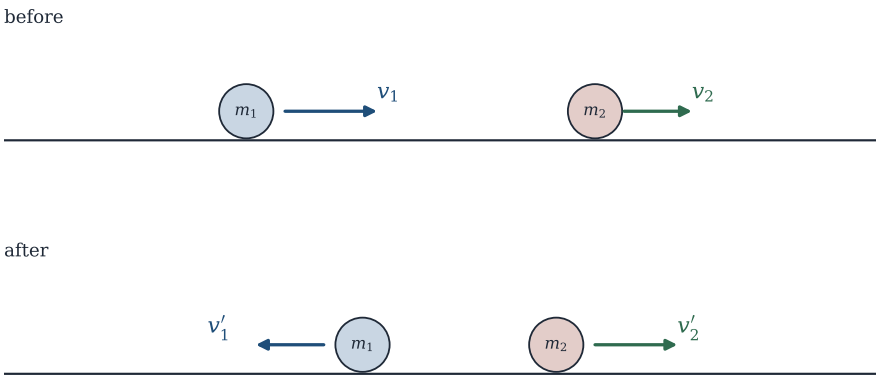


Figure 16.1: Two-particle collision in the laboratory frame. Total linear momentum $\vec{P}_{\text{tot}} = m_1\vec{v}_1 + m_2\vec{v}_2$ is conserved during the collision.

16.2 Classification of Collisions in 1D

In a brief collision between two bodies (m_1, m_2), the impact forces are so enormous that external forces (such as gravity or friction) are negligible during the collision duration $\Delta t \to 0$ (**impulsive approximation**). Total momentum is always conserved:

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}.$$

The elasticity of the collision is characterized by Newton's **coefficient of restitution** e:

$$\boxed{e = -\frac{v_{1f} - v_{2f}}{v_{1i} - v_{2i}} = \frac{v_{\text{rel, final}}}{v_{\text{rel, initial}}} \in [0, 1].}$$

Collision Type	Restitution e	Kinetic Energy ΔE_c	Outcome
Perfectly Elastic	$e = 1$	$\Delta E_c = 0$ (conserved)	$v_{1f} - v_{2f} = -(v_{1i} - v_{2i})$
Inelastic	$0 < e < 1$	$\Delta E_c < 0$ (partially lost)	Bodies bounce apart with reduced relative s
Perfectly Inelastic	$e = 0$	Maximum energy loss	Bodies stick together: $v_{1f} = v_{2f} = v_G$

For a 1D collision with known e, the final velocities are uniquely determined:

$$\boxed{v_{1f} = v_G - \frac{m_2}{m_1 + m_2} e(v_{1i} - v_{2i}), \qquad v_{2f} = v_G + \frac{m_1}{m_1 + m_2} e(v_{1i} - v_{2i}).}$$

16.3 Center-of-Mass Frame and Koenig's First Theorem

Let $\mathcal{R}^*$ be the center-of-mass frame (translating with velocity $\vec{v}_G$). In this frame, the total linear momentum is identically zero: $\vec{P}^* = \vec{0}$. The velocities in the laboratory and CM frames are related by $\vec{v}_i = \vec{v}_G + \vec{v}_i^*$. The total kinetic energy in the laboratory frame satisfies **Koenig's First Theorem**:

$$\boxed{E_c = \frac{1}{2} M v_G^2 + E_c^*,}$$

where for a two-body system, the internal kinetic energy in the CM frame simplifies to:

$$\boxed{E_c^* = \frac{1}{2}\mu\, v_{\text{rel}}^2, \quad \text{with reduced mass } \mu = \frac{m_1 m_2}{m_1 + m_2}.}$$

In a collision, the kinetic energy of the center of mass $\frac{1}{2}Mv_G^2$ can never be lost because $\vec{v}_G$ is constant! The maximum possible kinetic energy that can be converted into heat/deformation (in a totally inelastic collision) is precisely $E_c^* = \frac{1}{2}\mu v_{\text{rel}}^2$.

16.4 Worked Examples

Worked example 16.1 · The ballistic pendulum

A wooden block of mass $M = 1.98$ kg hangs from light strings of length $L = 1.50$ m. A bullet of mass $m = 20.0$ g $= 0.0200$ kg is fired horizontally into the block and embeds itself inside it. After impact, the block and bullet swing upward together to a maximum height $h = 8.50$ cm $= 0.0850$ m. Find the initial speed v_0 of the bullet and the mechanical energy lost in the collision.

Solution 16.1. The problem consists of two distinct stages:

1. **Stage 1: Inelastic impact (duration $\Delta t \to 0$).** External forces (gravity and string tension) do not have time to act horizontally. Total horizontal momentum is conserved:

$$mv_0 = (M+m)V \implies V = \frac{m}{M+m}v_0,$$

   where V is the speed of the combined mass immediately after impact.

2. **Stage 2: Conservative upward swing.** After the collision, the tension in the strings does no work ($\vec{T} \perp \vec{v}$), and only gravity acts. Mechanical energy is conserved:

$$\frac{1}{2}(M+m)V^2 = (M+m)gh \implies V = \sqrt{2gh}.$$

Equating the two expressions for V:

$$\frac{m}{M+m}v_0 = \sqrt{2gh} \implies \boxed{v_0 = \frac{M+m}{m}\sqrt{2gh}.}$$

Computing numerically:

$$\sqrt{2gh} = \sqrt{2(9.81)(0.0850)} = \sqrt{1.6677} \approx 1.291 \text{ m s}^{-1}.$$

Mass ratio: $\frac{M+m}{m} = \frac{1.98+0.0200}{0.0200} = \frac{2.00}{0.0200} = 100$. The initial bullet speed is:

$$v_0 = 100 \times 1.291 = \mathbf{129.1} \text{ m s}^{-1}.$$

Energy loss:

$$E_{c,\text{initial}} = \frac{1}{2}mv_0^2 = \frac{1}{2}(0.0200)(129.1)^2 \approx 166.7 \text{ J},$$
$$E_{c,\text{after impact}} = \frac{1}{2}(M+m)V^2 = \frac{1}{2}(2.00)(1.291)^2 \approx 1.67 \text{ J}.$$

The fraction of energy retained is only $\frac{m}{M+m} = \frac{1}{100} = 1\%$. The mechanical energy lost is $\Delta E = 166.7 - 1.67 = \mathbf{165.0}$ J (99% converted into heat, sound, and plastic deformation of the wood).

Worked example 16.2 · Glancing 2D elastic collision of equal masses

In a nuclear physics experiment, a proton of mass m moving with speed v_0 collides elastically with another proton of equal mass m initially at rest. Prove that after the collision, the two protons always fly apart at a right angle (90°), provided the collision is not head-on.

Solution 16.2. Let $\vec{v}_1$ and $\vec{v}_2$ be the final velocity vectors of the two protons after the collision.

- Conservation of linear momentum:

$$m\vec{v}_0 = m\vec{v}_1 + m\vec{v}_2 \implies \vec{v}_0 = \vec{v}_1 + \vec{v}_2.$$

- Conservation of kinetic energy (elastic collision):

$$\frac{1}{2}mv_0^2 = \frac{1}{2}mv_1^2 + \frac{1}{2}mv_2^2 \implies v_0^2 = v_1^2 + v_2^2.$$

Now take the dot product of the momentum vector equation with itself:

$$\vec{v}_0 \cdot \vec{v}_0 = (\vec{v}_1 + \vec{v}_2) \cdot (\vec{v}_1 + \vec{v}_2) \implies v_0^2 = v_1^2 + v_2^2 + 2\,\vec{v}_1 \cdot \vec{v}_2.$$

Subtracting the energy conservation equation ($v_0^2 = v_1^2 + v_2^2$):

$$0 = 2\,\vec{v}_1 \cdot \vec{v}_2 \implies \boxed{\vec{v}_1 \cdot \vec{v}_2 = 0.}$$

Since neither proton is at rest ($\vec{v}_1 \neq \vec{0}, \vec{v}_2 \neq \vec{0}$), their final velocity vectors are strictly orthogonal:

$$\theta_1 + \theta_2 = \mathbf{90.0}^\circ.$$

This iconic result is routinely observed in bubble chamber and cloud chamber photographs of proton-proton collisions.

16.5 Exercises and Step-by-Step Solutions

Exercise 16.1 · ⋆ Center of mass of a three-particle system

Three masses lie in the xy-plane: $m_1 = 1.00$ kg at $(1.0, 2.0)$ m, $m_2 = 2.00$ kg at $(4.0, 1.0)$ m, and $m_3 = 3.00$ kg at $(2.0, 5.0)$ m. (a) Calculate the coordinates (x_G, y_G) of the center of mass. (b) If particle m_1 moves with velocity $\vec{v}_1 = 6.0\,\hat{\imath}\ \mathrm{m\,s^{-1}}$ while m_2 and m_3 are stationary, find the velocity $\vec{v}_G$ of the center of mass.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 16.1. (a) Total mass: $M = m_1 + m_2 + m_3 = 1.00 + 2.00 + 3.00 = 6.00$ kg. Center of mass coordinates:

$$x_G = \frac{m_1 x_1 + m_2 x_2 + m_3 x_3}{M} = \frac{1.00(1.0) + 2.00(4.0) + 3.00(2.0)}{6.00} = \frac{1.0 + 8.0 + 6.0}{6.00} = \frac{15.0}{6.00} = \mathbf{2.50}\ \text{m}.$$

$$y_G = \frac{m_1 y_1 + m_2 y_2 + m_3 y_3}{M} = \frac{1.00(2.0) + 2.00(1.0) + 3.00(5.0)}{6.00} = \frac{2.0 + 2.0 + 15.0}{6.00} = \frac{19.0}{6.00} \approx \mathbf{3.17}\ \text{m}.$$

Thus $\vec{r}_G = 2.50\,\hat{\imath} + 3.17\,\hat{\jmath}$ m.
(b) Total linear momentum:

$$\vec{P}_{\text{tot}} = m_1\vec{v}_1 + m_2\vec{v}_2 + m_3\vec{v}_3 = (1.00)(6.0\,\hat{\imath}) + \vec{0} + \vec{0} = 6.0\,\hat{\imath}\ \text{kg m s}^{-1}.$$

The center of mass velocity is:

$$\vec{v}_G = \frac{\vec{P}_{\text{tot}}}{M} = \frac{6.0\,\hat{\imath}}{6.00} = \mathbf{1.00}\,\hat{\imath}\ \text{m s}^{-1}.$$

Exercise 16.2 · ⋆ 1D elastic collision and neutron moderation

In a nuclear reactor, fast fission neutrons of mass $m_1 \approx 1$ u moving at initial speed v_0 must be slowed down by head-on elastic collisions with stationary moderator nuclei of mass m_2. (a) For a head-on elastic collision with m_2 at rest, show that the final speed of the neutron is:

$$v_{1f} = \frac{m_1 - m_2}{m_1 + m_2} v_0.$$

(b) Show that the fraction of kinetic energy transferred to the moderator nucleus is:

$$\frac{\Delta E_c}{E_{c0}} = \frac{4m_1 m_2}{(m_1 + m_2)^2}.$$

(c) Evaluate this energy transfer fraction for a collision with ordinary hydrogen ($m_2 = 1$ u), carbon ($m_2 = 12$ u), and lead ($m_2 = 207$ u). Explain why water is an excellent neutron moderator while heavy metals are ineffective.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 16.2. (a) For an elastic collision ($e = 1$) with $v_{2i} = 0$:

Momentum: $m_1 v_0 = m_1 v_{1f} + m_2 v_{2f}$, Restitution: $v_{2f} - v_{1f} = v_0 \implies v_{2f} = v_0 + v_{1f}$.

Substituting into the momentum equation:

$$m_1 v_0 = m_1 v_{1f} + m_2(v_0 + v_{1f}) = (m_1 + m_2)v_{1f} + m_2 v_0.$$

Rearranging:

$$(m_1 - m_2)v_0 = (m_1 + m_2)v_{1f} \implies \boxed{v_{1f} = \frac{m_1 - m_2}{m_1 + m_2} v_0.}$$

(b) The final kinetic energy of the neutron is:

$$E_{c1,f} = \frac{1}{2} m_1 v_{1f}^2 = \frac{1}{2} m_1 v_0^2 \left(\frac{m_1 - m_2}{m_1 + m_2}\right)^2 = E_{c0} \left(\frac{m_1 - m_2}{m_1 + m_2}\right)^2.$$

The kinetic energy transferred to the moderator nucleus is:

$$\frac{\Delta E_c}{E_{c0}} = 1 - \left(\frac{m_1 - m_2}{m_1 + m_2}\right)^2 = \frac{(m_1 + m_2)^2 - (m_1 - m_2)^2}{(m_1 + m_2)^2} = \boxed{\frac{4m_1 m_2}{(m_1 + m_2)^2}.}$$

(c) Numerical evaluation:

- Hydrogen ($m_2 = 1$ u): $\frac{4(1)(1)}{(1+1)^2} = \frac{4}{4} = \mathbf{1.00} = \mathbf{100}\%$. In a head-on collision with a proton, the neutron transfers 100% of its kinetic energy and stops completely!
- Carbon-12 ($m_2 = 12$ u): $\frac{4(1)(12)}{(1+12)^2} = \frac{48}{169} \approx \mathbf{0.284} = \mathbf{28.4}\%$.
- Lead-207 ($m_2 = 207$ u): $\frac{4(1)(207)}{(1+207)^2} = \frac{828}{(208)^2} = \frac{828}{43264} \approx \mathbf{0.019} = \mathbf{1.9}\%$.

Because hydrogen nuclei (protons) match the neutron mass almost exactly, water (H_2O) slows down neutrons rapidly in very few collisions. Heavy nuclei like lead barely recoil, bouncing the neutron elastically with almost zero energy loss!

Exercise 16.3 · Bouncing ball with coefficient of restitution e

A rubber ball is dropped from rest from an initial height h_0 onto a rigid horizontal floor. The collision has a constant coefficient of restitution $e \in (0, 1)$. (a) Find the height h_n reached after the n-th bounce. (b) Find the time interval t_n between the $(n-1)$-th bounce and the n-th bounce. (c) Sum the infinite geometric series to compute the total time T_{total} until the ball stops bouncing, and the total distance D_{total} traveled.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 16.3. (a) The impact speed before the first bounce is $v_1 = \sqrt{2gh_0}$. After rebound, its speed is $u_1 = ev_1$. The height reached after the first rebound is $h_1 = \frac{u_1^2}{2g} = \frac{e^2 v_1^2}{2g} = e^2 h_0$. By induction, after n rebounds:

$$\boxed{h_n = e^{2n} h_0.}$$

(b) Time to fall from h_0 to the first bounce: $t_0 = \sqrt{\frac{2h_0}{g}}$. For each subsequent bounce $n \geq 1$, the ball goes up to height h_n and falls back down, so the time between bounces is:

$$t_n = 2\sqrt{\frac{2h_n}{g}} = 2\sqrt{\frac{2(e^{2n}h_0)}{g}} = 2e^n\sqrt{\frac{2h_0}{g}} = 2e^n t_0.$$

(c) Total time to stop bouncing:

$$T_{\text{total}} = t_0 + \sum_{n=1}^{\infty} t_n = t_0 + 2t_0\sum_{n=1}^{\infty} e^n = t_0\left(1 + 2\frac{e}{1-e}\right) = t_0\left(\frac{1-e+2e}{1-e}\right) = \boxed{\sqrt{\frac{2h_0}{g}}\left(\frac{1+e}{1-e}\right).}$$

Remarkably, although the ball undergoes infinitely many bounces, the total time is **finite**! Total distance traveled:

$$D_{\text{total}} = h_0 + 2\sum_{n=1}^{\infty} h_n = h_0 + 2h_0\sum_{n=1}^{\infty} e^{2n} = h_0\left(1 + \frac{2e^2}{1-e^2}\right) = \boxed{h_0\left(\frac{1+e^2}{1-e^2}\right).}$$

Exercise 16.4 · The Tsiolkovsky Rocket Equation

A rocket of total initial mass m_0 in deep space (zero gravity and zero drag) accelerates by ejecting burned propellant backward at a constant speed u_{ex} relative to the rocket. (a) In an inertial frame where the rocket has velocity v, analyze the momentum balance between time t and $t + \mathrm{d}t$ when mass $-\mathrm{d}m$ ($\mathrm{d}m < 0$) is ejected at speed $v - u_{\text{ex}}$ to derive the differential equation:

$$m\,\mathrm{d}v = -u_{\text{ex}}\,\mathrm{d}m.$$

(b) Integrate this equation to derive the famous **Tsiolkovsky Rocket Equation**:

$$\Delta v = v_f - v_0 = u_{\text{ex}}\ln\left(\frac{m_0}{m_f}\right).$$

(c) A rocket has an exhaust velocity $u_{\text{ex}} = 3000\ \mathrm{m\,s^{-1}}$. What mass ratio m_0/m_f is required to achieve Earth orbital velocity $\Delta v = 7.80\ \mathrm{km\,s^{-1}}$?

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 16.4. (a) At time t: the rocket and remaining fuel have mass m and velocity v. The total momentum is:

$$P(t) = mv.$$

At time $t + \mathrm{d}t$: fuel of mass $\mathrm{d}m_{\text{fuel}} = -\mathrm{d}m > 0$ has been expelled backward with velocity $v - u_{\text{ex}}$ relative to the inertial frame. The rocket has mass $m + \mathrm{d}m$ and velocity $v + \mathrm{d}v$. The momentum is:

$$P(t+\mathrm{d}t) = (m+\mathrm{d}m)(v+\mathrm{d}v) + (-\mathrm{d}m)(v-u_{\text{ex}}) = mv + m\,\mathrm{d}v + v\,\mathrm{d}m + \mathrm{d}m\mathrm{d}v - v\,\mathrm{d}m + u_{\text{ex}}\,\mathrm{d}m.$$

Dropping the second-order differential $\mathrm{d}m\mathrm{d}v$:

$$P(t + \mathrm{d}t) = mv + m\,\mathrm{d}v + u_{\text{ex}}\,\mathrm{d}m.$$

Since no external force acts ($F_{\text{ext}} = 0$):

$$P(t + \mathrm{d}t) - P(t) = 0 \implies m\,\mathrm{d}v + u_{\text{ex}}\,\mathrm{d}m = 0 \implies \boxed{m\,\mathrm{d}v = -u_{\text{ex}}\,\mathrm{d}m.}$$

(b) Dividing by m:

$$\mathrm{d}v = -u_{\text{ex}}\frac{\mathrm{d}m}{m}.$$

Integrating from initial state (m_0, v_0) to final empty propellant burnout state (m_f, v_f):

$$\int_{v_0}^{v_f} \mathrm{d}v = -u_{\text{ex}} \int_{m_0}^{m_f} \frac{\mathrm{d}m}{m} \implies v_f - v_0 = -u_{\text{ex}} \left[\ln m_f - \ln m_0\right].$$

Using $\ln(m_f/m_0) = -\ln(m_0/m_f)$:

$$\boxed{\Delta v = u_{\text{ex}} \ln\left(\frac{m_0}{m_f}\right).}$$

(c) Given $\Delta v = 7800\ \mathrm{m\,s^{-1}}$ and $u_{\text{ex}} = 3000\ \mathrm{m\,s^{-1}}$:

$$\ln\left(\frac{m_0}{m_f}\right) = \frac{\Delta v}{u_{\text{ex}}} = \frac{7800}{3000} = 2.60.$$

Taking the exponential:

$$\frac{m_0}{m_f} = e^{2.60} \approx \mathbf{13.46}.$$

This means the fuel must constitute $\frac{13.46-1}{13.46} \approx \mathbf{92.6}\%$ of the rocket's total launch mass! This exponential dependence explains why multi-stage rockets are essential for space exploration.

Exercise 16.5 · ★★★ Synoptic Problem: The Spring Launcher and Vertical Loop

A block of mass $m = 0.500$ kg is pushed against a horizontal spring of stiffness $k = 800\ \mathrm{N\,m^{-1}}$, compressing it by $\Delta x = 0.200$ m. Upon release, the block moves along a smooth horizontal track, crosses a rough horizontal patch of length $d = 0.800$ m with kinetic friction coefficient $\mu_k = 0.250$, and then enters a smooth vertical circular loop of radius $R = 0.500$ m. (a) Find the speed v_1 of the block immediately after separating from the spring. (b) Find the speed v_2 after traversing the rough patch. (c) Determine whether the block completes the loop without falling off, and calculate the normal reaction force N_{top} at the very top of the loop.

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 16.5. (a) From release to spring separation: only the conservative spring force acts. Mechanical energy conservation:

$$E_{m0} = \frac{1}{2}k(\Delta x)^2 = \frac{1}{2}mv_1^2 \implies v_1 = \Delta x\sqrt{\frac{k}{m}}.$$

Computing:

$$v_1 = 0.200\sqrt{\frac{800}{0.500}} = 0.200\sqrt{1600} = 0.200 \times 40.0 = \mathbf{8.00}\ \mathrm{m\,s^{-1}}.$$

(b) Across the rough patch of length d: friction does work $W_f = -\mu_k mgd$. Work-energy theorem:

$$\frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 = -\mu_k mgd \implies v_2^2 = v_1^2 - 2\mu_k gd.$$

Computing:

$$v_2^2 = (8.00)^2 - 2(0.250)(9.81)(0.800) = 64.0 - 3.924 = 60.076\ \mathrm{m^2\,s^{-2}}.$$

$$v_2 = \sqrt{60.076} \approx \mathbf{7.75}\ \mathrm{m\,s^{-1}}.$$

(c) In the smooth vertical loop: only gravity performs work as the block rises to height $2R = 2(0.500) = 1.00$ m. Work-energy theorem to the top of the loop:

$$\frac{1}{2}mv_{\text{top}}^2 - \frac{1}{2}mv_2^2 = -mg(2R) \implies v_{\text{top}}^2 = v_2^2 - 4gR.$$

Computing:

$$v_{\text{top}}^2 = 60.076 - 4(9.81)(0.500) = 60.076 - 19.62 = 40.456\ \mathrm{m^2\,s^{-2}} \implies v_{\text{top}} \approx 6.36\ \mathrm{m\,s^{-1}}.$$

The condition to complete the loop without leaving the track is $v_{\text{top}}^2 \geq gR$:

$$gR = 9.81 \times 0.500 = 4.905\ \mathrm{m^2\,s^{-2}}.$$

Since $v_{\text{top}}^2 = 40.46 \gg 4.91$, the block **comfortably completes the loop**! The normal force at the top is:

$$N_{\text{top}} = m\frac{v_{\text{top}}^2}{R} - mg = \frac{0.500 \times 40.456}{0.500} - 0.500(9.81) = 40.456 - 4.905 \approx \mathbf{35.55}\ \mathrm{N}.$$

Exercise 16.6 · ⋆⋆⋆ **Synoptic Problem: Gravitational Slingshot (Planetary Flyby)**

A spacecraft of mass m approaches Jupiter (mass $M \gg m$) which orbits the Sun at speed $V_J = 13.0\ \text{km}\,\text{s}^{-1}$. In the solar frame, the spacecraft moves with speed $v_0 = 10.0\ \text{km}\,\text{s}^{-1}$ directly toward Jupiter (head-on encounter). (a) Using a Galilean transformation, find the velocity u_i of the spacecraft in Jupiter's rest frame before the encounter. (b) Since gravitational deflection is an elastic scattering in Jupiter's rest frame, the spacecraft swings around Jupiter and emerges traveling in the opposite direction. What is its speed u_f in Jupiter's frame? (c) Transform back to the solar frame to find the final speed v_f of the spacecraft. How much kinetic energy did the spacecraft gain from Jupiter's orbital motion?

Tablet Workspace · Work out your solution, draw diagrams, and write your steps here

Solution 16.6. (a) Choose the x-axis in the direction of Jupiter's orbital motion: $\vec{V}_J = +V_J\hat{\imath} = +13.0\,\hat{\imath}\ \text{km}\,\text{s}^{-1}$. The incoming spacecraft travels toward Jupiter in head-on approach: $\vec{v}_0 = -v_0\hat{\imath} = -10.0\,\hat{\imath}\ \text{km}\,\text{s}^{-1}$. In Jupiter's reference frame $\mathcal{R}_J$, the relative velocity is:

$$\vec{u}_i = \vec{v}_0 - \vec{V}_J = -10.0\,\hat{\imath} - 13.0\,\hat{\imath} = -\mathbf{23.0}\,\hat{\imath}\ \text{km}\,\text{s}^{-1}.$$

The spacecraft approaches Jupiter at speed $u_i = 23.0\ \text{km}\,\text{s}^{-1}$.

(b) In Jupiter's frame, the spacecraft moves in a conservative gravitational hyperbola. By conservation of energy in $\mathcal{R}_J$, the outgoing asymptotic speed equals the incoming asymptotic speed:

$$u_f = u_i = 23.0\ \text{km}\,\text{s}^{-1}.$$

After completing a 180° swing around the planet, its outgoing velocity in Jupiter's frame is reversed:

$$\vec{u}_f = +u_i\hat{\imath} = +\mathbf{23.0}\,\hat{\imath}\ \text{km}\,\text{s}^{-1}.$$

(c) Transforming back to the Sun's inertial frame:

$$\vec{v}_f = \vec{u}_f + \vec{V}_J = +23.0\,\hat{\imath} + 13.0\,\hat{\imath} = +\mathbf{36.0}\,\hat{\imath}\ \text{km}\,\text{s}^{-1}.$$

The final speed is:

$$\boxed{v_f = v_0 + 2V_J = 10.0 + 2(13.0) = \mathbf{36.0}\ \text{km}\,\text{s}^{-1}!}$$

The spacecraft has gained $\Delta v = 26.0\ \text{km}\,\text{s}^{-1}$! Kinetic energy change per unit mass:

$$\Delta\left(\frac{E_c}{m}\right) = \frac{1}{2}v_f^2 - \frac{1}{2}v_0^2 = \frac{1}{2}(36.0^2 - 10.0^2) = \frac{1}{2}(1296 - 100) = \mathbf{598}\ \text{MJ}\,\text{kg}^{-1}.$$

This tremendous energy boost is taken directly from Jupiter's enormous orbital kinetic energy around the Sun (slowing Jupiter down by an utterly imperceptible $\Delta V_J \sim 10^{-21}\ \text{m}\,\text{s}^{-1}$). This is the gravity assist technique used by the Voyager, Cassini, and New Horizons probes!

Appendices

APPENDIX A

Mathematical Reference Toolbox

A.1 Coordinate Systems and Unit Vectors

A.1.1 Cartesian Coordinates (x, y, z)

- Position vector: $\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$
- Displacement: $\mathrm{d}\vec{r} = \mathrm{d}x\hat{\imath} + \mathrm{d}y\hat{\jmath} + \mathrm{d}z\hat{k}$
- Velocity: $\vec{v} = \dot{x}\hat{\imath} + \dot{y}\hat{\jmath} + \dot{z}\hat{k}$
- Acceleration: $\vec{a} = \ddot{x}\hat{\imath} + \ddot{y}\hat{\jmath} + \ddot{z}\hat{k}$

A.1.2 Plane Polar Coordinates (r, θ)

- Relation to Cartesian: $x = r\cos\theta, \quad y = r\sin\theta$
- Basis vectors: $\vec{u}_r = \cos\theta\hat{\imath} + \sin\theta\hat{\jmath}, \quad \vec{u}_\theta = -\sin\theta\hat{\imath} + \cos\theta\hat{\jmath}$
- Derivatives of basis vectors: $\frac{\mathrm{d}\vec{u}_r}{\mathrm{d}t} = \dot{\theta}\vec{u}_\theta, \quad \frac{\mathrm{d}\vec{u}_\theta}{\mathrm{d}t} = -\dot{\theta}\vec{u}_r$
- Position vector: $\vec{r} = r\vec{u}_r$
- Displacement: $\mathrm{d}\vec{r} = \mathrm{d}r\vec{u}_r + r\mathrm{d}\theta\vec{u}_\theta$
- Velocity: $\vec{v} = \dot{r}\vec{u}_r + r\dot{\theta}\vec{u}_\theta$
- Acceleration: $\vec{a} = (\ddot{r} - r\dot{\theta}^2)\vec{u}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\vec{u}_\theta = (\ddot{r} - r\dot{\theta}^2)\vec{u}_r + \frac{1}{r}\frac{\mathrm{d}(r^2\dot{\theta})}{\mathrm{d}t}\vec{u}_\theta$

A.1.3 Cylindrical Coordinates (ρ, θ, z)

- Position vector: $\vec{r} = \rho\vec{u}_\rho + z\hat{k}$
- Velocity: $\vec{v} = \dot{\rho}\vec{u}_\rho + \rho\dot{\theta}\vec{u}_\theta + \dot{z}\hat{k}$
- Acceleration: $\vec{a} = (\ddot{\rho} - \rho\dot{\theta}^2)\vec{u}_\rho + (\rho\ddot{\theta} + 2\dot{\rho}\dot{\theta})\vec{u}_\theta + \ddot{z}\hat{k}$

A.1.4 Spherical Coordinates (r, θ, ϕ)

- Relation to Cartesian: $x = r\sin\theta\cos\phi, \quad y = r\sin\theta\sin\phi, \quad z = r\cos\theta$
- Position vector: $\vec{r} = r\vec{u}_r$
- Velocity: $\vec{v} = \dot{r}\vec{u}_r + r\dot{\theta}\vec{u}_\theta + r\sin\theta\dot{\phi}\vec{u}_\phi$
- Acceleration:

$$
\begin{aligned}
a_r &= \ddot{r} - r\dot{\theta}^2 - r\sin^2\theta\dot{\phi}^2, \\
a_\theta &= r\ddot{\theta} + 2\dot{r}\dot{\theta} - r\sin\theta\cos\theta\dot{\phi}^2, \\
a_\phi &= r\sin\theta\ddot{\phi} + 2\sin\theta\dot{r}\dot{\phi} + 2r\cos\theta\dot{\theta}\dot{\phi}.
\end{aligned}
$$

A.1.5 Intrinsic Coordinates (Frenet-Serret Frame)

- Basis: $\vec{u}_t$ (tangent), $\vec{u}_n$ (principal normal directed toward center of curvature), $\vec{u}_b = \vec{u}_t \times \vec{u}_n$ (binormal)
- Velocity: $\vec{v} = v\,\vec{u}_t$ with $v = \frac{\mathrm{d}s}{\mathrm{d}t}$
- Acceleration: $\vec{a} = a_t\,\vec{u}_t + a_n\,\vec{u}_n = \frac{\mathrm{d}v}{\mathrm{d}t}\vec{u}_t + \frac{v^2}{R}\vec{u}_n$ where R is the local radius of curvature ($R = \frac{\mathrm{d}s}{\mathrm{d}\phi}$).

A.2 Vector Differential Identities

For scalar fields f, g and vector fields $\vec{A}, \vec{B}$:

- $\vec{\nabla}(fg) = f\vec{\nabla}g + g\vec{\nabla}f$
- $\vec{\nabla}\cdot(f\vec{A}) = f(\vec{\nabla}\cdot\vec{A}) + \vec{A}\cdot\vec{\nabla}f$
- $\vec{\nabla}\times(f\vec{A}) = f(\vec{\nabla}\times\vec{A}) + (\vec{\nabla}f)\times\vec{A}$
- $\vec{\nabla}\times(\vec{\nabla}f) = \vec{0}$ (**every gradient field is irrotational / conservative**)
- $\vec{\nabla}\cdot(\vec{\nabla}\times\vec{A}) = 0$ (**every curl field is divergence-free**)
- $\vec{\nabla}\times(\vec{\nabla}\times\vec{A}) = \vec{\nabla}(\vec{\nabla}\cdot\vec{A}) - \Delta\vec{A}$

A.3 Essential Taylor Series Approximations ($x \to 0$)

$$
\begin{aligned}
(1+x)^n &= 1 + nx + \frac{n(n-1)}{2}x^2 + \dots \\
\sqrt{1+x} &= 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \dots \\
\frac{1}{1+x} &= 1 - x + x^2 - x^3 + \dots \\
\frac{1}{\sqrt{1+x}} &= 1 - \frac{1}{2}x + \frac{3}{8}x^2 + \dots \\
e^x &= 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \dots \\
\ln(1+x) &= x - \frac{1}{2}x^2 + \frac{1}{3}x^3 + \dots \\
\sin x &= x - \frac{1}{6}x^3 + \dots \\
\cos x &= 1 - \frac{1}{2}x^2 + \frac{1}{24}x^4 + \dots \\
\tan x &= x + \frac{1}{3}x^3 + \dots
\end{aligned}
$$

APPENDIX B

Physical Constants and Astronomical Data

B.1 Fundamental Physical Constants

Constant	Symbol	Standard Value	SI Units
Gravitational constant	G	$6.67430(15) \times 10^{-11}$	$\mathrm{m^3\,kg^{-1}\,s^{-2}} = \mathrm{N\,m^2\,kg^{-2}}$
Speed of light in vacuum	c	299 792 458 (exact)	$\mathrm{m\,s^{-1}}$
Standard gravity near Earth	g	9.80665 (exact standard)	$\mathrm{m\,s^{-2}} = \mathrm{N\,kg^{-1}}$
Elementary charge	e	$1.602176634 \times 10^{-19}$ (exact)	C
Vacuum permittivity	ε_0	$8.8541878128 \times 10^{-12}$	$\mathrm{F\,m^{-1}} = \mathrm{C^2\,N^{-1}\,m^{-2}}$
Coulomb constant	$k_e = \frac{1}{4\pi\varepsilon_0}$	$8.9875517923 \times 10^{9}$	$\mathrm{N\,m^2\,C^{-2}}$
Planck constant	h	$6.62607015 \times 10^{-34}$ (exact)	$\mathrm{J\,s} = \mathrm{kg\,m^2\,s^{-1}}$
Avogadro constant	N_A	$6.02214076 \times 10^{23}$ (exact)	$\mathrm{mol^{-1}}$
Atomic mass unit	1 u	$1.660539066 \times 10^{-27}$	kg
Electron mass	m_e	$9.1093837 \times 10^{-31}$	kg
Proton mass	m_p	$1.6726219 \times 10^{-27}$	kg
Neutron mass	m_n	$1.6749275 \times 10^{-27}$	kg

B.2 Earth, Moon, and Solar Data

Quantity	Symbol	Value	SI Units
Earth's mass	M_E	5.9722×10^{24}	kg
Earth's equatorial radius	R_e	6.3781×10^{6}	m
Earth's polar radius	R_p	6.3568×10^{6}	m
Earth's mean volumetric radius	R_E	6.3710×10^{6}	m
Earth's angular rotation rate	Ω_E	7.292115×10^{-5}	$\mathrm{rad\,s^{-1}}$
Earth's sidereal day	T_{sidereal}	86 164.1 (23 h 56 m 4.1 s)	s
Moon's mass	M_M	7.342×10^{22}	kg
Moon's mean radius	R_M	1.7374×10^{6}	m
Earth-Moon mean distance	d_{EM}	3.844×10^{8}	m
Sun's mass	$M_\odot$	1.9885×10^{30}	kg
Sun's radius	$R_\odot$	6.957×10^{8}	m
Astronomical Unit (mean Sun-Earth distance)	1 AU	$1.495978707 \times 10^{11}$ (exact)	m
Solar gravitational parameter	$\mu_\odot = GM_\odot$	1.3271244×10^{20}	$\mathrm{m^3\,s^{-2}}$
Earth gravitational parameter	$\mu_E = GM_E$	3.9860044×10^{14}	$\mathrm{m^3\,s^{-2}}$

APPENDIX C

Comprehensive Formula Sheet and Problem-Solving Protocols

C.1 The Five-Step Mechanics Problem-Solving Protocol

Whenever you encounter a mechanics problem, systematically follow this standard five-step algorithm:

1. **Define the System and the Reference Frame**:
   - Explicitly specify the particle(s) or body being studied.
   - Choose an appropriate reference frame $\mathcal{R}$ (Galilean whenever possible; otherwise identify the non-inertial acceleration $\vec{a}_e$ and angular velocity $\vec{\Omega}$).
   - Select the most natural coordinate system (Cartesian for uniform fields; Polar/Cylindrical for circular/axial symmetry; Spherical for central fields; Frenet for known trajectories).
2. **Perform the Comprehensive Force Inventory (Free-Body Diagram)**:
   - Field forces (at a distance): Weight $\vec{P} = m\vec{g}$, Gravitation $\vec{F}_g = -G\frac{Mm}{r^2}\vec{u}_r$, Electrostatic $\vec{F}_e = q\vec{E}$.
   - Contact forces: Normal reaction $\vec{N}$ (orthogonal to surface), Friction $\vec{f}$ (tangential, static $f_s \leq \mu_s N$ or kinetic $f_k = \mu_k N$), Tension $\vec{T}$ (along string/rope), Spring restoring force $\vec{F}_s = -k(\ell - \ell_0)\vec{u}$.
   - Fictitious forces (if in non-inertial frame): $\vec{F}_{ie} = -m\vec{a}_e$ and $\vec{F}_{ic} = -2m\vec{\Omega} \times \vec{v}_r$.
3. **Select the Optimal Fundamental Principle**:
   - If seeking accelerations or forces at an instant → **Newton's Second Law**: $\sum \vec{F} = m\vec{a}$.
   - If relating speeds and positions without tracking time → **Work-Energy Theorem**: $\Delta E_c = W_{\text{all forces}}$ or **Conservation of Mechanical Energy**: $\Delta E_m = W_{\text{nc}}$.
   - If dealing with rotational, central, or orbital motion → **Angular Momentum Theorem**: $\frac{\mathrm{d}\vec{L}_O}{\mathrm{d}t} = \vec{\tau}_O$.
   - If dealing with sudden collisions, impacts, or explosions → **Conservation of Linear Momentum**: $\vec{P}_{\text{tot}} = \text{const}$.
4. **Project and Integrate**:
   - Project the vector equation onto the coordinate axes.
   - Separate variables and integrate with respect to time or space using initial conditions.
5. **Sanity Checks**:
   - **Dimensional check**: Verify that both sides have the same physical units!
   - **Limiting cases**: Check limits (e.g. $\theta \to 0$, $m_1 \gg m_2$, friction $\mu \to 0$, $t \to \infty$).
   - **Physical plausibility**: Is the sign reasonable? Is the numerical order of magnitude realistic?

C.2 Core Formula Quick-Reference Sheet

C.2.1 Kinematics

$$\begin{aligned}
\text{1D Constant Acceleration:} &\quad v(t) = at + v_0, \quad x(t) = \tfrac{1}{2}at^2 + v_0 t + x_0, \quad v^2 - v_0^2 = 2a(x - x_0) \\
\text{Frenet Frame:} &\quad \vec{v} = v\,\vec{u}_t, \quad \vec{a} = \dot{v}\,\vec{u}_t + \frac{v^2}{R}\vec{u}_n \\
\text{Polar Coordinates:} &\quad \vec{v} = \dot{r}\vec{u}_r + r\dot{\theta}\vec{u}_\theta, \quad \vec{a} = (\ddot{r} - r\dot{\theta}^2)\vec{u}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\vec{u}_\theta \\
\text{Composition of Velocities:} &\quad \vec{v}_a = \vec{v}_r + \vec{v}_e \quad (\vec{v}_e = \vec{v}(O') + \vec{\Omega} \times \vec{r}') \\
\text{Composition of Accelerations:} &\quad \vec{a}_a = \vec{a}_r + \vec{a}_e + \vec{a}_c \quad (\vec{a}_c = 2\vec{\Omega} \times \vec{v}_r)
\end{aligned}$$

C.2.2 Dynamics and Forces

$$\begin{aligned}
\text{Newton's 2nd Law (Inertial):} &\quad \sum \vec{F} = m\vec{a} = \frac{\mathrm{d}\vec{p}}{\mathrm{d}t} \\
\text{Newton's 2nd Law (Non-Inertial):} &\quad m\vec{a}_r = \sum \vec{F}_{\text{real}} - m\vec{a}_e - 2m\,\vec{\Omega} \times \vec{v}_r \\
\text{Coulomb Solid Friction:} &\quad \text{Static: } f_s \le \mu_s N, \quad \text{Kinetic: } f_k = \mu_k N \\
\text{Hooke's Law:} &\quad \vec{F}_s = -k(x - x_0)\hat{\imath}
\end{aligned}$$

C.2.3 Work, Energy, and Oscillations

$$\begin{aligned}
\text{Work-Energy Theorem:} &\quad \Delta E_c = W_{A\to B}(\sum \vec{F}) \\
\text{Conservative Force:} &\quad \vec{F} = -\vec{\nabla}E_p \iff \vec{\nabla} \times \vec{F} = \vec{0} \\
\text{Total Mechanical Energy:} &\quad E_m = E_c + E_p, \quad \frac{\mathrm{d}E_m}{\mathrm{d}t} = P_{\text{nc}} = \vec{F}_{\text{nc}} \cdot \vec{v} \\
\text{Small Oscillations around } x_0: &\quad \omega_0 = \sqrt{\frac{k_{\text{eff}}}{m}}, \quad k_{\text{eff}} = \left.\frac{\mathrm{d}^2 E_p}{\mathrm{d}x^2}\right|_{x_0} > 0 \\
\text{Damped Oscillator:} &\quad \ddot{x} + 2\lambda\dot{x} + \omega_0^2 x = 0, \quad \omega_d = \sqrt{\omega_0^2 - \lambda^2}, \quad Q = \frac{\omega_0}{2\lambda}
\end{aligned}$$

C.2.4 Angular Momentum and Central Forces

$$\begin{aligned}
\text{Angular Momentum Theorem:} &\quad \frac{\mathrm{d}\vec{L}_O}{\mathrm{d}t} = \vec{\tau}_O = \vec{r} \times \vec{F} \\
\text{Central Force Conservation:} &\quad \vec{L}_O = \text{const} \implies r^2\dot{\theta} = C = \text{const} \\
\text{Effective Potential:} &\quad U_{\text{eff}}(r) = U(r) + \frac{L^2}{2mr^2} \\
\text{Vis-Viva Equation (Kepler):} &\quad v^2 = GM\left(\frac{2}{r} - \frac{1}{a}\right), \quad T^2 = \frac{4\pi^2}{GM}a^3
\end{aligned}$$

C.2.5 Systems and Collisions

$$\begin{aligned}
\text{Center of Mass:} \quad & M\vec{a}_G = \sum \vec{F}_{\text{ext}} \\
\text{1D Restitution Coefficient:} \quad & e = -\frac{v_{1f} - v_{2f}}{v_{1i} - v_{2i}} \in [0, 1] \\
\text{Koenig's First Theorem:} \quad & E_c = \frac{1}{2} M v_G^2 + \frac{1}{2} \mu v_{\text{rel}}^2, \quad \mu = \frac{m_1 m_2}{m_1 + m_2} \\
\text{Tsiolkovsky Rocket:} \quad & \Delta v = u_{\text{ex}} \ln\left(\frac{m_0}{m_f}\right)
\end{aligned}$$